

# Flavor interactions, physical CP and exact constraint geometry

Finite symmetries, canonical quark matching, soft mediation and  
coupled one-loop evolution

PerfectPower research monograph  
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A declared Hermitian-source action supports finite hierarchical matching and physical weak CP with termwise one-loop mass-phase protection. Locally stable branches, exact invariant constraints, CP operator counts and coupled one-loop evolution are executable. A resolved lower CP-conserving competitor now establishes the selection boundary for the retained minimal branch.

Exact representation and polynomial results are distinguished from numerical vacuum calculations. The constructed interactions protect their stated structures; the golden CKM coefficient and physical 66-degree phase remain unpredicted.

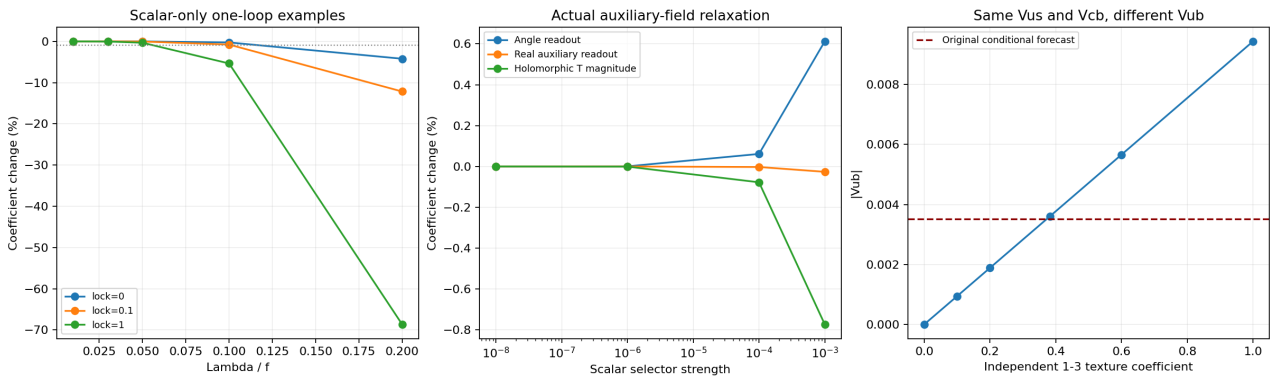
## What this push accomplishes

The subsequent shared-mediator investigation is documented in B5\_FLAVOR\_MEDIATOR.md and included as the second part of the reader monograph. It derives exact canonical tree matching, establishes a scoped low-degree alignment obstruction, and tests an adjoint alternative. It transmits an initialized golden frame but does not derive that frame from symmetry.

The candidate now has an explicit renormalizable supersymmetric singlet sector, a holomorphic golden-coefficient extension, an actual auxiliary-field relaxation calculation, and a one-loop calculation including every scalar and fermion in that sector. The 34-field extension has 68 real scalar modes and 34 Weyl fermions. In the declared numerical examples, a small scalar energy selector prefers the branches near plus and minus 66 degrees, and the one-loop corrections leave that preference intact among all sixteen continued branches. The construction produces a quantitatively controlled realization of the nominated vacuum. It does not derive the initial superpotential coefficients or the energy-selector pattern from an independently specified symmetry, and it does not derive the nominated exact CKM relation from quark interactions.

The strongest negative result is exact: the scalar-only version of the potential is not closed under its own one-loop divergent counterterm. Polynomial division yields a nonzero torque remainder, and a Bezout identity shows that the remainder vanishes at none of the primitive real order-60 roots. Thus there is a specific radiative protection problem, even before quarks or gauge fields are introduced. The strongest constructive result is an initialized holomorphic constraint circuit whose supersymmetric zeros are perturbatively protected. Soft selection then produces small calculable departures instead of an exact unchanged angle.

The quark calculation exposes another independent issue. The declared flavor charges allow the required hierarchy of operators, but their coefficients are independent. Exact diagonalization of a selected triangular texture, even with its 1-3 coefficient initialized to the golden value, gives a different observable mixing strength. This is a constructive counterexample to assuming that the right spurion powers automatically imply the exact CKM formula.



## The original exact carrier and energy winner

$$V'_{\text{lock}}(x) = P(x) [2\lambda P'(x) + \eta Q(x)]$$

Define  $x = 2 \cos(\theta)$ ,  $P(x) = x^8 - 7x^6 + 14x^4 - 8x^2 + 1$  and  $Q(x) = -1 + 3x - x^3$ . The previously constructed potential obeys  $V'(x) = P(x)Q(x)$ . Its six nonconstant angular terms are  $\cos(2\theta) + (2/3)\cos(3\theta) + (2/5)\cos(5\theta) + (1/4)\cos(8\theta) - (2/9)\cos(9\theta) - (1/6)\cos(12\theta)$ .  $V$  differs from that angular expression by the constant  $11/12$ . Exact rational Sturm isolation and energy enclosures selected  $x = 2 \cos(66^\circ)$  on  $[-2, 2]$ , with more than 0.060

separation from the next competing stationary energy in the polynomial normalization.

The new family  $V_{\text{lock}}(x) = \lambda P(x)^2 + \eta V(x)$ ,  $\lambda \geq 0$  and  $\eta > 0$ , obeys  $V_{\text{lock}}'(x) = P(x)[2\lambda P'(x) + \eta Q(x)]$ . Its global winner is unchanged: the added square is nonnegative, vanishes at the old winner, and the original potential already has a unique global winner in  $x$ . This statement covers the full declared parameter orthant without discretizing  $\lambda$  or  $\eta$ . The two full-circle CP partners remain degenerate. This is a mathematical family, not an assertion that the interaction generating  $\lambda$  or  $\eta$  has been identified.

The rational stationary-family constraint is stronger than stationarity under arbitrary real couplings at just one chosen angle. Rational coefficients force Galois-conjugate roots to share the polynomial factor. An isolated real linear torque condition at a prescribed angle is not equivalent to the eight rational relations found for the twelve-harmonic carrier. The field model must explain which of those structures it actually enforces.

## A precise scalar-loop obstruction

$$h(x) = (4 - x^2)V''(x) - xV'(x), \quad \gcd(P, (h^2)') = 1$$

Give the angle the kinetic term  $f^2 (\partial \theta)^2/2$  and the physical potential  $\Lambda^4 V(2 \cos(\theta))$ . With  $D(x) = 4 - x^2$ , its dimensionless angular curvature is  $h(x) = D(x)V''(x) - xV'(x)$ . The physical field-dependent squared mass is  $\Lambda^4 h(x)/f^2$ . The one-loop scalar divergence contains a coefficient proportional to this mass to the fourth power. Consequently the question of preserving the stationary family reduces to whether  $P$  divides the  $x$  derivative of  $h^2$ .

For the original potential, the exact remainder is  $R(x) = 5040 - 6880x - 31360x^2 + 43600x^3 + 24160x^4 - 30080x^5 - 4400x^6 + 4800x^7$ . Its gcd with  $P$  is one. The stored rational Bezout multipliers satisfy  $a(x)P(x) + b(x)R(x) = 1$ . Since  $\sin(\theta)$  is nonzero at every primitive order-60 angle, a nonzero  $x$  torque also means a nonzero angular torque. This establishes that the original family is not closed under this particular divergent scalar counterterm. It does not establish that every model containing additional fields must fail.

The coefficient field can also be included. Write  $c(x) = 1 - S_{12}(x)$ , where  $S_n(2 \cos(\theta)) = 2 \cos(n\theta)$ , and take  $F(\theta, C) = V_{\text{lock}}(2 \cos(\theta)) + \kappa[C - c(x)]^2$ . On its tree valley, with kinetic scales  $f$  and  $g$ , the polynomial proportional to  $\text{Tr}(M^4)/\Lambda^8$  is  $[h + 2\kappa D(c')^2]^2/f^4 + 8\kappa^2 D(c')^2/(f^2 g^2) + 4\kappa^2/g^4$ . The derivative is again reduced modulo  $P$ . All sixteen declared angular-only and two-scalar parameter cases give nonzero remainders. The calculation includes the coefficient scalar instead of silently integrating it out before evaluating the loops.

## Finite scalar-loop displacements

$$\Delta V = \frac{m^4}{64\pi^2} \left[ \log \frac{m^2}{\mu^2} - \frac{3}{2} \right], \quad \Delta\theta = -\frac{\Delta U'}{U''}$$

The scalar-only numerical calculation uses the local positive-mass  $\overline{\text{MS}}$  expression  $\Delta V = m^4 [\log(m^2/\mu^2) - 3/2]/(64\pi^2)$ . The renormalized additional finite Wilson coefficients are initialized to zero at the stated  $\mu$ . Every example declares  $\Lambda/f$ , the lock coefficient and  $\mu/m_{\text{tree}}$ . Resetting that boundary condition at another  $\mu$  defines another example; it is not a claim that a fixed physical theory depends on the arbitrary renormalization scale after consistent running.

At  $\theta_0=66$  degrees,  $U''=12.2092424208$  and  $U'''=-140.847974475$ . The first-order angular displacement is  $-\Delta U'(\theta_0)/U''(\theta_0)$ . For  $\Lambda/f=0.1$ , no added lock, and  $\mu=m_{\text{tree}}$ , the numerical stationary point is approximately 65.997446321 degrees. The angle-based coefficient changes by about -0.2663 percent. The small-ratio displacement scales as  $(\Lambda/f)^4$ . The derivative and curvature are independently checked against finite differences, and the nonlinear stationary solution is compared with its linear prediction.

The sweep retains sixty angular examples. One leaves the physical positive-depth CKM chart; it remains in the results with a null CKM forecast rather than being dropped. Eighty of eighty-one declared two-scalar examples produce a controlled local stationary solution with a positive numerical Hessian. The unsuccessful case is retained with its parameters and failure reason. These calculations certify neither a global quantum minimum nor a cosmological occupation probability.

At the tree portal  $C=1-2 \cos(12 \theta)$ ,  $dC/d\theta=22.8253563911$  at the selected angle. A one-percent coefficient tolerance corresponds locally to about 0.00958804 degrees. Adding a large classical locking term is not automatically a radiative solution: it changes the mass and its derivatives, which also changes its own loop torque. The stored sweeps illustrate this effect directly.

## A cubic supersymmetric constraint circuit

$$z_n = z^n, \quad \Phi_{60}(z) = z^{16} + z^{14} - z^{10} - z^8 - z^6 + z^2 + 1 = 0$$

Introduce sixteen gauge-singlet chiral fields  $Z_1, \dots, Z_{16}$ , with  $Z_1=Z$ , and sixteen drivers  $D_2, \dots, D_{16}, D_{\text{star}}$ . All  $Z$  fields have  $R$  charge zero; all drivers have  $R$  charge two. The declared CP action is complex conjugation and the chosen coefficients are real. A common real coupling can multiply the superpotential. With a real positive reference mass  $M$ , the base superpotential is  $W = \sum_{n=2}^{16} D_n (M Z_n - Z_{n-1}) + M D_{\text{star}} [Z_{16} + Z_{14} - Z_{10} - Z_8 - Z_6 + Z_2 + M]$ . Every term has mass dimension three.

At  $D=0$ ,  $F$ -flatness gives  $z_n = z^n$ , where  $z_n = Z_n/M$ , and  $\Phi_{60}(z) = z^{16} + z^{14} - z^{10} - z^8 - z^6 + z^2 + 1 = 0$ . There are sixteen simple complex roots. They have unit modulus and are the primitive order-60 roots. This is an explicit renormalizable circuit implementing the algebraic equation; it avoids introducing a degree-sixteen superpotential interaction in the fundamental variables.

The constraint Jacobian  $J$  has nonzero determinant at every root. Its determinant divided by  $\Phi_{60}'(z)$  is minus one in the declared ordering. Thus the chosen vacua are isolated. The full holomorphic mass matrix has blocks  $[[0, J^T], [J, 0]]$ , and every singular value of  $J$  occurs twice. With canonical kinetic terms, each pair has four real scalar modes and four fermionic spin modes with equal masses. The unbroken-SUSY one-loop contribution cancels at the  $F$ -flat zeros.

Perturbative protection here is the Wilsonian superpotential non-renormalization result for an initialized, globally supersymmetric sector, with a nonsingular positive Kahler metric. It does not force the initial coefficient choices. Kinetic corrections do not remove a zero of all  $F$  terms, but they change spectra and matter when supersymmetry is broken. This distinction follows the Wilsonian treatment of non-renormalization, rather than treating a generic 1PI potential as an unchanged superpotential [2].

## What the declared symmetry actually permits

For the base singlet sector, the complete holomorphic superpotential basis through field degree three has 2,448 monomials. R charge forces one driver, followed by zero, one or two neutral fields:  $16(1+16+136)$ . The selected circuit uses 37 monomials. CP and the declared R symmetry allow many other coefficients. They protect neither the chosen sparse pattern nor its relative integer values by themselves. The general CP-even quadratic Kahler metric contains 272 real parameters in the two neutral/driver blocks; the numerical mass calculations initialize it canonically.

A real change of the  $D_{\text{star}}$  tadpole changes  $\Phi_{60}(z)$  to  $\Phi_{60}(z)+\epsilon$ . This deformation is allowed by the declared symmetries and changes both phase and radius. It demonstrates freedom in the initialized theory. It is not a claim that perturbative supersymmetric loops generate that absent term in the Wilsonian superpotential.

For a separate one-complex-field comparison, all CP-even operators  $\text{Re}(z^a \bar{z}^b)$ ,  $a \geq b$ ,  $1 \leq a+b \leq 12$ , are enumerated. There are 48 operators when no rotation symmetry is imposed; 42 have a nonzero carrier torque. The six desired harmonic numbers have gcd one. Among all cyclic rotations tested, orders two through sixty cannot permit every desired harmonic. This is a precise obstruction for that one-field CP-plus-cyclic class, not for arbitrary multi-field finite groups.

## A holomorphic golden-coefficient extension

$$T = 2 - z^4 - z^6 + z^{14}, \quad T^2 - 3T + 1 = 0$$

The inverse phase needed for the original coefficient can be represented in the carrier without complex conjugation. Since  $z^{60}=1$  on  $\Phi_{60}=0$ ,  $z^{-12}=z^{48}$ . Exact reduction gives  $T=1-z^{12}-z^{48}=2-z^4-z^6+z^{14}$  modulo  $\Phi_{60}$ . It obeys  $T^2-3T+1=0$ . At the root with phase 66 degrees,  $T=(3-\sqrt{5})/2$ , with zero imaginary part.

Two additional chiral fields  $T$  and  $D_{\text{gold}}$  implement this linearly:  $W_{\text{gold}}=M D_{\text{gold}}[T-2M+Z_4+Z_6-Z_{14}]$ .  $T$  has R charge zero and  $D_{\text{gold}}$  has charge two. The extended model has 34 chiral fields and remains renormalizable. Its general allowed singlet superpotential through degree three contains 2,907 monomials. The extension fixes a golden spurion value in the selected initialized vacuum, while preserving the sixteenfold supersymmetric degeneracy. It does not fix a quark Wilson coefficient or explain why the twelfth power is the desired coefficient operator.

## Actual field relaxation under energy selection

To lift the degeneracy, the worked scalar deformation is  $V_{\text{soft}}/M^4=g_W^2 \epsilon \sum a_n \text{Re}(z_n)$ , with the six nonconstant Fourier coefficients  $a_n$ . This is an explicitly chosen supersymmetry-breaking scalar tadpole pattern. At exact circuit roots its leading energy is the original angular potential times  $g_W^2 \epsilon$ . The full fields are then allowed to relax; they are not constrained to remain equal to powers of a unit-modulus number.

For fixed  $z_1$ , the other fifteen base neutral fields enter the F-term potential quadratically. They are eliminated by linear least squares, including the scalar tadpoles. The remaining complex  $z_1$  is optimized in both its real and imaginary coordinates. All sixteen branches are continued for  $\epsilon=10^{-8}, 10^{-6}, 10^{-4}, 10^{-3}$ . The branches descended from root powers eleven and forty-nine have the lowest energies in every declared continuation. Every retained branch has a positive reduced tree Hessian. This is a comparison of local continued vacua, not a proof that the

full polynomial field potential has no distant competing vacuum or runaway.

At  $\epsilon=10^{-4}$ , the positive branch has phase 66.0005874177 degrees and radius 1.00002214970. The angle-only readout gives  $C=0.382200029974$ . The nonholomorphic real auxiliary readout  $1-2 \operatorname{Re}(z_{12})$  gives 0.381955432748. The holomorphic extension gives  $T=0.381669772701-0.000178406720 i$ . These are different operators once the fields relax. A quark interaction must specify which operator it actually uses. Choosing a real auxiliary readout in a holomorphic superpotential would require a different construction; it cannot be substituted silently.

## Full-circuit leading quantum calculation

$$A = J^\dagger J, \quad B = \sum_k \overline{F_k} H_k$$

The full quantum calculation includes the 34 chiral fields of the golden extension. On the R-symmetric  $D=0$  slice, let  $F$  be the vector of driver constraints,  $J$  their Jacobian with respect to the seventeen neutral fields, and  $H_k$  their constant quadratic Hessians. Define  $A=J^\dagger J$  and  $B=\sum \operatorname{conjugate}(F_k) H_k$ . In canonical  $\sqrt{2}$  real and imaginary neutral coordinates, the scalar squared-mass matrix is  $[[\operatorname{Re}(A+B), -\operatorname{Im}(A)-\operatorname{Im}(B)], [\operatorname{Im}(A)-\operatorname{Im}(B), \operatorname{Re}(A-B)]]$ . The driver scalar block has the corresponding unsplit singular-value spectrum.

The driver scalar trace cancels half of the fermion trace. The complete singlet one-loop potential therefore equals  $g_W^4/(64 \pi^2)$  times  $[\sum_{(34 \text{ neutral real modes})} b_i^2 (\log(g_W^2 b_i/\mu^2)-3/2)-2 \sum_{(17 \text{ Jacobian modes})} a_i^2 (\log(g_W^2 a_i/\mu^2)-3/2)]$ . This expression accounts for all 68 real scalars and 34 Weyl fields, not just the light angular mode. It is evaluated in a DRbar example with canonical renormalized Kahler metric and specified scalar tadpoles at the reference scale. There are no gauge fields in this singlet-sector calculation.

The gradient uses eigenvector contractions with the analytic derivatives of  $A$  and  $B$ . It is independently checked against potential differences. At exact F-flat roots, the potential and gradient cancel to numerical roundoff. At the softly selected vacuum, the gradients need not cancel. The quantum stationary equations are solved in all thirty-four real neutral coordinates; the corrected neutral-sector Hessian is numerically positive. An independent expression for the complete 68-scalar, 34-Weyl effective potential also computes the quantum driver-block Hessian by finite differences. The unbroken common driver R phase makes the neutral-driver block vanish at zero drivers. Both corrected blocks have positive minimum eigenvalues in all nine declared cases. Step-size comparison checks the driver Hessian, while the agreement of the independent full and reduced potentials checks the mode counting. These are numerical local stability checks at one loop, rather than exact certificates or a higher-loop theorem.

Nine epsilon/coupling examples are retained:  $\epsilon=10^{-6}, 10^{-4}, 10^{-3}$  and  $g_W=0.1, 0.3, 1$ . All produce controlled numerical neutral-sector minima. Leading tree-plus-one-loop energy comparisons among all sixteen continued branches retain root powers eleven and forty-nine. At  $\epsilon=10^{-4}$  and  $g_W=0.3$ , the phase becomes 66.0005860473 degrees, a quantum displacement of approximately  $-1.37037$  times  $10^{-6}$  degrees relative to the relaxed tree minimum. The coefficient field becomes  $T=0.381670086337-0.000178010587 i$ . The largest actual neutral-coordinate quantum displacement is about 4.67 times  $10^{-7}$ . These are chosen weak-coupling examples, not experimental uncertainties or a derived cosmological scale.

## Quark operators: hierarchy without an enforced coefficient

The declared holomorphic quark sector uses two flavor charges. The left-handed doublets  $Q_i$  have charges (1,1),(0,1),(0,0), while  $u_i^c$  and  $d_i^c$  have the opposite charges.  $H_u, H_d$  and all neutral circuit fields have zero flavor charge.  $A$  has charge (-1,0),  $B$  has charge (0,-1), and  $Q, f^c$  have  $R$  charge one. The scalar  $A$  and  $B$  backgrounds remain free inputs. No anomaly-free gauged flavor completion or mechanism fixing their magnitudes is claimed.

All allowed operators  $Q_i H f_j^c A^a B^b$  times neutral monomials are enumerated through superpotential field degree six. There are 6,460 operators with the sixteen base neutral fields, and 7,560 when  $T$  is included. The up and down sectors have independent real coefficients before spontaneous CP breaking. The pure singlet enumeration and this Yukawa enumeration are complete within their declared sectors and degree bounds; they are not an enumeration of all Standard Model, lepton or higher-Kahler interactions.

In particular,  $Q_1 H d_2^c A$ ,  $Q_2 H d_3^c B$ , and  $Q_1 H d_3^c A B$  carry independent coefficients. Allowing their spurion powers does not relate those coefficients. A  $T$  insertion can multiply the third operator, but its independent coefficient and the lower-degree  $A B$  operator remain allowed. Neutral fields can also appear in diagonal and up-sector entries. The phase of the scalar vacuum therefore does not automatically become the standard CKM phase. Higher-Kahler interactions such as  $Q_1^\dagger Q_2 A^\dagger/M$  are allowed and can alter canonical normalization.

## A derived texture counterexample

$$V_{\text{CKM}} = U_u^\dagger U_d, \quad U_f^\dagger Y_f Y_f^\dagger U_f = \text{diag}(y_f^2)$$

The worked trial uses  $Y_u = \text{diag}(u_1, u_2, u_3)$  and an upper-triangular  $Y_d$  with entries  $Y_{12} = a d_2$ ,  $Y_{23} = b d_3$  and  $Y_{13} = c a b \exp(-i \theta) d_3$ . The diagonal spectra are declared positive numerical inputs. This is a selected member of the allowed operator family. Up-sector off-diagonal terms and most neutral insertions are initialized to zero; the symmetry does not forbid them.

The matrices are diagonalized through the eigenvectors of  $Y Y^\dagger$ . For each independent  $c$ , only  $a$  and  $b$  are matched to the two existing anchors  $|V_{us}| = 0.22431$  and  $|V_{cb}| = 0.0411$ . The values of  $|V_{ub}|$  and CP observables are then consequences of the matrix, without fitting them. At  $c = \phi^{-2}$  and  $\theta = 66$  degrees, the texture gives  $|V_{ub}| = 0.00360299463$  and effective depth 0.390811666690, rather than the nominated chart's 0.00352144218 and 0.381966011250. Its standard CKM phase is approximately 65.9786293 degrees. Finite diagonalization effects matter even in this simple hierarchical matrix.

Changing the independent coefficient while retaining the same two anchors changes  $|V_{ub}|$  substantially. The stored six-coefficient sweep supplies explicit examples. This is a mathematical obstruction to inferring the exact relation  $s_{13} = C s_{12} s_{23}$  from charge arithmetic alone. A stronger quark construction must relate Wilson coefficients, kinetic normalization and matrix diagonalization, or predict the correction instead of absorbing it into a fit.

## Scale evolution as a separate matching test

$$16\pi^2 \frac{dY_u}{d \log \mu} = \left[ \frac{3}{2}(H_u - H_d) + \alpha_u I \right] Y_u$$

The implementation also integrates one-loop Standard Model matrix Yukawa equations using  $t=\log(\mu)$ , left-row Yukawa convention and SU(5)-normalized  $g_1$ . In this convention  $\beta_u=[(3/2)(H_u-H_d)+T_{\text{gauge},u}]Y_u/(16\pi^2)$ , with the corresponding down and lepton equations and the one-loop gauge equations. The source's  $\log(\mu^2)$  convention is converted explicitly [3]. Rephasing invariance, a common-left-basis covariance check and the top-only limit independently check the implementation.

The example initializes the original exact CKM chart with declared illustrative spectra and gauge couplings. At  $\mu/\mu_0=10^8$ , its effective depth differs from  $\phi^{-2}$  by about -8.62 times  $10^{-5}$  relatively, and its standard phase differs by about -0.00349036 degrees. The absolute  $|V_{ub}|$  also changes as the anchors run. No measured matching scale is supplied. This is a standalone SM evolution diagnostic, not a threshold-matched evolution of the supersymmetric circuit. It shows that a claim of an exact flavor relation must identify the scale at which it is imposed.

## What is now supported and what remains physical

The exact supports are the carrier algebra, the stationary-family identities, the global-winner preservation argument, the scalar-counterterm remainder and Bezout witness, and the complete declared finite operator spaces. The supersymmetric protection is a conditional application of the established Wilsonian non-renormalization result to the supplied globally supersymmetric circuit. The mass spectra, relaxed fields, one-loop extrema, quark diagonalizations and SM evolution are numerical computations with explicit parameters and limitations. No new Lean theorem is claimed.

The constructed model answers whether an initialized protected equation can be implemented by renormalizable fields and whether a specified weak selector produces a controlled nearby vacuum. It does so. The outstanding physics is the origin of the sparse initialized coefficients and selector terms, a quark interaction enforcing or replacing the nominated exact mixing rule, consistent kinetic and threshold matching, higher-loop stability, and a vacuum history. The branch comparison does not choose the positive CP sign. A complete theory of strong CP or baryogenesis has not been supplied.

## Reproduction and retained calculations

Run `PYTHONPATH=python OPENBLAS_NUM_THREADS=1 python python/develop_flavor_completion.py` from the repository root, then the corresponding `plot_flavor_completion.py`. Run `unittest discovery with test_flavor_completion.py` for the new mathematical and numerical checks, or `test_flavor*.py` for the combined flavor checks. NumPy and SciPy support numerical spectra, extrema and running; Matplotlib supports the scientific figure. Exact polynomial identities and operator labels use rational arithmetic and the existing PerfectPower algebra.

The receipts retain all declared cases, including one uncontrolled two-scalar continuation and one angular example outside the positive-depth CKM chart. The two exhaustive quark-operator tables are stored as deterministic gzip-compressed JSON; ordinary gzip decompression recovers the complete tables. The manifest covers the scientific JSON outputs, including those compressed tables. Fresh-copy reproduction checks those outputs in the same numerical runtime. A byte-identical replay is evidence of reproducibility in that environment; the tests provide separate mathematical and numerical checks. PDF generation uses ReportLab and standard installed fonts. Source code, prior repository work, raw receipts, diagrams and the monograph are included in the complete handoff archive.

The preceding vacuum-completion run completed 841 tests successfully, with four skips. Its 25 new completion tests cover the exact carrier obstruction, circuit elimination and mode pairing, independent loop gradients, corrected driver Hessian, complete declared operator enumeration, texture counterexamples and scale-evolution covariance. A fresh archive extraction reproduced all twenty scientific receipt, manifest and figure files byte for byte in the declared runtime. No new Lean theorem was added in that push.

## Primary references

- [1] S. Coleman and E. Weinberg, Radiative Corrections as the Origin of Spontaneous Symmetry Breaking, *Physical Review D* 7, 1888 (1973). <https://doi.org/10.1103/PhysRevD.7.1888>. This supplies the effective-potential framework; the worked scalar and circuit schemes are stated explicitly above.
- [2] N. Seiberg, Naturalness Versus Supersymmetric Non-renormalization Theorems (1993), <https://arxiv.org/abs/hep-ph/9309335>. The relevant distinction is protection of a chosen Wilsonian superpotential, with separate kinetic and supersymmetry-breaking effects. It does not derive the chosen flavor coefficients.
- [3] A. V. Bednyakov, A. F. Pikelner and V. N. Velizhanin, Three-loop SM beta-functions for matrix Yukawa couplings (2014), <https://arxiv.org/abs/1406.7171>. The one-loop matrix conventions and scale conversion are the parts used here.
- [4] I. de Medeiros Varzielas, D. Emmanuel-Costa and P. Leser, Geometrical CP violation from non-renormalisable scalar potentials (2012), <https://arxiv.org/abs/1204.3633>. This is a comparison for symmetry-constrained higher-order operators preserving particular vacua; it does not establish the present order-60 construction.

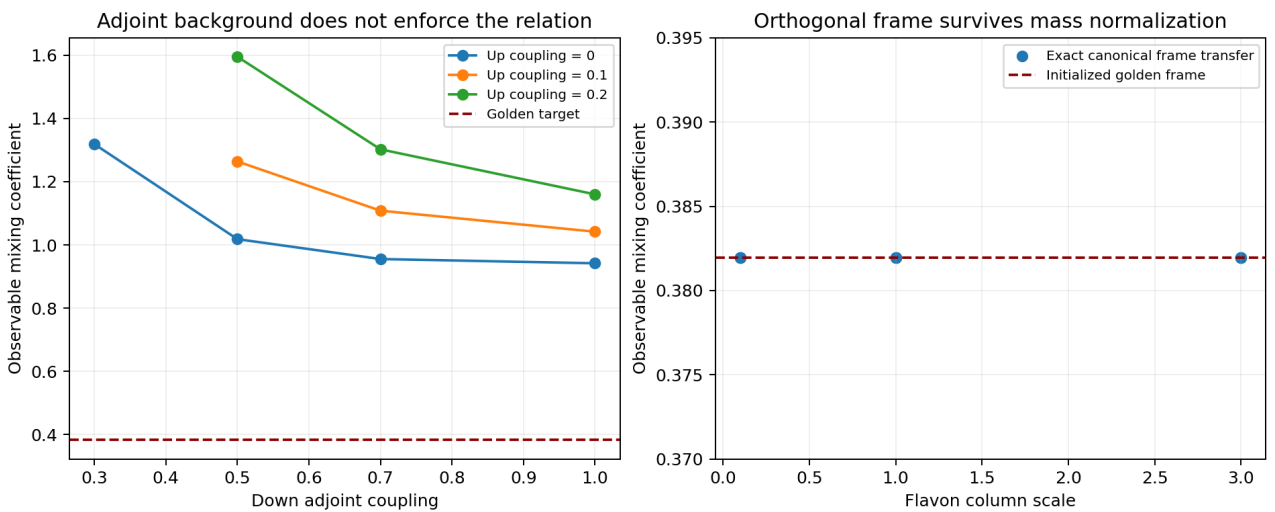
# Shared mediators and the alignment boundary

## What this mediator push establishes

The next step now has an explicit non-Abelian quark mediator construction and an exact tree matching calculation that includes the light right-handed kinetic normalization. The family symmetry makes each heavy mediator mass proportional to the identity and gives one contraction for each flavon column. Separately orthogonal flavon frames then determine the CKM matrix independently of the six column magnitudes and the two flavor-universal mediator masses. This is a constructive quark interaction that transmits an initialized mixing frame without the finite-mass diagonalization errors of the previous triangular texture.

It does not independently derive that frame. The accompanying scalar analysis gives a precise obstruction to the simplest alignment route. For six fundamental flavons with separate phase symmetries, fixed positive norms and exactly orthogonal up and down frames, every CP-even orientation term through scalar degree six reduces to a constant plus a linear function of  $|V_{ij}|^2$ . An exact rational assignment certificate proves that a permutation always attains its global minimum on the frame manifold. A generic nondegenerate cost therefore cannot select the required hierarchical CP-violating mixing. Degenerate costs can leave mixed flat directions; those are not a unique prediction.

I also tested a more favorable alternative in which an adjoint mediator mass contains initialized neighboring mixings, a golden 1-3 entry and the 66-degree phase. Both up and down adjoint couplings are allowed independently by the symmetry. Exact tree matching and canonical normalization were performed before extracting CKM quantities. Thirty of thirty-six declared cases satisfy the numerical control criteria. Every accepted case fails the golden target by more than one percent, despite matching the two existing anchors. The six unsuccessful or uncontrolled cases are retained with their reasons. The construction is therefore a tested candidate and a boundary result, not a completed predictive theory of flavor.



## The exact observable target

$$C_{\text{eff}} = \frac{w(1-w^2)}{uv} = \phi^{-2}$$

Write  $u=|V_{us}|$ ,  $v=|V_{cb}|$  and  $w=|V_{ub}|$ . The nominated physical relation is  $C_{\text{eff}}=w(1-w^2)/(uv)=\phi^{-2}$ , where  $\phi=(1+\sqrt{5})/2$ . This follows from the original standard-angle rule  $s_{13}=C s_{12} s_{23}$  because  $u=s_{12} c_{13}$ ,  $v=s_{23} c_{13}$  and  $w=s_{13}$ . It is already rephasing invariant. Relating a Yukawa matrix entry to other entries is insufficient unless its diagonalization yields this observable equation.

For  $p=u^2$ ,  $q=v^2$  and  $r=w^2$ , define  $A=r(1-r)^2$ . Eliminating  $C$  from  $C^2-3C+1=0$  gives  $A^2-7pqA+(pq)^2=0$ . The positive physical branch  $C_{\text{eff}}<1$  selects  $\phi^{-2}$ ; the other positive root is  $\phi^2$ . The equation does not fix a CP phase. In raw fixed-norm flavon overlaps its formal maximum scalar degree is twenty-four. Squaring it into an energy term would encode the desired result by construction and would not explain its origin.

The existing anchors remain  $u=0.22431$  and  $v=0.0411$ . The small-branch target is  $w=0.00352144218276$ . No new experimental data or fit to  $|V_{ub}|$  or CP measurements was introduced. All numerical examples use declared illustrative quark spectra and dimensionless mediator units. A claim of an exact relation still needs a specified matching scale and subsequent threshold and running calculations.

## An exact Abelian obstruction for the initialized circuit

Consider an additional ordinary diagonal Abelian phase action preserving every monomial of the initialized singlet superpotential, with its mass  $M$  and numerical coefficients neutral. The recursion terms force  $q(Z_n)=n q(Z_1)$  and  $q(D_n)=-n q(Z_1)$ . The constant  $D_{\text{star}}$  term forces  $q(D_{\text{star}})=0$ . Its remaining terms require  $n q(Z_1)=0$  for  $n=2,6,8,10,14,16$ . Their greatest common divisor is two, so the general character relation is  $2 q(Z_1)=0$ .

The golden extension has a constant  $D_{\text{gold}}$  term, forcing  $q(D_{\text{gold}})=0$ . Its linear  $D_{\text{gold}} T$  term then forces  $q(T)=0$ . Hence an operator  $O$  and  $O T$  have the same additional charge. Such a charge rule cannot forbid  $Q_1 H d_3^c A B$  while allowing  $Q_1 H d_3^c A B T$ . The implementation derives the monomial constraints from the existing circuit rather than a separate assumed polynomial, and checks cyclic groups of orders one through 120. The focused test independently extends those checks through order 240.

This result is deliberately restricted. Charged mass or coupling spurions, symmetries mixing fields, and non-Abelian representations can change the premises. Treating the initialized coefficients as transforming spurions would supply a different model and would require its own alignment and coupling audit. The obstruction does not rule out flavor symmetry in general.

## The declared non-Abelian mediator sector

Use the Standard Model gauge group, a formal global  $SU(3)_F$  family symmetry, six independent  $U(1)$  flavon phase symmetries and CP.  $Q$  is a family anti-triplet. The three right-handed-conjugate up fields and the three right-handed-conjugate down fields are family singlets. There are six complex gauge-singlet fundamental flavons  $F_{ui}$  and  $F_{di}$ . Each carries its own phase charge  $+1$ , while its associated right-handed-conjugate quark carries  $-1$ . The Higgs and  $Q$  have zero phase charge.

Introduce heavy  $X_u$  and  $X_d$  family anti-triplets and  $X_u^c$  and  $X_d^c$  family triplets.  $X_u$  and  $X_u^c$  are vectorlike color-triplet weak-singlet fermions with hypercharges  $+2/3$  and  $-2/3$ .  $X_d$

and  $X_d^c$  have hypercharges  $-1/3$  and  $+1/3$ . Each family triplet gives three heavy Dirac species, for six species overall. These pairs are vectorlike under the Standard Model. The global family group is not claimed to be an anomaly-free gauged completion; its Goldstone and cosmological questions are outside this calculation.

In ordinary left-handed Weyl notation, the allowed core interactions are  $M_f X_f \cdot X_f^c$ ,  $h_f Q \cdot H_f X_f^c$  and  $\sum_i \lambda_{fi} X_f \cdot F_{fi} f_i^c$ , plus their Hermitian conjugates.  $H_u$  is the Higgs doublet and  $H_d$  is its conjugate doublet. This notation is an ordinary non-SUSY mediator model. It is not a combined softly broken completion of the previously constructed 34-chiral-field singlet sector.

The operator audit exhausts fermion mass bilinears and two-fermion-one-scalar Yukawa invariants in this declared quark sector. It checks Standard Model gauge representations, family representations and all six phase charges. With the seventeen neutral scalar backgrounds  $Z_1$  through  $Z_{16}$  and  $T$ , including their conjugates, there are 78 independent invariant coefficients. Two are bare universal masses, two are Higgs vertices, six are flavon-column vertices and 68 are neutral-scalar universal-mass vertices. CP permits real coefficients; it does not equate them. Lepton interactions and a complete scalar potential for all neutral backgrounds are outside this finite audit.

Adding one complex phase-neutral family adjoint  $\Sigma$  and its conjugate adds four invariants, giving 82. They are  $X_u \Sigma X_u^c$ ,  $X_u \Sigma^\dagger X_u^c$ ,  $X_d \Sigma X_d^c$  and  $X_d \Sigma^\dagger X_d^c$ . Up and down coefficients are independent. In the Hermitian background used below the two coefficients in each sector combine into one real number, but the symmetry does not impose any relation between the sectors or between an adjoint coupling and a singlet mass.

## Canonical tree matching without a small-flavon expansion

$$A = M^{-1}L, \quad K = I + A^\dagger A, \quad Y = -hAK^{-1/2}$$

Let  $L_f$  be the 3 by 3 matrix of flavon columns multiplied by their  $\lambda_{fi}$  couplings, and let  $M_f$  be the invertible heavy mediator mass matrix. At zero Higgs background the heavy fermion row block is  $[L_f, M_f]$ . Define  $A_f = M_f^{-1} L_f$ ,  $K_f = I + A_f^\dagger A_f$  and  $R_f = K_f^{-1/2}$ . A normalized basis for the three light right-handed fields is  $N_f = [R_f; -A_f R_f]$ . Direct multiplication gives  $[L_f, M_f]N_f = 0$  and  $N_f^\dagger N_f = I$ .

The Higgs vertex projected into that normalized null space gives  $Y_f = -h_f A_f R_f$ . This is exact at tree order at unbroken electroweak matching, to all orders in the ratio of flavon mixing to heavy mass. It is not the unnormalized expression  $-h_f M_f^{-1} L_f$ . No loop threshold is included. At finite Higgs vacuum value, higher-dimensional Higgs operators and mixing with weak-singlet heavy fermions also affect the charged current. The dimension-four effective CKM matrix is unitary; the finite-Higgs full theory need not give an exactly unitary light 3 by 3 submatrix.

The focused checks verify the normalized null frame and Yukawa projection directly. They also compare against independent singular values of the complete 6 by 6 fermion mass matrix  $[[0, h_f v_H I], [L_f, M_f]]$ . The three light singular values divided by  $v_H$  converge to those of the matched  $Y_f$ . A separate finite-Higgs test compares the light charged-current overlaps with the matched CKM limit and checks their small departure from unitarity. Noncommuting mass matrices are tested under a common family-basis rotation, so the checks do not rely only on diagonal mediator masses.

## What universal masses can and cannot transmit

For a universal mass  $M_f = m_f I$  and a general singular decomposition  $L_f = U_f \text{diag}(s_f) W_f^\dagger$ , the canonical Yukawa matrix has the same left eigenvectors  $U_f$ . Its singular values are  $|h_f| s_f / \sqrt{|m_f|^2 + s_f^2}$ . This map is strictly increasing for positive  $s_f$ , preserving nondegenerate mass ordering. Thus changing a flavor-universal mediator mass, even by a complex golden or cyclotomic background, cannot change the mixing angles or physical CKM phase of fixed  $L_u$  and  $L_d$ . This statement does not require orthogonal flavon columns.

With separately orthogonal columns,  $L_f = U_f \text{diag}(\ell_f)$ , the stronger separation holds: changing each individual column magnitude only changes its associated quark singular value. While retaining the assigned nondegenerate mass ordering, the CKM matrix is  $U_u^\dagger U_d$  and is independent of those magnitudes and the universal masses. Twelve declared examples initialize  $U_u = I$  and  $U_d$  from the original golden CKM chart, then vary column magnitudes and real or complex universal masses. They retain the nominated coefficient and 66-degree standard phase to numerical precision. These are conditional transfer checks. The down frame was copied from the target chart and has not been derived from a symmetry-controlled potential.

A separate three-case test uses nonorthogonal random columns and changes only universal masses. It confirms the general unchanged-eigenvector result while the spectra change. Both results narrow the physical role a golden field can play: placing it only in a universal heavy mass does not orient the CKM matrix. Placing its phase as an overall Yukawa-column multiplier also fails, because  $(Y \text{diag}(\exp(i \theta_j)))(Y \text{diag}(\exp(i \theta_j)))^\dagger = Y Y^\dagger$ . A physical CP phase must influence relative left-handed directions or interfering flavor structures.

## The complete low-degree triplet alignment basis

For the six complex triplets alone, write  $G_{ab} = F_a^\dagger F_b$ . Independent phase invariance requires each triplet label to have equal occurrences with and without conjugation. At scalar degree two there are six norms. At degree four there are 21 products of norms and fifteen squared overlaps  $|G_{ab}|^2$ , giving 36 quartic invariants. Including the quadratic terms, the renormalizable triplet potential has 42 independent real coefficients. One Standard Model Higgs adds its quadratic term, its quartic term and six Higgs-norm/flavon-norm portals, giving fifty terms in that restricted scalar sector.

At degree six the CP-even basis has 56 triple products of norms, ninety norm-times-squared-overlap terms and twenty real Gram triangles  $\text{Re}(G_{ab} G_{bc} G_{ca})$  with distinct labels. Products of balanced SU(3) epsilon tensors are Gram determinants and reduce to this basis. Unbalanced epsilon contractions are forbidden by the six separate phase symmetries. The first Gram-rank determinant relation for six triplets in three dimensions has degree eight, so it does not remove any of these degree-six contraction types.

Now restrict to positive fixed norms and exactly orthogonal  $U_1, U_2, U_3$  and  $D_1, D_2, D_3$  frames. Overlaps within each frame vanish. Every three-label triangle contains two labels from one frame, so every Gram triangle vanishes. Norm products are constant. The surviving squared cross overlaps are proportional to  $B_{ij} = |U_i^\dagger D_j|^2 = |V_{ij}|^2$ . Norm factors at degree six only modify their coefficients. Therefore the complete orientation dependence through degree six is  $E = \text{constant} + \sum_{ij} K_{ij} B_{ij}$ .

Fixed norms and exact separate orthogonality are assumptions of this theorem. Arbitrary finite quartic coefficients in the full potential can distort the frames, and a model outside this regime requires a new calculation. An adjoint introduces new orientation tensors and is also outside this theorem. Neutral singlets can change norm-dependent parameters; multiplying a quartic

overlap by a singlet would exceed renormalizable degree and does not create a new degree-four orientation tensor.

## A rational global certificate for the orientation obstruction

$$E = \sum_i \alpha_i + \sum_j \beta_j + \sum_{ij} R_{ij} B_{ij} \geq E_{\text{permutation}}$$

For any unitary relative frame,  $B$  is doubly stochastic: its entries are nonnegative and each row and column sums to one. For each declared rational cost matrix  $K$ , the implementation finds row and column numbers  $\alpha_i$  and  $\beta_j$  with  $\alpha_i + \beta_j \leq K_{ij}$ . It also finds a permutation attaining the same sum  $\sum_i \alpha_i + \sum_j \beta_j$ . Define reduced costs  $R_{ij} = K_{ij} - \alpha_i - \beta_j$ , which are exactly nonnegative rational numbers.

Then  $E = \sum_i \alpha_i + \sum_j \beta_j + \sum_{ij} R_{ij} B_{ij}$  is bounded below by the permutation energy. A permutation matrix is itself unitary, so the bound is attained within the physical frame manifold. The receipt supplies the rational dual numbers, reduced costs and all winning permutations. Tests check eighty further rational cost matrices independently against all six assignments and compare the identity against nontrivial random unitaries.

This proves that a permutation is always a global minimizer of the restricted linear orientation energy. With a unique minimum assignment, no nonpermutation  $B$  attains the same energy. For degenerate costs, a mixed CP-violating frame can instead belong to a flat minimum set. The constant-cost example explicitly retains such a direction. The correct conclusion is that this potential cannot uniquely select the nominated nonpermutation CKM matrix in the stated regime, not that every possible minimum must have zero CP violation.

## The first higher-degree CP capability example

Scalar degree eight permits  $|F_{ui}^\dagger F_{dj}|^4$  and real four-overlap cycles. With equal fixed norms, the chosen positive sum of fourth powers reduces to  $\sum_{ij} B_{ij}^2$ . Each row obeys  $\sum_j B_{ij}^2 \geq 1/3$ , so the total is at least one. Equality requires every  $B_{ij} = 1/3$ . The unitary three-dimensional Fourier frame reaches the bound and has  $|J| = 1/(6 \sqrt{3})$ , with a conjugate CP partner.

This is an explicit CP-even orientation interaction that can select nonzero physical  $|J|$  when paired with nondegenerate quark column magnitudes. It selects democratic mixing, with every CKM magnitude  $1/\sqrt{3}$ , and fails the hierarchical target. Its coefficient pattern is chosen; other degree-eight invariants are independently allowed under the declared symmetry. No renormalizable UV generation of that pattern or connection to the protected 66-degree singlet vacuum has been provided. The example demonstrates what new orientation dependence becomes possible at degree eight, while leaving the actual golden alignment unresolved.

## The favorable adjoint candidate and its failure

The numerical candidate uses a Hermitian traceless adjoint background  $\Sigma$  with entries  $\Sigma_{12}=a$ ,  $\Sigma_{23}=b$  and  $\Sigma_{13}=\phi^{-2} a b \exp(-i \theta)$ , together with their conjugates and zero diagonal.  $\theta$  is initialized at 66 degrees. The mediator masses are  $M_u = I + g_u \Sigma$  and  $M_d = I + g_d \Sigma$ . The columns before filtering are diagonal, using the same illustrative hierarchies as the preceding quark calculation. This gives the golden factor and phase a direct place in a flavor-dependent mediator, which a universal mass could not supply.

The assumed adjoint background is deliberately favorable and explicitly initialized. No scalar potential producing it is claimed. Both  $g_u$  and  $g_d$  are independent allowed couplings. Canonical tree matching is performed for each sector, followed by exact numerical diagonalization of  $YY^\dagger$ . Only  $a$  and  $b$  are matched to  $|V_{us}|$  and  $|V_{cb}|$ . They are not tuned to  $|V_{ub}|$  or any CP observable.

The declared sweep uses  $g_u=0,0.1,0.2$ ;  $g_d=0.3,0.5,0.7,1$ ; and column scales 0.1,1,3. A case is accepted only when the two anchor residuals are below  $10^{-9}$ , both mediator matrices have minimum singular value at least 0.2 and  $\max(a,b)\leq 1$ . Thirty cases are accepted and six fail the matching or control criteria. Accepted  $C_{\text{eff}}$  ranges from approximately 0.941676 to 1.595694, against the golden target 0.381966. All accepted cases fail the target at the one-percent level. The input phase also does not become the standard CKM phase automatically. Four conjugate-pair checks retain magnitudes and reverse  $J$ .

Unnormalized Yukawa predictions are retained for the same backgrounds so that the kinetic correction can be inspected. They are not substituted for the canonical result. The failure is not a theorem against all adjoint models; it demonstrates that this favorable entry pattern and the allowed shared-mediator couplings do not enforce the desired observable relation. A successful adjoint construction would need a derived vacuum and additional relations among its allowed couplings, followed by the same normalization and mixing checks.

## Reproduction, validation and the next interaction target

Run `PYTHONPATH=python OPENBLAS_NUM_THREADS=1 python python/develop_flavor_mediator.py`, followed by `python python/plot_flavor_mediator.py` with the same environment. The runner uses only the existing  $|V_{us}|$  and  $|V_{cb}|$  means. It retains exact charge, operator and rational dual certificates, the conditional transfer cases, all adjoint attempts and the scientific figure. NumPy and SciPy support the numerical matching; the exact charge and assignment statements use integer or rational arithmetic. There are no new Lean theorems or newly acquired experimental datasets.

The twenty focused tests cover the charge obstruction, independent operator-list construction, exact rational assignment witnesses, canonical null-space matching, the complete fermion mass comparison, family-basis covariance, CP conjugation, universal-mass invariance and the failed golden enforcement. They accompany the repository-wide suite and a fresh-extraction byte comparison of the retained scientific artifacts. The integrated repository-wide run completed 879 tests successfully, with four skips. A fresh archive extraction reproduced all 32 scientific receipt, manifest and figure files from both flavor investigations byte for byte. The full test and replay results are recorded in the validation receipt for this push.

The resulting research target is more specific than adding a golden Yukawa coefficient. The quark mediator interaction can already transmit an orthogonal frame exactly, so the remaining positive construction is a symmetry-controlled relative frame that produces the golden amplitude relation and the physical CP phase. Within the tested phase-symmetric triplet class, a uniquely selected nonpermutation frame requires new orientation dependence beyond degree six or a change of fields or assumptions. Higher-degree interactions, additional non-Abelian representations or nonorthogonal-frame dynamics are possibilities to test; none is asserted here to solve the problem. Threshold matching, running, a consistent combined singlet/flavor theory and a vacuum history remain separate requirements.

## Primary comparisons and their limits

S. Antusch, S. F. King, M. Malinsky and M. Spinrath, Quark mixing sum rules and the right unitarity triangle, <https://arxiv.org/abs/0910.5127>. Its texture-zero sum-rule route motivates deriving mixing from both Yukawa sectors. It does not establish the present golden relation.

R. Alonso, M. B. Gavela, L. Merlo and S. Rigolin, On The Potential of Minimal Flavour Violation, <https://arxiv.org/abs/1103.2915>. Its analysis provides context for restrictions of low-degree dynamical flavor potentials. The six-triplet fixed-frame obstruction above has separate explicit premises and its own rational certificate; no claim of duplicating every model in that paper is made.

J. Gehrlein, J. P. Oppermann, D. Schafer and M. Spinrath, An  $SU(5) \times A_5$  Golden Ratio Flavour Model, <https://arxiv.org/abs/1410.2057>. It provides an example of an explicit renormalizable flavor and messenger construction. Its golden mixing result concerns the neutrino sector, not our nominated CKM coefficient. Only the primary abstracts and metadata were used for these comparisons.

# Recovered M22 geometry and the flavor frame

## What was recovered

The repository preserves selected original files from WILSON\_FORMATION\_FORGE v0.26.0, including its M22/A5 Wilson algebra, Schur-selector theorems, transport-category results, and a 1,023-word exact integer ledger. The provenance manifest is `galois_merge/receipts/vendor_sources.json`. This is a verified subset of the old work, not recovery of every later Atlas or EQ-LAB release. The saved ledger has SHA-256 `1351e514fcb5f23832f65665e0a12d2e172f8131c34875de0e03cddc5aec3093`. The large original orbital tensor is absent from the subset; the ledger is nevertheless sufficient to reconstruct its two-generator algebra faithfully.

The carrier is  $V = C[M22/A5]$ , of dimension 7,392. Its commutant  $H = \text{End}_{M22}(V)$  has dimension 152. The complex simple block sizes are 1,3,3,2,5,5,5,3,3,6. Their squares sum to 152. Thus there are four three-dimensional multiplicity carriers available for a frame calculation. They are multiplicity spaces of larger M22 representations, rather than four nontrivial three-dimensional representations of M22.

The recovered Q192 generators are  $X = A49$  and  $Y = -A12 - A48 - A59 + A95 + A116 + A125$ .  $X$  is self-adjoint and  $Y$  is skew-adjoint. This convention differs from the earlier Q7 signed operator; no Q192 calculation here is silently identified with Q7 or with its exceptional 154 block. Likewise, the three-dimensional minus-context factor in the separate 486-dimensional transport algebra is not identified with a Wilson multiplicity carrier or a physical family triplet.

## Exact reconstruction from the word ledger

The new verifier `python/develop_m22_frame.py` selects 152 independent ledger rows through word degree eight, inverts their integer coordinate matrix over the rationals, and reconstructs left and right multiplication by  $X$  and  $Y$ . It also reconstructs the orbital transpose permutation from reversed words and their skew signs. No CKM number, target phase, or golden coefficient enters this reconstruction.

Every available one-step left and right extension of all 511 words through degree eight is checked exactly against the original ledger. The transpose squares to the identity and is an actual orbital permutation. All four left/right generator commutators vanish exactly. Adjointness recovers positive integral orbital subdegrees, normalized at the identity; their sum is exactly 7,392. The recovered regular matrices are integral, not numerical fits.

The exact characteristic polynomial of  $X$  is  $(t-30)^9(t-3)^{51}(t+2)^{65}(t+5)^{27}$ . The exact  $Y$  factorization is recorded in `receipts/m22_frame/frame_audit.json`. In particular it contains the factors  $t^2+1452$  and  $t^2+2732$  relevant to the two real three-dimensional carriers, and a degree-six factor for the conjugate complex pair. Regular-module multiplicities and physical permutation-module multiplicities are different; only the shared root sets are being used here.

Unitary one-copy carrier extraction is a separate numerical step. A generic Hermitian left-multiplication selector commutes with right multiplication, so its eigenspaces supply right-invariant carriers. Word traces through degree five distinguish the conjugate complex blocks. Even trace data alone would merge that pair incorrectly. The extracted block sizes

reproduce all ten expected simple blocks. The largest reported invariance residual is below  $3.3\text{e-}11$ . Carrier coordinates and numerical overlap matrices are explicitly not promoted to exact algebraic coordinates.

## Why equivariance alone cannot select the frame

Write  $V$  as the direct sum of  $R_\lambda$  tensor  $M_\lambda$ , where  $R_\lambda$  is an irreducible  $M_{22}$  carrier and  $M_\lambda$  is its multiplicity space.  $M_{22}$  acts as  $\rho_\lambda(g)$  tensor  $I$  on this summand. Every operator  $I$  tensor  $A_\lambda$  commutes with  $M_{22}$ , for any matrix  $A_\lambda$  on  $M_\lambda$ . In a three-dimensional multiplicity carrier the full  $M_3(C)$  algebra is therefore allowed by  $M_{22}$  equivariance.

For any desired unitary frame  $W$ , the two positive operators  $D_u$  and  $W D_d W^\dagger$  are both permitted on such a multiplicity space. Their eigenvalues can be distinct. Their relative eigenframe is  $W$ . Consequently  $M_{22}$  equivariance alone accommodates every CKM frame and cannot force the golden relation or a particular CP phase. Canonical orbital geometry supplies distinguished candidate operators, but selecting specific orbital coefficients requires an additional interaction or residual symmetry. Algebra-generation and transport-isomorphism theorems do not perform that selection.

This is a scoped obstruction to using the unbroken commutant as the complete flavor principle. It is not a no-go theorem for  $M_{22}$  breaking, decorated incidence fields, projective carriers, or a more restrictive interaction category.

## The canonical C3/C7 frame fails

The first frame test takes the up-sector Hermitian operator to be  $X$  and the down-sector operator to be  $iY$ . Positive shifts, positive overall rescalings, and scalar functions that preserve the nondegenerate eigenprojectors do not change this frame test. Both ordinary real three-dimensional carriers have simple spectra for  $X$  and  $iY$ .

On the first real carrier the  $X$  eigenvalues are  $-5, 3, 30$  and the  $iY$  eigenvalues are  $-\sqrt{1452}, 0, \sqrt{1452}$ . Its squared overlap matrix, in increasing spectral order, has rows approximately  $(0.422078, 0.155844, 0.422078)$ ,  $(0.196970, 0.606061, 0.196970)$ , and  $(0.380952, 0.238095, 0.380952)$ . On the second real carrier  $X$  has eigenvalues  $-5, -2, 3$  and  $iY$  has eigenvalues  $-\sqrt{2732}, 0, \sqrt{2732}$ . Its rows are approximately  $(0.00512445, 0.98975110, 0.00512445)$ ,  $(0.49897511, 0.00204978, 0.49897511)$ , and  $(0.49590044, 0.00819912, 0.49590044)$ .

The equality of the outer probability columns is structural. In a real carrier  $X$  has a real eigenbasis, while a real skew  $Y$  has conjugate eigenvectors for the opposite eigenvalues of  $iY$ . Their squared overlaps with any real  $X$  eigenvector are equal. The numerical column differences are below  $6\text{e-}16$ . Row and column permutations retain the identical-column obstruction. Two identical probability columns cannot each concentrate near one in different rows, so neither frame has the CKM hierarchy. This statement does not require interpreting a convention-dependent eigenvector phase as the CKM phase.

The other two three-dimensional carriers are a conjugate complex pair, so the preceding real-carrier argument must not be applied to them individually. Both have  $X$  spectrum  $-2, -2, 3$ . The repeated eigenvalue leaves a  $U(2)$  freedom and prevents  $X$  from selecting a unique family frame. The unique  $X$  eigendirection has overlaps approximately  $0.0381554, 0.5982882, 0.3635564$  with the  $iY$  eigenlines, in reverse order on the conjugate carrier. These overlaps are unchanged by rotations in the repeated eigenspace. Their maximum

is below 0.599, so even that fixed direction cannot be a nearly unmixed CKM row under any permutation. The receipt omits a full mixing matrix for these degenerate cases because it would depend on arbitrary eigenbasis choices.

Thus all four available three-dimensional blocks fail for this canonical pair. This does not exclude pairs built from additional unsigned channels, higher words, decorated C5 incidence, or another vacuum-selected Hermitian operator.

## The twelvefold cover supplies a phase carrier

The primary ATLAS data gives M22 Schur multiplier of order twelve and lists its twelvefold cover. The recovered Formation work already tracks separate 2-primary and 3-primary cover restrictions. The full universal cover has central C12. An order-five element of M22 can be lifted to an order-five element of the cover: its fifth power lies in C12, and the fifth-power map is invertible on C12, so a central correction cancels that power. This lift commutes with the center. Their subgroup is C12 times C5, hence C60.

This is a group-theoretic inference from the cover data. It avoids the ordinary-M22 restriction that a single group element has no order sixty. It is not a claim that the old untwisted 7,392-state permutation module sees the central character; that module factors through M22 and its center acts trivially.

A character of this C60 subgroup has phase residue  $n=5a+12b$  modulo 60, with  $a$  modulo 12 and  $b$  modulo 5. All sixty residues occur. The primitive positive phases are 6,42,66,78,102,114,138,174 degrees. The residue eleven, corresponding to 66 degrees, can be written with  $a=7$  and  $b=3$ . These are possible character labels, not dynamically selected quantum numbers.

If one retains the previously nominated readout  $C=1-2 \cos(12 \theta)$ , the primitive residues  $n$  congruent to plus or minus one modulo five give  $C=(3-\sqrt{5})/2=\phi^{-2}$ . Its exact polynomial is  $C^2-3C+1$ . The positive angles on this branch are 6,66,114,174 degrees. Thus the cover supplies both the relevant cyclotomic arithmetic and several competing branches. It neither selects 66 degrees nor derives the equality of this singlet readout with the observable CKM coefficient. The other primitive residues give the larger golden root.

Central phases do not select a residual eigenframe. Multiplying a residual group matrix by a central scalar changes its eigenvalues but leaves all its spectral projectors unchanged. Such scalar factors therefore cancel from projector overlaps. Similarly an overall Yukawa or Yukawa-column phase cancels from YY-dagger. A primitive eigenphase of 66 degrees is not evidence for a physical CKM phase of 66 degrees. Multiple charge-compatible interfering contractions and a nonzero rephasing invariant must be demonstrated explicitly.

## The next interaction calculation

The useful continuation is a decorated order-five incidence construction coupled to the recovered signed C7 channels. First derive the allowed cover characters and local contractions from the actual Schur restrictions. A faithful central character cannot simply be attached to an ordinary A5 triplet without checking its extension and charge constraints. Build the resulting twisted or decorated carrier and identify two Hermitian operators with simple three-dimensional eigenframes. Their coefficients must follow from the permitted contractions or a solved vacuum, rather than from a fitted CKM matrix.

Then compute the relative spectral-projector overlaps and a CP-sensitive closed path invariant. Check whether any central phase survives these neutral observables. Fit only the existing two anchor magnitudes, if the construction admits two independently justified scales; evaluate the remaining CKM magnitude, effective golden coefficient, and physical phase as predictions. Enumerate every allowed orientation and character branch, including CP conjugates and the four golden-angle candidates. Finally enumerate symmetry-allowed counterterms to test whether the frame survives coefficient changes.

The recovered algebra has enough structure to make this calculation concrete. What it currently establishes is an exact operator reconstruction, a failure of the simplest canonical frame, and an order-60 cover route with an explicit branch and rephasing obstruction. A symmetry-protected quark frame has not been derived.

## Reproduction and sources

Run `python python/develop_m22_frame.py` from the repository root with NumPy, SciPy, SymPy and python-flint installed. The output operator matrices and transpose/subdegree data are in `receipts/m22_frame/recovered_operators.json`. The diagnostic, exact polynomial factors and phase enumeration are in `frame_audit.json`. The trusted original object-array archive is the same one used by the byte-preserved Forge verifier; do not replace it with an untrusted NumPy pickle.

Recovered source theorems include `RAMANUJAN_WILSON_FAMILY.md`, `WEDDERBURN_35_RAMANUJAN_REDUCTION.md`, `RAW_FIVE_BLOCK_STAR_TRANSPORT_v11.md`, `CHAR0_TRANSPORT_GLUE_M5_v13.md`, `M22_PRIMARY_SCHUR_SIGNATURES_v11.md` and `L34_MATCHED_SELECTOR_V24.md` under `galois_merge/vendor/forge/THEOREMS`. The primary external cover reference is the ATLAS M22 page, <https://brauer.maths.qmul.ac.uk/Atlas/spor/M22/>. Its multiplier, cover and conjugacy-class data were checked on 5 October 2026. The CKM target and the mediator matching remain those of the existing protected-flavor chapters.

# Triplet transport and exact residual obstructions

Research manuscript and continuation of the protected-flavor monograph. Prepared 5 October 2026. Status: computationally certified mathematical results with proofs; publication novelty and the proposed particle-physics interpretation require independent review. No golden CKM coefficient, CKM phase, or experimental mixing anchor enters the group, connection, or overlap enumeration.

## Abstract and the claim being tested

We reconstruct a canonical decorated connection on a 7,392-cap M22/A5 geometry recovered from the Wilson work. Its order-five scalar adjacency  $B$  satisfies  $B^2 = X + 6I$  in the preserved 152-dimensional commutant, so it cannot supply eigenframes independent of  $X$ . The cap graph consists of 176 distance-regular components with 42 vertices, with intersection array  $\{6,5,1;1,1,6\}$ . Each cap has five A4 subgroups. Matching them across a common D5 by their common involutions defines a parameter-free five-letter edge permutation. The resulting holonomy is S5. On the exterior square of the standard four-dimensional hyperplane representation, every edge exchanges the two local golden A5 triplets. There is no parallel rank-three projector on the base graph. Its connected 84-vertex orientation cover has A5 holonomy and admits exactly two such projectors; a positive interaction selects them. Independently, an exact census of the genuine complex triplet of 3.A6 gives 151 complete Abelian residual frames and 111 squared entry values. Every nonzero entry has magnitude at least 0.0759431758699. Two involution residuals also have an exact fixed-entry obstruction. These results identify concrete failures and a precise alignment repair, rather than deriving the nominated golden CKM relation.

The central question is whether the recovered symmetry can enforce  $C_{\text{eff}} = |\text{Vub}|(1-|\text{Vub}|^2)/(|\text{Vus}||\text{Vcb}|) = \phi^{(-2)}$  and a physical CKM phase of 66 degrees. The results below do not establish either equality. In particular, geometric transport, a representation's character field, a central scalar phase, and the physical CKM invariant are different objects.

## Exact scalar identity and the 42-cap component

$$B^2 = X + 6I, \quad B^3 = 4X + 11B + 6D$$

The trusted source is `galois_merge/vendor/forgedATA/RELATION/TASK_SELECTED_WORD_PROGRAMS_V23/exact_prefix_1023.npz`, SHA256

1351e514fcb5f23832f65665e0a12d2e172f8131c34875de0e03cddc5aec3093. It records the original integral Hecke coordinates of 1, X, Y and their binary words. The original X is orbital A49, and the original Y is -A12-A48-A59+A95+A116+A125. The sole valency-six orbital is B = A134; the sole valency-five orbital is D = A139. Solving against 152 independent preserved words and checking all 511 available extensions reconstructs multiplication by B and D without CKM input.

Theorem 1. In the full reconstructed regular commutant representation,  $B^2 = X + 6I$  and  $B^3 = 4X + 11B + 6D$ . Consequently  $B^4 - 4B^3 - 17B^2 + 24B + 36I = 0$ . Its regular characteristic polynomial is  $(t-6)^9(t-2)^{65}(t+1)^{27}(t+3)^{51}$ . These are exact integral matrix identities, not rounded

spectral matches.

Proof. Express the orbitals B and D in the selected exact word basis using rational matrix inversion. Their recovered multiplication matrices have integral entries. Direct integer matrix products give the stated identities. The characteristic polynomial is computed independently from the integral matrix. The complete matrices and replay script are supplied with the receipts. The four distinct roots occur, so the displayed quartic is the minimal polynomial.

An independent degree-22 permutation construction uses the ATLAS standard M22 generators. A subgroup generated by an involution and an element of order three with product of order five has order 60. The selected cap class has subdegree counts  $1^1, 5^1, 6^1, 10^1, 15^2, 20^4, 30^{42}, 60^{100}$ , matching the recovered action. Enumerating its cosets accounts for all 443,520 group elements and 7,392 caps. Each cap contains six C5 subgroups; a C5 lies in exactly two caps in this class. The corresponding undirected cap graph has valency six. The implementation finds a 42-vertex component with distance layers 1,6,30,5 and the stated intersection array; transitivity gives 176 components.

The component graph is a known Hoffman-Singleton second-subconstituent graph. Its spectrum is  $6^1, 2^{21}, (-1)^6, (-3)^{14}$ . Neither the graph nor its intersection array is claimed as a new discovery. The five distance-three neighbors together with a vertex form an antipodal six-cap block; the component has seven such blocks. The exact component stabilizer acts faithfully as all 2,520 even permutations of the seven blocks, identifying it as A7. A six-cap block stabilizer acts faithfully as all 360 even permutations of its six caps, identifying an ordinary A6 containing the cap A5 with index six.

The identity  $B^2 = X + 6I$  means every X eigenspace is B-invariant and conversely B selects only branches already lying in these spaces. Since the recovered X eigenvalues correspond to four distinct  $B^2$  values, B and X have the same eigenspace decomposition. The order-five scalar operator therefore does not repair the earlier failure to obtain a new three-family mixing frame.

## A canonical decorated transport

$$\Omega = \frac{EG_2}{\sqrt{5}}, \quad \Omega^2 = I, \quad \Omega T = -T\Omega$$

Each A5 cap has five A4 subgroups, equivalently the normalizers of its five Klein four subgroups. Adjacent caps intersect in D5. Each local A4 meets this D5 in a single order-two subgroup, and the five A4 subgroups exhaust its five involutions. Match the source and target A4 subgroups with the same actual involution. This defines a unique bijection of the five local letters, hence a permutation edge transport. The construction is covariant under M22 conjugation and has no continuous coupling or chosen CKM angle.

Theorem 2. For this incidence-matching connection, every edge permutation is odd and the holonomy of a 42-cap component is S5. On  $W = \text{exterior-square } R$ , where R is the four-dimensional standard hyperplane representation of the local five letters, transport exchanges the self-dual and anti-self-dual three-dimensional subspaces.

Proof. The supplied cap permutations identify all local subgroups and all 252 directed edge transports by exact set intersections. Every edge is odd. Products around loops, after a spanning-tree trivialization, generate all 120 permutations of five letters. An independently written verifier recomputes every transport from conjugated A4 and cap intersections, uses a depth-first rather than breadth-first tree, and again obtains the entire S5. One explicit five-cycle is cap 0, cap 1083, cap 2333, cap 3074, cap 5702, cap 0. Its loop permutation is (2,0,4,3,1) in zero-based image notation, an odd permutation. Vertex labels refer to the supplied deterministic

coset enumeration.

For the representation-theoretic statement, use the hyperplane basis  $e_i - e_5$ ,  $i=1,\dots,4$ , with metric  $G = I + 11^T$ , of determinant five. Let  $G_2$  be its induced exterior-square metric and  $E$  the matrix of wedge pairing in the six ordered bivector coordinates. The Hodge operator is  $\Omega = E G_2 / \sqrt{5}$ , and  $\Omega^2 = I$ . Its two projectors  $(I \pm \Omega)/2$  have rank three. For any five-letter permutation  $p$ , its induced bivector transport  $T$  satisfies  $\Omega T = \text{sign}(p) T \Omega$ . Every edge is odd, hence maps one Hodge branch to the other. The local  $A_5$  preserves the branches. Their order-five characters are  $(1+\sqrt{5})/2$  and  $(1-\sqrt{5})/2$  in opposite classes, identifying the familiar two real golden triplets. This is a real representation-theoretic branch exchange, not a CP phase calculation.

For an independent irreducibility check, the bivector character is  $\chi(p) = ((\text{fix}(p)-1)^2 - (\text{fix}(p^2)-1))/2$ . Its squared character sum is 120 over  $S_5$ , giving inner product one, and is 120 over  $A_5$ , giving inner product two. The full  $S_5$  commutant is therefore scalar, while the  $A_5$  commutant is spanned by  $I$  and  $\Omega$ . Exact commutator-rank calculations reproduce dimensions one and two.

Corollary. There is no parallel Hermitian rank-three projector on this connected base connection. Any such projector would commute with every loop at the root. The irreducible  $S_5$  action forces it to be a scalar idempotent, hence either zero or identity, of rank zero or six. The theorem excludes parallel projectors for the specified connection; it does not prohibit homogeneous  $A_5$  triplet bundles or all possible  $M_{22}$  models.

## What edge symmetry actually enforces

$$\dim \text{Hom}_{D_5}(3, 3^t) = 1, \quad \dim \text{Hom}_{D_5}(3, 3^{\bar{t}}) = 2$$

The  $D_5$  intersection reveals an additional coefficient issue. The restriction of one golden triplet to  $D_5$  is a one-dimensional sign axis plus a two-dimensional plane carrying  $C_5$  weights  $\pm 1$ ; its Galois partner has the same axis and plane weights  $\pm 2$ . Transport between adjacent caps exchanges the two five-cycle character classes. Taking the exact character inner products over the ten common elements gives dimension one for a same-branch edge intertwiner and dimension two for an opposite-branch edge intertwiner.

Thus a same-branch covariant map has rank at most one. An opposite-branch map can be invertible, but its axis and plane couplings are independent. The particular incidence matching above picks a unitary map on both channels;  $D_5$  symmetry alone does not force their relative strengths. It would be incorrect to turn this particular geometric isometry into a general coupling-protection theorem. A particle interaction must explain why it uses the incidence matching rather than the full symmetry-allowed two-channel family.

Multiplying edges by central scalar phases, including phases from a central cover, leaves  $\Omega T = -T \Omega$  unchanged. A phase factor also cancels in  $T P T^{-1}$ , so it cannot repair the parallel-projector obstruction. A noncentral transport or different fiber content could do so; neither is excluded here. No identification of a cover eigenphase with the CKM phase is made.

## An explicit protected alignment interaction

$$J_{V,s} = (-1)^s \Omega_V, \quad P_{\pm} = \frac{l \pm j}{2}$$

The orientation cover has vertices  $(v,s)$ , with  $s=0,1$ , and lifts an edge  $v-w$  to  $(v,s)-(w,1-s)$ . It is connected because the base has odd loops. Each component has 84 vertices. The lifted

holonomy consists exactly of the 60 even permutations,  $A_5$ . The field  $J_{(v,s)} = (-1)^s \Omega_v$  is parallel because edge transport anticommutes with  $\Omega$ . The deck exchange changes the sign of  $J$ .

Theorem 3. On this cover the only parallel Hermitian rank-three projectors are  $P_+ = (I+J)/2$  and  $P_- = (I-J)/2$ . For dimensionless Hermitian endomorphism order parameters  $P_a$ , an explicit potential selecting exactly these configurations is the sum of three nonnegative terms: a positive coefficient  $\lambda$  times the sum of squared norms of  $P_a^2 - P_a$ , a positive coefficient  $\mu$  times the sum of  $(\text{Tr } P_a - 3)^2$ , and a positive coefficient  $\kappa$  times the sum over edges of squared norms of  $P_b - T_{ba} P_a T_{ba}^{-1}$ . The norms use the positive fiber metric.

Proof. Zero energy enforces a rank-three projector and parallel transport. At a root the  $A_5$  commutant is  $\text{span}\{I, J\}$ . A Hermitian idempotent in this commutant has eigenvalues  $a+b$  and  $a-b$  on the two inequivalent triplets. Rank three forces these to be 1 and 0 in either order. This gives exactly  $P_+$  and  $P_-$ . Conversely, their parallelism and rank make all terms vanish. On the base the identical constraints have no zero-energy configuration, by the scalar  $S_5$  commutant. The cover is therefore a concrete repair of this alignment problem, rather than a choice of a fitted frame.

The selected projectors remain exact for arbitrary positive  $\lambda$ ,  $\mu$  and  $\kappa$  within this stated interaction. This is a classical robustness result for the declared potential family. Symmetry allows additional interactions; no general protection against counterterms or radiative corrections is claimed.

This establishes a local rank-three subspace. It does not by itself yield exactly three light global quark modes, nondegenerate quark masses, a relative up/down frame, a UV origin for the interaction, or a cosmological selection of its two branches. In particular, the two real Hodge sectors are not the positive and negative physical CP vacua. Additional physical fields and a light-mode analysis are necessary before treating the construction as a flavor model.

## Exact complex-triplet residual gap

$$|V_{ij}| \in \{0\} \cup [0.0759431758\dots, 0.9566183482\dots] \cup \{1\}$$

The identified  $A_6$  block supplies a concrete place to investigate a genuine complex triplet. Ordinary  $A_6$  has no faithful ordinary triplet; its triple cover  $3.A_6$ , the Valentiner group  $\text{Sigma}(360 \times 3)$ , does. We use the published generators of Hagedorn, Meroni and Vitale, equations (7) and (33), and close them exactly in  $Q(\sqrt{5}, \omega)$ , with  $\omega^2 + \omega + 1 = 0$ . The closure has 1,080 elements and a center of order three, and its real  $A_5$  subgroup has order 60. A consistent embedding of this triple cover into a particular central  $M_{22}$  cover is a separate requirement, not an assumption silently imposed by this calculation.

The complete minimal Abelian residual frames comprise 40 order-three frames, 45 order-four frames, 36 order-five frames, and 30 joint Klein frames, totaling 151. The order-three frame count is smaller than the number of cyclic order-three subgroups because central multiplication gives the same projectors. Exact spectral-projector enumeration checks coverage of all 474 noncentral elements of orders three, four and five. Order-twelve and order-fifteen lifts have the same frames as their cubes; order-two and order-six residuals leave a two-dimensional degeneracy. Higher Abelian residuals add the center or already contain one of the complete frames. This classification agrees with the published subgroup analysis; it is not presented as a newly discovered flavor group.

**Theorem 4.** Every entry probability between the 151 complete frames belongs to the supplied exact set of 111 values. The smallest positive value is  $p_{\min} = 0.005767365961210174\dots$ , and the largest value strictly below one is  $p_{\max} = 0.9151186641666433\dots$ . Consequently every nonzero magnitude is at least  $0.07594317586992\dots$ , and every subunit magnitude is at most  $0.956618348228092\dots$ . Rephasing and all row and column mass permutations preserve these bounds.

**Proof.** Construct cyclic projectors as the finite Fourier sums of exact group powers, retaining the three projectors of trace one. Construct Klein projectors by their four characters, retaining the three projectors of trace one. For all 11,476 unordered frame pairs with repetition, evaluate all nine traces  $\text{Tr}(P_i Q_j)$ , giving 103,284 exact comparisons. Computation is in  $\mathbb{Q}(\zeta_{60})$ , reduced modulo  $\Phi_{60}$ . All probabilities are exactly fixed by complex conjugation. Convert them to degree-seven rational polynomials in  $t = \zeta_{60} + \zeta_{60}^{-1}$ . Its minimal polynomial is  $t^8 - 7t^6 + 14t^4 - 8t^2 + 1$ , and the physical embedding is its largest real root,  $2 \cos(\pi/30)$ . Rational root isolation and interval Horner evaluation give strictly ordered, disjoint rational enclosures for all 111 values. This proves both extrema and their ordering without using floating point to choose the winner. The 60th root is a common arithmetic field for orders 3,4,5; its appearance is not a prediction of 66 degrees.

Explicitly,  $p_{\min} = -t^{7/10} - t^{6/20} + 11t^{5/20} + 3t^{4/10} - 3t^{3/5} - 9t^2/20 - t/5 + 1/2$ . The upper endpoint below one is  $p_{\max} = 2t^{6/9} - 4t^{4/3} + 2t^2 + 1/9$ . The full rational polynomial table and certified intervals are in `receipts/m22_interactions/valentiner_exact_gap.json`. A separate unitary numerical census of all 22,801 ordered frame pairs independently reproduces the extrema; 36 mass label permutations per pair correspond to 820,836 labeled pairs, and no labeling changes the entry spectrum.

The nominated values  $|V_{cb}|=0.0411$  and  $|V_{ub}|=0.00352144218276$  fall inside the forbidden nonzero gap. Thus complete Abelian residual symmetries cannot give that CKM hierarchy in this triplet without symmetry breaking. These numbers are used only after the exact spectrum is constructed, as a comparison to the existing nominated model, not as a current experimental global fit. The result excludes the specified exact residual construction; it does not exclude 3.A6 as a broken flavor symmetry.

## Partial residuals and the fixed-line obstruction

$$|V_{ij}|^2 \in \left\{ 0, \frac{3-\sqrt{5}}{8}, \frac{1}{4}, \frac{1}{2}, \frac{3+\sqrt{5}}{8}, 1 \right\}$$

Leaving a two-dimensional degenerate sector in each quark mass matrix permits continuous rotations, so the preceding complete-frame gap does not apply directly. Nevertheless, retaining an involution in both sectors fixes one line in each. There are 45 order-two elements. Their rank-one projectors are  $(I+g)/2$ . Evaluating all 2,025 ordered projector overlaps exactly gives squared fixed entries  $0, (3-\sqrt{5})/8, 1/4, 1/2, (3+\sqrt{5})/8$ , and  $1$ . Their magnitudes are  $0, 0.309016994\dots, 0.5, 0.707106781\dots, 0.809016994\dots$ , and  $1$ .

**Theorem 5.** If both nondegenerate quark Hermitian mass matrices commute with nontrivial involutions in the stated triplet, at least one CKM entry is in this fixed set, irrespective of the two otherwise arbitrary  $U(2)$  rotations and irrespective of any added compatible CP constraint. Distinct fixed lines cannot give the nominated hierarchical CKM matrix. Identical fixed lines give an exactly decoupled family and hence a two-family mixing block. A zero fixed overlap also supplies an exact zero entry. All these cases require symmetry breaking to realize the nominated matrix with nine nonzero entries.

Proof. Commutation with an involution preserves its one-dimensional  $+1$  eigenspace and its two-dimensional  $-1$  eigenspace. Nondegenerate mass eigenstates therefore include the fixed line in each sector. Their overlap is a CKM entry. Rotations confined to the two-dimensional complements do not change this overlap. The exact projector calculation gives the stated list. The nominated matrix's off-diagonal magnitudes are below 0.3090 and its diagonal magnitudes exceed 0.8091; none is zero or one. Adding CP cannot change the representation-fixed line overlap. This argument does not assume a specific CP automorphism or a chosen mixing phase.

## Publication scope and the next physical problem

The result suitable for scrutiny is a conjunction of exact transport obstruction, explicit cover alignment mechanism, and exact quark residual gaps. It is not a derivation of the golden CKM relation. Known ingredients include the graph, the two A5 triplets, the Valentiner generators and residual-subgroup classification, and the Hodge transformation law. The documented contribution is their explicit reconstruction on the recovered cap system, the canonical involution-matching connection with certified holonomy, the scoped interaction repair, and exact gap certificates applied to the nominated quark mechanism. A literature search did not establish novelty of this combination; independent specialist review remains necessary before submitting a novelty claim.

The remaining viable problem is now narrower. A proposed theory must specify a noncentral, charge-compatible interaction that produces three light quark modes, breaks the exact residual obstruction in a controlled way, and correlates the up/down breaking operators after canonical normalization. It must then derive the coefficient  $\phi^{-2}$  and a neutral CP observable corresponding to 66 degrees without inserting either target. Central phases alone, scalar order-five adjacency alone, and exact complete or two-involution residual frames cannot supply that result. Small free corrections that merely fit the missing entries would not resolve the original coupling-protection problem.

## Reproduction and sources

Install `python/m22_frame_requirements.txt`. From the repository root run `python python/certify_m22_c5.py`, `python python/develop_m22_transport.py`, `python python/verify_m22_transport.py`, `python python/develop_valentiner_frames.py`, and `python python/certify_valentiner_gap.py`. The first three scripts distinguish the preserved Wilson regular algebra from the independently enumerated cap geometry. The independent verifier reconstructs subgroup intersections directly and does not reuse the connection's cocycle code or centralizer solver. The exact residual-gap script checks its entire field and real embedding using rational intervals. Logs and JSON receipts are included. A cold replay compares JSON outputs byte for byte; the compressed permutation file is validated by its array contents, not by ZIP container metadata. No CKM input enters any of these five computations.

The standard degree-22 generators are from the ATLAS of Finite Group Representations, <https://brauer.maths.qmul.ac.uk/Atlas/spor/M22/>, generator files `gap/M22G1-p22B0.g1` and `gap/M22G1-p22B0.g2`. Thomas Connor and Dimitri Leemans, The subgroup lattice of M22 (2012), <https://leemans.dimitri.web.ulb.be/atlaslat/m22.pdf>, independently documents the relevant A5, A6 and A7 subgroup classes; the selected class is recorded by explicit permutations rather than inferred only from a class name. Claudia Hagedorn, Aurora Meroni and Lorenzo Vitale, Mixing Patterns from the Groups  $\Sigma(n, \phi)$ , arXiv:1307.5308, <https://arxiv.org/abs/1307.5308>, supplies the Valentiner generators and prior residual analysis. The known 42-vertex graph is described in Wei Jin and Cheryl E. Praeger, Normal quotients of

diameter at most two of finite three-geodesic-transitive graphs, *Journal of Combinatorial Theory A* 178 (2021), 105349, <https://doi.org/10.1016/j.jcta.2020.105349>. These sources establish antecedents; they do not independently verify the new receipts or confer a publication claim.

# Valentiner interactions and the protection boundary

## Result and scope

The recovered cap geometry has a supported central-cover route to a genuine complex triplet. The triplet's first new symmetric invariant is a unique sextic, derived here directly from its exact generators. A minimal vectorlike mediator model has exactly three light modes per charge sector before electroweak breaking, and its canonical matching transmits the flavon frame. Its complete scalar operator space nevertheless contains independent cross-sector quartics. A target-free numerical local-vacuum experiment shows one such operator changing the observable CKM coefficient. Neither the golden coefficient nor the 66-degree physical phase is derived.

These results replace an unspecified symmetry search with a concrete representation, a fixed invariant tensor, specified field content and a counterterm audit. The sextic itself is the classical Valentiner/Wiman invariant, not a newly discovered polynomial. Publication novelty of the combined construction and obstruction requires comparison with the existing invariant-theory and flavor literature.

## The central-cover connection

The earlier cap computation established a component stabilizer  $A_7$  and a six-cap block stabilizer  $A_6$  of index seven in that  $A_7$ . Both  $A_7$  conjugacy classes in  $M_{22}$  lift to  $3.A_7$  in  $3.M_{22}$ . The GAP Character Table Library lists the index-seven maximal subgroup of  $3.A_7$  as  $3.A_6$ , of order 1080. Thus the full preimage of the cap-block  $A_6$  is the nonsplit Valentiner cover, rather than the split product  $C_3 \times A_6$ . This establishes the abstract central-cover chain  $3.A_6 < 3.A_7 < 3.M_{22}$  for the recovered stabilizer geometry.

This conclusion uses the independently computed ordinary stabilizers together with published subgroup classifications. We have not constructed an explicit matrix representation of  $3.M_{22}$  or lifted every recovered permutation generator. The local triplet does not extend to a three-dimensional representation of the whole cover by this argument. A global bundle, a global mass operator on 7392 caps, and decoupling of its other modes remain separate obligations.

Primary sources:

<https://www.math.rwth-aachen.de/~Thomas.Breuer/ctbllib/ctbltoc/data/3.M22.html> ;

<https://www.math.rwth-aachen.de/~Thomas.Breuer/ctbllib/ctbltoc/data/3.A7.html> ;

<https://www.math.rwth-aachen.de/homes/sam/ctbllib/ctbltoc/data/3.A6.html> . The exact local  $SU(3)$  generators are those of Hagedorn, Meroni and Vitale, <https://arxiv.org/abs/1307.5308> , equations (7,33).

## An exact tensor boundary

Let  $R$  be the faithful complex triplet and  $G=3.A_6$ . We closed its four generators exactly in  $Q(\sqrt{5}, \omega)$ , with  $\omega$  squared plus  $\omega$  plus one equal to zero, obtaining all 1080 elements. For every pair  $p, q$  with total degree at most eight, we evaluated the character average  $\dim \text{Inv}(R \text{ tensor } p \text{ times } \bar{R} \text{ tensor } q) = \sum_g \text{tr}(R(g))^p \text{conjugate}(\text{tr}(R(g)))^q / 1080$ . We independently computed the corresponding  $SU(3)$  tensor multiplicities from partitions and hook lengths.

**Theorem.** At total degree at most five the  $G$ -invariant tensor space equals the  $SU(3)$ -invariant tensor space, for arbitrarily labeled triplet fields. At degree six, the  $(3,3)$  space still has dimension six for both groups. The  $(6,0)$  space has dimension six for  $G$  and five for  $SU(3)$ , leaving exactly one new tensor. The conjugate statement holds for  $(0,6)$ ; the other degree-six bidegrees add none.

**Proof.**  $SU(3)$  invariants form a subspace of  $G$  invariants. The exact character averages and independent hook-length counts coincide in the stated bidegrees. Equality of dimensions therefore proves equality of spaces. In degree six, the symmetric-power average gives  $\dim \text{Inv}(\text{Sym}^6 R) = 1$ . Its tensor cannot be an  $SU(3)$  symmetric invariant, because there is no positive-degree  $SU(3)$ -invariant polynomial on one fundamental triplet. It exhausts the one-dimensional quotient. All character coefficients cancel exactly in the four-dimensional number field. The full counts are in `receipts/m22_interactions/valentiner_invariants.json`.

For one commuting triplet, through degree six the only nonconstant invariant bidegrees are  $(1,1)$ ,  $(2,2)$ ,  $(3,3)$ ,  $(6,0)$  and  $(0,6)$ , each with dimension one. The first three are powers of the norm. Consequently every renormalizable single-triplet potential has an accidental continuous  $SU(3)$  orientation symmetry. A sextic is the first possible finite-group anisotropy. This is an effective interaction in four spacetime dimensions, requiring a suppression scale or a specified mediator completion.

## The unique sextic in the recovered basis

$$F_6 = x^6 + y^6 + z^6 + A(x^4 y^2 + y^4 z^2 + z^4 x^2) + B(x^2 y^4 + y^2 z^4 + z^2 x^4) + D x^2 y^2 z^2$$

Writing the triplet as  $(x, y, z)$ , normalize the sextic by its  $x^6$  coefficient. It is  $F_6 = x^6 + y^6 + z^6 + A(x^4 y^2 + y^4 z^2 + z^4 x^2) + B(x^2 y^4 + y^2 z^4 + z^2 x^4) + D x^2 y^2 z^2$ , with  $A = [15 - 9\sqrt{5} + i(15\sqrt{3} + 3\sqrt{15})]/8$ ,  $B = [15 + 9\sqrt{5} + i(-15\sqrt{3} + 3\sqrt{15})]/8$  and  $D = 15 - 3i\sqrt{15}$ .

The coefficients arise from solving invariant linear equations, without a selected angle or golden CKM coefficient. Sparse monomial generators first reduce the degree-six space; the remaining generator leaves a single invariant. Each of the four polynomial identities  $F_6(g \phi) = F_6(\phi)$  is then checked exactly in the number field. The receipt records every monomial coefficient.

A potential contains a free complex overall coefficient multiplying  $F_6$  plus its conjugate. Uniqueness fixes internal component ratios; it does not fix the coefficient relative to other invariants. Its complex coefficients also do not establish physical CP violation. A consistent generalized CP action on all fields and its allowed couplings must still be constructed. Ordinary componentwise conjugation is not assumed to be a symmetry in this basis.

## Specified quarks, flavons and mediator interactions

Take  $Q_L$  in  $R$ , the Higgs in the trivial family representation, three right-handed up quarks and three right-handed down quarks in family singlets, and one vectorlike quark triplet  $F_u$  and one  $F_d$  in  $R$ . These vectorlike fields are Standard Model weak singlets with the respective up or down electric charge. Introduce six Standard Model singlet flavons  $\phi_i$  in  $R$ . This is a local four-dimensional effective model after selecting the cap stabilizer; it is not a derived global cap spectrum.

Add a specified shaping group  $C_6$  to the sixth power. Each right-handed singlet  $f_R$  has charge +1 under its own factor, its flavon has charge -1, and all other listed fields are neutral under that factor. The family and shaping symmetries permit, per sector,  $M_f F_{\bar{L}} F_R$ ,  $h_f Q_{\bar{L}} H_f F_R$ , and  $\sum_i \lambda_i F_{\bar{L}} \phi_i f_R$ , plus Hermitian conjugates. Here  $H_u$  is the conjugate Higgs doublet and  $H_d$  the ordinary one. Each family contraction is unique, by  $\dim \text{Hom}_G(R, R) = 1$ . The mediator mass is therefore flavor-universal. Cross-column flavon couplings are forbidden by the specified shaping charges. A bare  $Q_{\bar{L}} H_f f_R$  coupling has no family singlet. A mediator-triplet bilinear with one triplet flavon is forbidden by the nontrivial central charge. These exhaust renormalizable mass and Yukawa structures for this listed field content.

The symmetry allows separate  $M_f$ ,  $h_f$  and three  $\lambda_i$  in each sector. It does not correlate the up and down parameters. The shaping group is an explicit model assumption, not something derived from  $M_{22}$ . Its gauging, anomalies, a scalar messenger completion and quantum thresholds are not claimed.

## Three light modes and canonical projectors

$$K_f = I + A_f^\dagger A_f, \quad Y_f = -h_f A_f K_f^{-1/2}, \quad |V_{ij}|^2 = \text{Tr}(P_i^u P_j^d)$$

At  $H=0$  define  $C_f$  with columns  $\lambda_i$  times the flavon expectation values. The heavy row block is  $[C_f, M_f I]$ , with three left-handed rows and six right-handed columns. For nonzero  $M_f$  it has rank exactly three, regardless of  $C_f$ . Its right kernel therefore has dimension three; the three  $Q_L$  components are the corresponding left chiral light modes. The three heavy singular values are  $\sqrt{|M_f|^2 + \sum_i \lambda_i^2}$ . There are exactly three light families per charge sector in this specified local model, without interpreting a rank-three fiber as a global family count.

Set  $A_f = C_f / M_f$ ,  $K_f = I + A_f^\dagger A_f$ . The normalized light right frame is  $[I; -A_f] K_f$  to the power  $-1/2$ , and the matched Yukawa matrix is  $Y_f = -h_f A_f K_f$  to the power  $-1/2$ . These are exact tree identities at  $H=0$ , with no small-flavon expansion. They imply  $H_f = Y_f Y_f^\dagger = |h_f|^2 S_f (|M_f|^2 I + S_f)$  inverse, where  $S_f = C_f C_f^\dagger$ .

The positive spectral function is strictly increasing. Thus the left spectral projectors of  $H_f$  and  $S_f$  agree, including the mass ordering for a nondegenerate spectrum. Quark mixing is computed as  $|V_{ij}|^2 = \text{Tr}(P_i^u P_j^d)$ , and  $J = \text{Im Tr}(P_1^u P_1^d P_2^u P_2^d)$ . Arbitrary flavon frames remain arbitrary physical quark frames; canonical normalization neither creates nor protects the golden relation. At finite Higgs expectation value, heavy-light mixing supplies higher-dimensional EFT corrections and potential charged-current nonunitarity. Those corrections are not included in this matching statement.

## The complete scalar space through degree six

For the six specified flavons and C6 shaping factors, a degree smaller than six has equal numbers of each flavon and its conjugate. Consequently the scalar operators coincide with balanced Gram contractions: six quadratic norms; 36 quartics consisting of 21 norm products and 15 absolute overlap squares. With one Higgs, add its quadratic norm, its quartic norm and six Higgs-norm/flavon-norm products, for 50 real renormalizable scalar couplings in total.

At degree six the general Hermitian flavon space has 198 real parameters: 56 cubic norm products, 90 norm-times-overlap squares, the real and imaginary parts of 20 Gram triangles, and the real and imaginary parts of six individual  $F6(\phi_i)$ . Balanced epsilon contractions are Gram determinants and add no independent operators. Gram rank relations first occur at degree eight. The shaping charges forbid mixed polarizations of the new sextic at degree six: an unbalanced field must occur six times by itself. Under a specified CP action the count can decrease, but no CP classification has been assumed here. Every absolute overlap square remains CP-even under a common unitary anti-linear CP action on the triplets.

In particular,  $\epsilon |\phi_{u1}^\dagger \phi_{d1}|^2$  squared is an independent allowed quartic. Neither the group nor the shaping charges constrain its coefficient to the chosen sextic coefficient. This is a concrete obstruction to claiming protected alignment merely from uniqueness of F6. It is the interaction-level counterpart of the earlier allowed linear angular deformation, although no equality between those two coordinate descriptions is assumed.

## A target-free local-vacuum deformation experiment

We minimized the phase of each sextic analytically, fixed each flavon norm to one, and considered the reduced potential  $\sum_i -|F6(\phi_i)| + \epsilon |\phi_{u1}^\dagger \phi_{d1}|^2$  squared. This is the fixed-norm reduction of polynomial sextic terms plus their conjugates, with independent phases minimized. Radial stability requires separate radial terms; no global minimum or cosmological vacuum selection is certified.

Six independently generated sextic stationary directions were selected from the numerical group orbit, and their column strengths set to 0.07, 0.31 and 0.9 in each sector. None is a measured CKM anchor or the golden target. Analytic gradients are checked against independent central finite differences, and each numerical branch must have gradient norm below  $1e-7$ . The complete matrices, CP invariant and effective golden coefficient are recorded in `receipts/m22_interactions/valentiner_alignment_response.json`. For  $\epsilon=0$ ,  $C_{\text{eff}}$  is 0.4830422217 and  $J$  is 0.01663167860. At  $\epsilon=0.01$  they become 0.4831824152 and 0.01663578380. The intermediate positive and negative 0.001 cases also give distinct values. The  $\epsilon=0$  and  $\epsilon=0.01$  branches have different  $C_{\text{eff}}$ ; this change survives the exact canonical mediator matching theorem. The mixing is not phenomenologically realistic, which is appropriate for an allowed-deformation counterexample rather than a fitted proposal.

The found single-flavon extrema form a numerical 60-line orbit, and sixteen independent searches reach modulus approximately 1.0886621079. These are numerical observations, not an exact orbit classification or a proof of the global modulus bound. Optimizer success flags and final gradients are retained, including precision-loss flags when a sufficiently small verified gradient remains.

## What the next derivation must add

The sextic supplies a real representation-derived coefficient pattern, and the mediator calculation supplies a concrete light-family and normalization result. Their combination does not protect the observable golden relation. The next justified search must introduce interactions that correlate the sectors while controlling all 15 independent quartic overlaps, permit useful mixed sextic contractions without opening arbitrary singlet-copy matrices, and define a consistent generalized CP action. A proposed mechanism must then demonstrate its chosen spectrum and vacuum, test all remaining free parameters, and match the observable projectors. No choice of the desired CKM angle enters the invariant derivation above.

## Reproduction

Run `python python/develop_valentiner_invariants.py`, then `python python/develop_valentiner_alignment.py` from the repository. The first command produces an exact invariant receipt and 24 canonical-matching scenarios. The second produces a numerical local-branch receipt with an independent gradient check. The exact group code and dependencies are shared with `python/develop_valentiner_frames.py` and `python/m22_frame_requirements.txt`. The monograph renderer includes this complete chapter. No new Lean theorem or experimental measurement is asserted.

# Product-family CP and the global spectrum

The subsequent [input audit](FLAVOR\_SEARCH\_INPUT\_AUDIT.md) distinguishes these separate constructions and checks their missing matching steps. In particular, the supersymmetric messenger superpotential is not the supplied positive scalar alignment potential, and the nonsupersymmetric quark example's conjugate-link reverse coupling cannot be copied unchanged into a superpotential with just one chiral link. The audit also retains updated experimental comparisons without changing the historical inputs.

## Concrete advance and remaining boundary

This push constrains the problematic cross-sector interaction by changing the specified family group to  $G_u \times G_d$ , with each  $G_f = 3.A6$ , and joining the sectors with a bifundamental link. The bare absolute overlap quartic is forbidden. The first dressed overlap is degree six. The link has a unique new holomorphic sextic tensor, whose component ratios come from the previously derived Valentiner invariant. A renormalizable supersymmetric messenger chain generates this contraction at leading tree order, with its complete allowed holomorphic operator audit retained. A positive alignment interaction has exact zero-energy states at every group matrix, without inserting a CKM angle.

We also construct an exact anti-linear involution on the specified triplet and singlet fields. It leaves the sextic conjugate as required, and a canonical quark calculation produces a pair with opposite nonzero Jarlskog invariants. This establishes physical CP violation in a conditional matched example; it does not derive a unique phase or select the physical vacuum. Finally, the canonical cap connection is certified to have no zero modes. A fully specified modified global mass-squared operator has exactly three zero modes among 44,352 modes, with a gap of one half. Its required symmetry breaking and its chiral quark completion remain independent assumptions.

## Exact CP on the elementary field content

$$X\overline{R(g)}X^{-1} = R(u(g)), \quad X\overline{X} = I$$

Use the published outer automorphism that sends the faithful triplet to its conjugate:  $a$  maps to  $a^2$ ,  $h$  to  $q$ ,  $f$  to  $f a^2 h f$ ,  $h$  to  $q$ , and  $q$  to  $a$ . We evaluate these words on the exact generators, propagate the map through all 1080 elements, and check that every repeated group word has the same image. We solve  $X \text{conjugate}(R(g)) = R(u(g)) X$  as exact linear equations over  $Q(\sqrt{5}, \sqrt{-3})$ . The published representative can square to an inner action. An explicitly recorded inner adjustment yields a symmetric unitary  $X$  with  $X \text{conjugate}(X) = I$ .

The receipt gives the matrix entries, its normalization and the adjustment word. Before normalization,  $X_0^\dagger X_0 = (6-2\sqrt{5}) I$ . The sextic identity is checked coefficient by coefficient:  $F_6(X_0 \phi) = (6-2\sqrt{5})^3 \text{conjugate}(F_6)(\phi)$ . Equivalently  $F_6(X \text{conjugate}(\phi)) = \text{conjugate}(F_6(\phi))$ . Thus real coefficients of  $F_6$  plus its conjugate define an invariant interaction on this specified field content. Ordinary componentwise conjugation in the original basis would not establish this result.

There is an important scope restriction. We enumerate all 17 conjugacy classes and the four outer-class representatives supplied by the known Klein outer automorphism group. Every representative fails to send at least one class to its inverse class. For the representative interchanging this triplet with its conjugate, two classes fail. Inner adjustments do not change these class permutations. There is therefore no class-inverting automorphism that could define the same physical CP operation for arbitrary irreducible field content. The anti-linear operation constructed here is restricted to the elementary triplet, singlet and bifundamental fields that pass the explicit consistency and Lagrangian checks. Adding a different elementary representation requires a fresh audit.

Primary sources: Hagedorn, Meroni and Vitale, <https://arxiv.org/abs/1307.5308> , equation (34), for the outer maps and their classification; Chen et al., <https://arxiv.org/abs/1402.0507> , for the class-inversion requirement in generic finite-group CP; <https://doi.org/10.1140/epjc/s10052-025-15050-8> , for the representation-content dependence of conjugation in Sigma(1080). Our exact class enumeration and intertwiner calculation are independent computations using the recovered generators.

## Cross-sector sequestering and its exact operator boundary

$$G_u \times G_d, \quad |\phi_{ui}^\dagger \wedge \phi_{dj}|^2 \quad (\text{degree } 6)$$

Take  $\phi_{ui}$  in (3,1),  $\phi_{dj}$  in (1,3), and  $\Lambda$  in (3,3bar). Retain one independent C6 shaping factor for each right-handed singlet and its flavon. Before a link is inserted,  $|\phi_{ui}^\dagger \phi_{dj}|^2$  squared is not an invariant of the product family group. There is no invariant identification of its independent family indices. Its proposed replacement  $|\phi_{ui}^\dagger \Lambda \phi_{dj}|^2$  squared is invariant and has field degree six. Its nine coefficients remain independent. Arbitrary additional phase charges could not have forbidden the original absolute overlap square in the diagonal-group model: its charges cancel identically.

The bifundamental Molien calculation averages over the 1080 squared product group using exact power-trace class signatures and Newton identities. Through total degree six it has nonzero spaces (1,1) of dimension one, (3,0) and (0,3) of dimension one, (2,2) of dimension two, (4,1) and (1,4) of dimension one, (3,3) of dimension three, and (6,0) and (0,6) of dimension two. This is the complete self-operator count through that degree, not only a selected contraction.

The degree-three invariant is  $\det \Lambda$ . The two holomorphic degree-six invariants are its square and  $I_6(\Lambda)$ , obtained by contracting the unique symmetric sextic tensor with six link matrices and its conjugate tensor. Consequently there is only one genuinely new holomorphic orientation tensor at this degree. Balanced link-only sextics are the three symmetric polynomials of the singular values of  $\Lambda$ . At degree five the invariant is  $\det \Lambda \times \text{Tr}(\Lambda \dagger \Lambda)$ . Real parts are selected by the field-content CP involution; none of these counts supplies a relation between independent coupling constants.

The complete renormalizable scalar count for the listed six flavons, one link and one Higgs is 58 real parameters with this CP restriction. There are eight quadratic norms, one real cubic determinant coefficient, and 49 quartics. The quartics consist of 21 flavon norm products, six within-sector absolute overlaps, two link self-contractions, six link-norm/flavon-norm products, six  $\phi \dagger \Lambda \Lambda \dagger \phi$  or conjugate-side contractions, and eight Higgs quartics and norm portals. The nine direct up/down overlap quartics of the earlier diagonal-group model are absent. Without CP the determinant coefficient has two real components, giving 59 parameters. The counts refer to this explicitly assumed product family and shaping content, not to every possible UV theory.

On a unitary link,  $\Lambda^\dagger \Lambda$  is proportional to  $I$ . The six link/flavon orientation quartics then become norm terms and cannot rotate a source frame. Thus this construction removes the previous quartic angular perturbation in that regime. At degree six the dressed overlaps return, and a general Wilsonian effective theory permits them even if a chosen tree potential sets their coefficients to zero. They must be controlled by an additional mechanism before an exact CKM prediction could be protected.

## A representation-derived positive link interaction

$$I_6(U) = \langle T, T \circ U \rangle, \quad \text{Re } I_6(U) \leq \|T\|^2 = 16$$

Let  $T$  be the symmetric coefficient tensor defined by  $F_6(\phi) = T$  contracted with six copies of  $\phi$ . Tensor norms require multinomial weights; the exact polynomial coefficients give  $\text{norm}(T)^2 = 16$ . Define  $I_6(\Lambda) = \text{inner product of } T \text{ with } T \text{ composed with } \Lambda$ . For unitary  $U$ ,  $\text{norm}(T \text{ composed with } U) = \text{norm}(T)$ , so  $\text{Re } I_6(U)$  is at most 16 by Cauchy-Schwarz. Equality is equivalent to  $T \text{ composed with } U = T$ . All 1080 exact group matrices satisfy that tensor identity, independently of the chosen quark model or CKM target.

A globally nonnegative polynomial alignment energy is  $V_L = a \text{Tr}[(\Lambda^\dagger \Lambda - \text{ell } I)^2] + b |\det \Lambda - \text{ell } I|^2 + c \text{norm}[T \text{ composed with } \Lambda - \text{ell } I \text{ to the sixth } T]^2$ , for positive  $a, b, c$  and  $\text{ell}$ . Every  $\Lambda = \text{ell } R(g)$  is an exact zero-energy state. On the unitary, determinant-fixed manifold its last term is proportional to  $16 - \text{Re } I_6(U)$ . The last norm square contains degree-six and degree-twelve terms; the interaction is not a wholly renormalizable potential. Its coefficients and its higher-degree completion are explicit choices. We establish these zero-energy states and their positivity, without claiming an independently derived UV origin or a complete classification of all zero-energy matrices.

The link transforms under the elementary CP operation as  $\Lambda$  maps to  $X$  conjugate( $\Lambda$ )  $X^\dagger$ .  $I_6$  and  $\det \Lambda$  then map to their conjugates. The positive interaction is therefore invariant. It correlates the sectors through a fixed representation tensor rather than through an arbitrary fitted matrix. Its degenerate group-related states do not select a physical CKM phase by themselves.

## A renormalizable messenger origin for the sextic

The sextic also has an explicit interaction-level origin. Symmetric squared of the faithful triplet is an irreducible complex six, and symmetric cubed is an irreducible real ten. The latter has a unique invariant symmetric bilinear  $Q_{10}$ . Polarizing the sextic across two groups of three indices gives this bilinear, with  $F_6(\phi) = \text{Sym}^3(\phi)$  transpose  $Q_{10} \text{Sym}^3(\phi)$ . In an orthonormal symmetric-cube basis its exact identity is  $Q_{10}^\dagger Q_{10} = (8/5) I$ . We certify all 100 entries of that matrix identity in the coefficient field. This establishes nondegeneracy and fixes the internal component coefficients independently of a desired root.

For the product-family link introduce supersymmetric chiral messengers  $A$  in  $(\bar{6}, 6)$ ,  $A_\text{tilde}$  in  $(6, \bar{6})$ , and  $B$  in  $(10, 10)$ , with dimensions 36, 36 and 100. These are Standard Model singlets and neutral under the six shaping factors. The superpotential chain contains  $M_6 A \cdot A_\text{tilde}$ ,  $M_{10} B \cdot B$  divided by two,  $y_2 A \cdot (\Lambda \Lambda)$  projected to the 36, and  $y_3 B \cdot (A_\text{tilde} \Lambda)$  projected to the 100. Every displayed family contraction is unique. The CP action on these messenger fields is inherited from the symmetric-square and symmetric-cube actions of the already checked  $X$ ; no arbitrary new CP matrix is introduced.

Eliminating the massive fields at leading tree order generates a holomorphic operator proportional to  $-(y_2 y_3)^2 I_6(\Lambda)/(2 M_6^2 M_{10})$ , in compatible invariant normalizations. Thus the sextic is generated by renormalizable cubic vertices whose representation coefficients fix its component pattern. This is stronger than simply inserting a high-degree polynomial into the scalar potential. It is a supersymmetric messenger-sector construction, not a completed softly broken scalar model: the F-term scalar potential from the effective superpotential, soft terms, canonical kinetic corrections and physical vacuum selection must be matched separately.

The full audit counts all 34 possible monomial multisets of degrees one through three for these four chiral field species. There are no invariant linear terms, two independent mass contractions, and 15 cubic contractions: the determinant of  $\Lambda$ ; the two unique chain vertices; and four self-cubic contractions each for  $A$ ,  $A_{\tilde{}}$  and  $B$ . The unused self-cubics are not silently excluded.  $A_{\tilde{}}$  self-cubics can contribute at degree six to the same two-dimensional link-invariant space. The  $A$  and  $B$  self-cubics first affect higher orders in the small-link expansion for this massive branch. Hence the generated orientation tensor is fixed, while its coefficient and the determinant-squared coefficient remain independently adjustable. Other branches and the regime of large link expectation values require the full messenger equations.

The extra messenger fields do not couple holomorphically to the charged source flavons at renormalizable degree under the listed shaping charges. This controls a concrete set of interactions, while higher-degree Kahler and effective scalar terms can still generate the dressed cross-sector overlaps. No radiative absence theorem for those overlaps is asserted.

## Complete mediator matching with the reverse interaction included

Assign  $Q_L$  to  $(3,1)$ . The up-sector mediator remains a triplet of  $G_u$ . For the down sector introduce  $D_u$  in  $(3,1)$  and  $D_d$  in  $(1,3)$ , both vectorlike Standard Model weak singlets of down electric charge. The right-handed down singlets couple to  $D_d$  through the three down flavons. The allowed link Yukawa couplings run in both directions:  $g \bar{D}_u L \Lambda D_d R$  and  $g_{\text{reverse}} \bar{D}_d L \Lambda^\dagger D_u R$ . Both are allowed by the same mass and shaping charges. Omitting the reverse contraction would incorrectly restrict the operator space.

At  $H=0$  the down heavy block is  $[B, M]$ , with  $B=[0; C_d]$  and  $M=[M_u I, g \Lambda; g_{\text{reverse}} \Lambda^\dagger, M_d I]$ . The light normalized right frame is  $[I; -A] K$  to the power  $-1/2$ , where  $A=M^{-1} B$  and  $K=I+A^\dagger A$ . The Higgs couples  $Q_L$  to  $D_u R$ , giving  $Y_d=-h A_{\text{top}} K$  to the power  $-1/2$ . For an invertible  $M$  there are exactly three right light modes and six massive down mediator modes.

For  $\Lambda=\ell U$  and real parameters, write  $D=M_u M_d - g g_{\text{reverse}} \ell^2$ . Then  $A_{\text{top}}=-g \Lambda C_d/D$  and  $A_{\text{bottom}}=M_u C_d/D$ . Consequently  $K=I+[(g^2 \ell^2 + M_u^2)/D^2] C_d^\dagger C_d$ , and  $Y_d$  is a scalar times  $U C_d$  times that canonical factor. The left frame is exactly  $U$  times the left singular frame of  $C_d$ . This includes the reverse link coupling and all orders in flavon mixing. The heavy masses must remain nonzero, requiring  $D$  nonzero. Finite-Higgs and quantum threshold corrections are separate calculations.

## Physical CP established through quark invariants

$$J \neq 0, \quad \text{Im det}[H_u, H_d] \neq 0, \quad J_{\text{CP}} = -J$$

The numerical example uses a unitary  $W$  spanning the real fixed space of the anti-linear map, so  $X \text{ conjugate}(W)=W$ . Its source matrices are  $C_u=W \text{ diag}(0.07,0.31,0.9)$  and  $C_d=W \text{ diag}(0.08,0.27,0.85)$ . All listed mediator couplings are real. We evaluate every group link  $U=R(g)$ , select the largest absolute  $J$  without a target phase or mixing anchor, and calculate the conjugate link  $X \text{ conjugate}(U) X \text{ dagger}$ .

After canonical matching the pair has  $|J|$  approximately 0.0703201663, identical mixing magnitudes and opposite  $J$  signs. The chart phases are approximately plus and minus 94.6111562 degrees, and  $C_{\text{eff}}$  is approximately 1.50313413. An independent weak-basis check gives nonzero  $\text{Im det}[H_u, H_d]$ , approximately  $7.09e-9$  in absolute value. Nonzero values with nondegenerate quark spectra demonstrate physical CP violation; complex representation entries or flavon phases alone would not do so.

This example is conditional on the chosen CP-real source frames and on a selected group link vacuum. The CP-real basis itself has a real orthogonal freedom, and the scalar model has not selected those frames. The large mixing is not phenomenologically viable. Real, independently allowed column-coupling changes in a nonorthogonal source example alter both  $C_{\text{eff}}$  and the physical phase. The receipt retains all cases. Thus physical CP is possible within a consistent restricted elementary symmetry, but the observed phase and the golden relation remain unpredicted.

## The canonical global spectrum is gapped

The recovered six-dimensional real cap fiber is wedge squared of the standard four-dimensional  $A_5$  representation, with its stored positive metric. Construct the covariant graph Laplacian using the actual 252 directed incidence transports on a 42-cap component. In integral coordinates its characteristic polynomial is  $(t-2)15 (t-3)55 (t-5)21 (t-6)70 (t-7)21 (t-9)55 (t-10)15$ . We compute and factor this polynomial exactly over the integers. The numerical metric-whitened Hermitian matrix independently reproduces the spectrum.

The base connection has dimension 252 and zero-dimensional kernel. Its orientation double cover has dimension 504 and the same spectral levels with doubled multiplicities: the other block is obtained by  $t$  mapping to  $12-t$ . Both have exact mass-squared gap two. On all 176 isomorphic cap components the base space contains 44,352 modes and has no zero mode. A parallel rank-three projector therefore does not yield three parallel vectors. The  $A_5$  cover holonomy has no invariant vector in either irreducible golden triplet, consistent with the spectral result.

## A global three-mode operator with explicit breaking

$$\dim \ker M^2 = 3, \quad \lambda_{\text{next}}(M^2) = \frac{1}{2}, \quad N = 44\,352$$

To exhibit exactly what an additional assumption can accomplish, choose one 42-cap component and a spanning tree, transport a root frame along that tree, and replace every edge matrix by the corresponding pure-gauge transport  $R_w R_v$  inverse. This keeps the tree transports and changes 160 of the 252 directed edge matrices of the component. It trivializes the holonomy and is an explicit change to the canonical connection. It is not invariant under the full recovered  $M_{22}$  incidence action.

Transport the root rank-three Hodge projector by the same frames, and add a one-half penalty on its orthogonal complement. The scalar component Laplacian has exact spectrum 0 with multiplicity one, 4 with multiplicity 21, 7 with multiplicity six, and 9 with multiplicity 14. The

selected six-dimensional bundle therefore has three zero modes, and levels one half, 4, 4.5, 7, 7.5, 9 and 9.5 with multiplicities 3, 63, 63, 18, 18, 42 and 42 respectively. Gap every other cap fiber by an onsite mass squared of two, adding 44,100 massive modes.

This explicitly defined global mass-squared operator has exactly three zero modes among 44,352 modes and gap one half. Their normalized wavefunctions are transported root vectors divided by  $\sqrt{42}$ ; the numerical mass and normalization residuals are below  $2e-15$ . The spectral statement is exact from the scalar characteristic polynomial and pure-gauge similarity. Selecting the component breaks M22, and flattening its connection breaks further symmetry. We have not derived either selection dynamically. The operator is a positive Hermitian flavor mass-squared construction, not a gauge-invariant chiral quark UV completion. Its real A5 transport and the local complex Valentiner link model have not been unified into one global particle theory.

## What remains after this push

The bare quartic deformation is now excluded by specified product symmetry, but the degree-six dressed operators survive. The CP action is explicit on the listed fields, and physical CP is demonstrated in canonically matched matrices, but its vacuum and numerical phase are not selected. The global spectrum is calculated, including an exact obstruction for the canonical connection and a conditional three-mode alternative, but the required breaking interactions and chirality have not been derived. These boundaries are sharper than another constructed golden root. A complete predictive theory must connect the product-family link, its mixed operator coefficients and CP-compatible scalar vacua to a dynamically selected global chiral spectrum.

## Reproduction and receipts

Run `python python/develop_valentiner_cp.py`, `python python/develop_valentiner_link.py` and `python python/develop_m22_global_modes.py`, and `python python/develop_valentiner_messenger.py`. The computations use the existing exact triplet generators, sextic receipt and frozen cap transports. The new receipts are `valentiner_cp.json`, `valentiner_link.json`, `m22_global_modes.json` and `valentiner_messenger.json` under `receipts/m22_interactions`. The CP and polynomial identities, class map propagation, bifundamental invariant counts and characteristic-polynomial factors use exact arithmetic. The quark examples, matrix whitening and normalization residuals are numerical cross-checks. No experimental data or new Lean theorem is claimed.

# Audit of the flavor search inputs

Audited baseline: `c7cdb4cf9814a7be006c17965678084b85211612`. Date: 6 October 2026. This audit covers the flavor search and its B5, M22, Valentiner, messenger and CP inputs. It is not an audit of every unrelated arithmetic application in the repository. Historical data and numerical examples remain pinned; newer measurements are compared separately.

The mathematical constructions are useful, but their combination is not yet a derived particle theory. The decisive unsupported step remains the identification of a selected scalar or representation invariant with a relation between the canonically normalized up and down quark eigenframes. The audit also identifies a concrete incompatibility in directly combining the ordinary quark example with the proposed supersymmetric messenger sector, and a missing scalar-potential calculation that can already be tested exactly.

## Input ledger

Here “computed” means checked for the specified input object, not established as a law of nature. “Chosen” identifies an additional model assumption. “Missing” means no calculation in this research establishes the required physical identification.

| Input or inference                                                                                           | Audit status                                               | Consequence for the search                                                                                        |
|--------------------------------------------------------------------------------------------------------------|------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------|
| Exact three-generation unitary CKM parametrization                                                           | Correct within the three-light-generation decoupling limit | Vectorlike finite-Higgs effects require separate charged-current matching.                                        |
| Relation $s_{13}=C*s_{12}*s_{23}$                                                                            | Chosen phenomenological hypothesis                         | CKM unitarity alone does not imply it.                                                                            |
| Golden value $C=(3-\sqrt{5})/2$                                                                              | Training-nominated candidate                               | Neither B5 nor M22 has derived its identification with the observable coefficient.                                |
| $\delta=66$ degrees                                                                                          | Training-nominated phase                                   | A scalar angle, a root-of-unity eigenphase and the standard CKM phase are different quantities until matched.     |
| Original $V_{us}=.22431$ , $V_{cb}=.0411$                                                                    | Pinned historical measured central values                  | They have uncertainties; the 2026 combined $V_{cb}$ central value is $.0407$ .                                    |
| $\Gamma$ and $V_{td}/V_{ts}$ training anchors                                                                | Explicit measured inputs with assumptions                  | $\Gamma$ helps select 66 degrees; the mixing-ratio extraction assumes SM dynamics.                                |
| Independent Gaussian/split-normal likelihood                                                                 | Chosen summary-data approximation                          | Missing cross-input covariance prevents treating it as a precision global fit.                                    |
| Basis $\alpha$ , $\alpha_s$ , $\pi$ , $\phi$ , $m_p/m_e$ , rational alphabet, support penalty and phase grid | Chosen search domain and prior                             | A MAP formula in this domain is not a symmetry theorem or a prior-independent discovery.                          |
| Excluded observable keys in the training API                                                                 | Correct software safeguard                                 | Previously published observables were known; the exercise is retrospective, not blinded.                          |
| B5 action on five exponent axes                                                                              | Correct combinatorics                                      | Swapping physically labelled constants is not a physical symmetry. B5 supplies no primitive 60th-root eigenphase. |

| Input or inference                                                                      | Audit status                                                       | Consequence for the search                                                                                                                                   |
|-----------------------------------------------------------------------------------------|--------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Golden extremum of positive-definite symmetric integral 2-by-2 determinant-one matrices | Correct conditional spectral statement                             | Integrality, determinant, matrix choice, canonical kinetic metric and quark coupling are physical assumptions.                                               |
| Six-harmonic potential selecting plus/minus 66 degrees                                  | Constructed and exactly checked                                    | Its coefficients were found with the desired root known; their physical origin is absent. An additive constant in the implementation does not affect minima. |
| Eight rational stationarity constraints; four on the six-term support                   | Correct over the rational coefficient field                        | They are not eight independent equations for arbitrary real couplings at one angle.                                                                          |
| CP symmetry of the scalar potential                                                     | Correct                                                            | It permits CP-even deformations and pairs the two signs; it does not select the observed sign.                                                               |
| Generic 1% coupling sensitivity                                                         | Numerical study of the declared perturbation ensemble              | Its interval is not an experimental confidence interval or an invariant measure of naturalness.                                                              |
| 20% attraction basin and low-temperature occupation                                     | Conditional dynamics with specified measure                        | Angular kinetic metric, initial measure, thermal history and CP-sign selection are not derived.                                                              |
| Recovered Wilson ledger and M22 algebra                                                 | Verified selected archival subset and exact reconstruction         | The recovery is not the whole lost archive; provenance checks do not validate all historical physical claims.                                                |
| Three-dimensional M22 multiplicity spaces                                               | Correct                                                            | M22 acts trivially on their multiplicity factors, allowing arbitrary frame matrices. They are not faithful family triplets.                                  |
| Canonical decorated A5 transport and orientation cover                                  | Computed geometry and projector splitting                          | Rank-three parallel projectors do not imply three light quark modes; the canonical connection has zero kernel.                                               |
| Central C60 subgroup in the twelvefold cover                                            | Supported group-theoretic construction                             | Central phases cancel from neutral spectral-projector overlaps; they do not select a CKM frame.                                                              |
| 3.A6 inside 3.A7 inside 3.M22                                                           | Abstract subgroup/cover route supported by character-table data    | The specific global twisted quark bundle and its explicit lifted matrices have not been built.                                                               |
| Exact Valentiner triplet and low-degree invariant census                                | Computed for the specified generators and polynomial/tensor spaces | The unique sextic is a classical invariant. Its uniqueness fixes tensor components, not all independent Wilson coefficients.                                 |
| Failure of complete residual frames and the tested two-involution route                 | Correct scoped obstructions                                        | They exclude those unbroken constructions, not every model with controlled symmetry breaking.                                                                |
| Product family symmetry $G_u \times G_d$                                                | Added model assumption                                             | It is not a consequence of the recovered single M22 action.                                                                                                  |
| Six separately charged source flavons, singlet right-handed quarks and chosen mediators | Added field/charge assumptions                                     | The field content already contains three column labels. Their presence is not a derivation of three families.                                                |
| Global versus gauged family and shaping symmetries                                      | UV realization unspecified                                         | A gauge completion and its anomaly constraints have not been established. Formal selection rules alone do not supply that completion.                        |
| Product symmetry forbids bare cross-sector overlap quartics                             | Correct for those representations                                  | It still permits nine dressed degree-six overlaps and other source/link couplings.                                                                           |
| 58 CP-even renormalizable scalar parameters                                             | Specified nonsupersymmetric scalar-sector count                    | It is not the parameter count of a supersymmetric completion including messenger scalars, Kähler terms and soft breaking.                                    |

| Input or inference                                                | Audit status                                                         | Consequence for the search                                                                                                                                                  |
|-------------------------------------------------------------------|----------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Positive link alignment at scaled group matrices                  | Correct for the chosen polynomial potential                          | Unitarity, determinant stabilization and the degree-twelve completion were supplied; uniqueness and physical vacuum selection were not proved.                              |
| Renormalizable holomorphic sextic messenger chain                 | Representation counts and bilinear support the leading contraction   | Other allowed self-cubics remain independent. The 17 invariant coefficients belong to the four-species messenger sector, not the whole theory.                              |
| Supersymmetric messenger superpotential versus quark link example | Separate constructions; direct amalgamation invalid                  | The quark reverse coupling uses the conjugate link. It cannot be promoted unchanged to a holomorphic superpotential with only the same chiral link field.                   |
| Source frames and column hierarchies                              | Chosen backgrounds and couplings                                     | The CP-real basis has real orthogonal freedom. No joint scalar minimum selects its orientation or realistic mass hierarchy.                                                 |
| Canonical tree matching at zero Higgs background                  | Correct within the specified invertible heavy blocks                 | Thresholds, loop-induced kinetic corrections and finite-Higgs current normalization remain separate.                                                                        |
| Scale at which the golden relation is imposed                     | Unspecified physical matching input                                  | CKM parameters run above the weak scale. The standalone SM running diagnostic is not a threshold-matched evolution of the messenger construction.                           |
| Anti-linear involution on the chosen elementary fields            | Exactly computed with a representation-content restriction           | 3.A6 has no class-inverting automorphism valid for arbitrary irreducible content. All added fields and contractions need compatible CP action.                              |
| Nonzero J and commutator invariant in the quark example           | Correct conditional demonstration of weak CP violation               | The group element was selected for large absolute J, not by a scalar vacuum; it gives unrealistic mixing and no prediction of the physical phase. Strong CP is unaddressed. |
| Modified 44,352-mode operator with three zero modes               | Correct conditional mass-squared spectrum                            | Component choice, connection flattening and rank-three penalty are engineered breaking inputs. A chiral mass operator and the complex local model are not unified.          |
| Observable polynomial relations                                   | Exact consequences of the nominated CKM chart with branch conditions | Elimination creates a test of the hypothesis; it does not supply an independently justified interaction.                                                                    |
| Reproducible artifacts and passing tests                          | Evidence of implementation consistency                               | Neither byte reproduction nor test count establishes experimental truth, radiative protection or novelty.                                                                   |

## The rational-coupling qualification

Set  $x=2\cos(\theta)$ . The minimal polynomial of  $2\cos(66^\circ)$  has degree eight. A rational derivative vanishing at that root must be divisible by this polynomial, so it also vanishes at all its conjugates. The independent Chebyshev calculation reproduces rank eight for the constant plus twelve harmonics, and rank four on harmonics 2, 3, 5, 8, 9 and 12.

For unrestricted real coefficients, stationarity at one fixed interior angle is one nonzero linear equation. For example,  $\cos(2\theta)+b\cos(3\theta)$  is stationary at 66 degrees when  $b=-2\sin(132^\circ)/(3\sin(198^\circ))$ . That example does not claim a minimum or a protecting symmetry. It isolates the coefficient-field assumption. Representation coefficients may be algebraic and independent physical couplings may be real, so the appropriate coefficient

field must be derived before promoting the rational rank to a model-building requirement. The exact CP-even deformation remains a valid obstruction in the scalar model where it is allowed.

## The superpotential is not the chosen scalar alignment energy

In global supersymmetry with canonical kinetic terms, the scalar F-term potential is  $\sum |dW/d\Lambda_{ij}|^2$ ; it is not  $-\text{Re } W$ . The audit reconstructs the fully symmetric sextic tensor from its exact polynomial coefficients. Its norm squared is 16 and its one-index reduced tensor is exactly  $(16/3) I$ . Hence, for  $I6(\Lambda) = \langle T, T \text{ composed with } \Lambda \rangle$ ,

$$I6(I) = 16, \quad dI6/d\Lambda_{ij} \text{ at } I = 32 \delta_{ij}, \quad \text{and} \quad \sum \Lambda_{ij} dI6/d\Lambda_{ij} \text{ at } I = 96.$$

With only  $W = \kappa I6$ , the radial line  $\Lambda = rI$ , for real positive  $r$ , has  $VF = 3072 |\kappa|^2 r^{10}$ . A nonzero scaled group matrix is therefore not radially stationary in this isolated model, while the origin has zero energy. Generating the sextic in  $W$  does not by itself generate the previously chosen nonzero alignment vacuum.

This is a limitation of the isolated term, not a no-go theorem for the allowed messenger EFT. The independently allowed determinant term can balance the gradient: for  $W = a \det(\Lambda) + \kappa I6(\Lambda)$ , a nonzero  $rI$  is F-flat when  $a + 32 \kappa r^3 = 0$ . Its group transforms are then F-flat as well. The origin remains F-flat, and the complete allowed terms, kinetic metric, source couplings and soft breaking must determine stability and occupation. This audit does not assert that those nonzero branches solve the CKM problem.

The ordinary down-quark example contains both a link and its Hermitian conjugate in its heavy mass block. That is allowed in the nonsupersymmetric theory. Holomorphy prevents copying both entries unchanged into a renormalizable superpotential with just one chiral link. A separate conjugate chiral link, a modified coupling chain, or a nonholomorphic effective interaction would require new fields or matching and a fresh operator audit. A hybrid construction may be possible; none was supplied by the prior calculations.

## Updated data and the interpretation of the target

The 2026 PDG CKM review retains  $|V_{us}| = .22431 \pm .00085$ , and quotes combined  $|V_{cb}| = .0407 \pm .0013$ ,  $|V_{ub}| = .00389 \pm .00016$ , and  $\sin(2\beta) = .710 \pm .011$ . Inclusive and exclusive determinations remain distinct, and the review's correlated global fit must not be treated as independent data added to these inputs.

Keeping the golden coefficient and 66-degree phase fixed while substituting only the updated central  $V_{cb}$  gives  $|V_{ub}| = .00348716940$  and  $\sin(2\beta) = .690881841$ . These remain conditional chart predictions. The rounded combined measurement centers instead give  $C_{\text{eff}} = .426088669$ . First-order independent-error propagation gives about  $.0222480$ , putting the golden coefficient approximately 1.98 nominal standard deviations below that central combination. This diagnostic is not a joint likelihood or an exclusion significance. It establishes that the measured inputs do not currently fix the coefficient to the golden value. No new fit or posterior selection was run.

The existing retrospective search already reports that its leading formula has only .1603% posterior probability in the support-three family, its B5/smooth training evidence ratio is about .800, and removing  $\phi$  barely changes its broad forecast. Those findings must remain inputs to our judgment. The narrow forecast conditional on choosing the golden formula does not include uncertainty over choosing that formula. The 66-degree phase was favored using measured  $\gamma$ ; its appearance in later constructed potentials is not independent evidence for it.

## What the audit changes

The search should retain the recovered exact algebra, genuine triplet invariant tensors, canonical quark matching, explicit CP consistency calculation and the scoped obstructions. It should stop treating the golden target, product symmetry, CP-real source frames, unitary link, and three-mode modified spectrum as deductions from one common starting symmetry. They are different added inputs, and the physical maps between them remain missing.

A defensible next result would specify one field content and one Lagrangian, derive its scalar and kinetic terms, solve its joint source/link vacuum, and extract the two canonically normalized quark eigenframes without inserting the target. Allowed variations then test whether any observable relation is forced. A failed relation would still be an informative result. Recovering more elaborate arithmetic that only re-encodes the nominated angle would not repair the audited gaps.

## Verification and sources

Run ``PYTHONPATH=python OPENBLAS_NUM_THREADS=1 python python/audit_flavor_inputs.py``. The deterministic receipt records the independent rational ranks, exact reduced tensor and sextic gradient, updated central-input calculation, and SHA-256 hashes of eleven audited implementation/input files. The historical training receipt is unchanged. The audit reran 65 flavor unit tests and eight Valentiner invariant/CP/global checks successfully. These checks do not independently reconstruct every archival M22 object or certify a complete UV theory.

Primary references: PDG, *\*CKM Quark-Mixing Matrix\**, revised March 2026, <https://pdg.lbl.gov/2026/reviews/rpp2026-rev-ckm-matrix.pdf>, equations 12.8, 12.11, 12.12 and 12.20; S. P. Martin, *\*A Supersymmetry Primer\**, <https://arxiv.org/abs/hep-ph/9709356>, sections 3.2 and 4.7; M.-C. Chen et al., *\*CP Violation from Finite Groups\**, <https://arxiv.org/abs/1402.0507>, especially section 4's restricted-representation qualification. Existing character-table and generator references remain in the M22 and Valentiner chapters. The sextic-gradient and coefficient-field checks are independent calculations in this audit, rather than claims taken from these references.

# Supersymmetric vacua and the CKM boundary

This calculation connects the previously separate link, scalar and quark constructions in a specified effective field theory. It derives an isolated group-aligned supersymmetric link branch and its complete local mass spectrum. It then proves that the minimal exact-supersymmetry source sector has only zero flavon vacua. A specified spurion-breaking extension supplies locally stable nonzero sources and a joint vacuum with physical weak CP violation. Finally an exact finite census excludes every orthogonal axis branch from realistic CKM mixing, and a permitted cross-sector interaction moves the solved quark frame.

Neither the golden coefficient nor 66 degrees enters the superpotential, source directions, breaking terms or vacuum parameters. The positive result is an explicit dynamical realization of discrete alignment and weak CP on a locally stable branch. The negative result is a sharp limitation of that branch, not a no-go theorem for broken Valentiner flavor models. The global family spectrum from M22 is not used as a physical input or claimed to be unified here.

## One specified flavor effective theory

Take  $G_u \times G_d$ , with each factor the exact  $3.A_6$  triplet group. The chiral link  $K$  transforms as  $(\bar{3}_u, 3_d)$ . There are three source flavons  $\Phi_{ui}$  in  $(\bar{3}_u, 1)$  and three  $\Phi_{di}$  in  $(1, \bar{3}_d)$ . Each source has its own  $C_6$  shaping charge; its corresponding right-handed conjugate quark has the opposite charge. All other family messengers in the displayed quark chain are neutral under these factors. The left-handed quark doublet  $Q$  transforms as  $(3_u, 1)$ . Two Higgs doublets and vectorlike quarks carry the usual Standard Model gauge representations. Matter parity and baryon conservation are assumed. The canonical kinetic metric, finite-order EFT truncation and eventual gauge/anomaly completion remain separate assumptions.

Write  $F_6$  for the complex-conjugate polynomial coefficients of the previously derived Valentiner sextic, and  $I_6$  for the corresponding holomorphic link invariant. On the link-and-source scalar sector, all nonconstant holomorphic invariants through field degree six give

$$W_{\text{flavor}} = a \det(K) + b I_6(K) + c \det(K)^2 + \sum_i \kappa_i F_6(\Phi_i).$$

There are nine independent contractions: the link determinant, its square, the new link sextic and six source sextics. Individual shaping charges exclude mixed link/source and mixed-source holomorphic terms at these degrees. This is a complete statement about this scalar superpotential sector, not a complete Kähler, breaking or higher-degree quark EFT enumeration. CP permits real coefficients in the checked convention. The source coefficients are independently nonzero; no golden relation among them is assumed. With fields of mass dimension one,  $a$  is dimensionless and  $b$ ,  $c$  and  $\kappa$  have dimension minus three.

The physical Dirac row link is  $L = \text{conjugate}(K)$ . The quark superpotential contains  $Q.H_u U^c$ ,  $u_u U^c$ , source terms  $u_i^c \Phi_{ui}^T U$ ,  $Q.H_d D_u^c$ ,  $u_d D_u^c$ ,  $md D_d D_d^c$ , the forward term  $g D_u^T K D_d^c$ , and source terms  $d_i^c \Phi_{di}^T D_d$ . All column couplings are independent real inputs. A Higgs  $\mu$  term and positive Higgs breaking masses keep the Higgs backgrounds zero in this flavor-scale calculation; electroweak breaking is not derived.

The earliest holomorphic reverse vertex is  $\eta D_d^\dagger \text{adj}(K) D_u^c$ . It has field degree four, with an independent coefficient of mass dimension minus one. The adjugate is a quadratic polynomial in  $K$ , and is defined even at singular  $K$ . It replaces the invalid attempt to put  $K^\dagger$  in a superpotential with one chiral link. The corresponding physical heavy down-quark block is  $M = [[\mu, I, g, L], [\eta \text{adj}(L), m_d, I]]$ . Canonical tree matching is applied to this block, rather than copying the earlier nonsupersymmetric result. Additional higher-degree quark terms are not asserted absent.

## The group-aligned branch and all nine link masses

$$r^3 = -\frac{a}{32b+2c}, \quad m_1 = |96b+6c| |r|^4, \quad m_8 = |72b| |r|^4$$

The exact symmetric tensor  $T$  has norm squared 16. Its two-index reduced tensor is  $(4/3)(\delta_{ik} \delta_{jl} + \delta_{il} \delta_{jk})$ ; all 81 entries are checked in the coefficient field. This gives  $d^2 I_6(I)[E, F] = 40(\text{Tr}(E)\text{Tr}(F) + \text{Tr}(EF))$ . Combined with the determinant derivatives, it establishes the  $F$ -flat branch

$r^3 = -a/(32b+2c)$ , with  $K = r \text{conjugate}(R(g))$ .

At a nonzero  $r$ , every group transform has zero  $F$  energy. On that branch the superpotential Hessian is  $r^4[(8b+2c)\text{Tr}(E)\text{Tr}(F) + 72b \text{Tr}(EF)]$ . The nine complex link modes split into a singlet with fermion mass  $|96b+6c||r|^4$  and eight modes with mass  $|72b||r|^4$ . Canonical scalar squared masses are their squares, with real multiplicities two and sixteen. Thus a nonzero branch is locally isolated in the link sector when  $b$  and  $16b+c$  are nonzero, in addition to the existence conditions. No continuous family group is gauged here, so these eight massive directions are not Goldstone modes.

For  $a = -.0324$ ,  $b = 1$  and  $c = .2$ , the real branch has  $r = .1$ . Its fermion masses are .00972 and .0072. Four independent coefficient variations retain the nonzero branch and positive masses. The radius follows the coupling ratio; it is not a mixing-angle input. The origin also remains  $F$ -flat. Other  $F$ -flat link solutions and global occupation have not been classified.

There is a local protection statement worth separating from a CKM prediction. The diagonal family stabilizer of  $rI$  fixes only the scalar matrices, because  $3 \text{ tensor } 3\text{bar}$  decomposes as  $1+8$ . An invariant superpotential gradient at  $rI$  must therefore lie in that one-dimensional fixed subspace. With a nonzero singlet Hessian, small invariant holomorphic deformations preserve a nearby scalar-radius branch by the implicit function theorem. They do not need the specially chosen coefficients of the original angular potential. This protects a local link alignment branch; source-dependent operators can remove that stabilizer and change the relative quark frame.

## What exact supersymmetry does to the sources

$$\nabla F_6(\phi) = 0 \iff \phi = 0$$

The minimal source superpotential does not stabilize nonzero sources. The exact computation establishes  $\text{grad } F_6(\phi) = 0$  if and only if  $\phi = 0$ . To verify this, write  $u = x^2$ ,  $v = y^2$  and  $w = z^2$ , turning  $F_6$  into a cyclic cubic  $H$ . One-coordinate support has a nonzero derivative. The two-coordinate derivative resultant is  $(67797-35235i\sqrt{15})/32$ , which is nonzero. On three-coordinate support, the two-variable Gröbner basis of  $H_u$ ,  $H_v$  and  $H_w$  after  $u=1$  is [1]. Cyclicity covers every projective patch. The independent test works directly with the original quintic derivatives, without the squared-coordinate reduction, and also obtains [1]. Conjugating the representation leaves the obstruction unchanged.

Consequently all six flavons vanish in the exact-SUSY, zero-matter, zero-Higgs vacuum slice for nonzero  $\kappa_i$ . Both Yukawa matrices then vanish. There is no CKM frame to extract. The eighteen complex source modes have zero quadratic mass, but their potential is lifted at degree ten; they are not flat moduli. This is an obstruction to this minimal field-content/truncation, not to models with driving fields, different charges or breaking interactions.

## A specified breaking extension with stable nonzero sources

$$V_\phi = \kappa^2 \|\nabla F_6\|^2 - m^2 \|\phi\|^2 + 2A\kappa \text{Re } F_6$$

Add for each source  $V_i = \kappa_i^2 \text{norm}(\text{grad } F_6)^2 - m_i^2 \text{norm}(\Phi_i)^2 + 2A_i \kappa_i \text{Re } F_6(\Phi_i)$ . These are explicitly specified symmetry-respecting spurion-breaking EFT terms. They are not claimed to exhaust the allowed breaking action, and their origin or common coefficients are not derived. The independent column labels and their quark couplings are already field-content inputs, not a derivation of three families.

There is a stationary axis branch  $\Phi = \rho e_j$  when  $m_i^2 = \kappa_i^2 \rho^8 (180 + 6t)$ , where  $t = A_i / (\kappa_i \rho^4)$ . The two transverse tensor coefficient magnitudes are exactly  $\sqrt{45/2}$ . In canonical real coordinates the six squared masses, divided by  $\kappa_i^2 \rho^8$ , are  $1440 + 24t$ ,  $-36t$ , and each of  $-90 - 6t$  plus/minus  $(3\sqrt{10}/2)|48 + 2t|$  repeated twice. They are positive with positive  $m_i^2$  throughout

$-30 < t < -(90 + 72\sqrt{10})/(6 + 3\sqrt{10})$ .

At  $t = -24$  they are 864, 864 and four copies of 54. The source vacuum energy is  $-48 \kappa_i^2 \rho^{10}$ , below the source origin. Every group transform of an axis has the same energy and spectrum. The  $t = -26$ ,  $-24$  and  $-22$  cases are retained; direct six-dimensional real Hessians independently confirm their positive spectra. These are locally stable vacua. Other source extrema and the global winners are not classified.

Nonzero source  $F$  backgrounds also induce bilinear terms in the quark scalar potential. They cannot be ignored in a stability claim. At  $\rho = .1$ ,  $\kappa = 1$  and the stated real quark column couplings, their operator norm is at most  $5.4e-5$ . Adding a specified positive universal quark scalar breaking mass squared of  $1e-4$  gives a rigorous lower bound  $4.6e-5$  after including the nonnegative supersymmetric mass block. Explicit up and down scalar matrices have smallest squared masses approximately  $9.99965e-5$  and  $9.99985e-5$ . Positive Higgs masses complete local stability in the declared zero-Higgs, zero-matter slice. This is not a threshold- or loop-matched complete Standard Model vacuum.

## CP on a link and on a joint vacuum

For real CP-compatible coefficients,  $r^3$  is real. The three cube-root phases differ by a central element of  $3A_6$ . The group link states form one family orbit. Acting with the elementary CP involution sends  $rR(g)$  to  $r^*R(u(g))$ ; a family transformation and, if needed, a central element restore the vacuum. The group link by itself therefore conserves a CP operation combined with family symmetry. Complex entries alone do not establish weak CP.

The nonzero source vacua change that conclusion. Exact field arithmetic identifies 45 axis rays, 30 unordered orthogonal frames, and two family orbits of fifteen frames each. The generator action on exact rank-one projectors, their orthogonality and the orbit partition are checked without clustering tolerances. Their 900 ordered up/down frame pairs cover every group link choice, because a group link permutes the ray and frame sets. Mass-order permutations only permute the displayed magnitudes or reverse the sign of  $J$ .

A joint stationary vacuum from the retained frame pairs has  $J = -\sqrt{3}/32$ , with an exactly computed invariant-product imaginary part. Canonically normalized quark matrices reproduce  $J = -.0541265877$  and their CP conjugates give the opposite sign, with identical spectra and magnitudes. An independent weak-basis commutator determinant is nonzero. The worked pair has  $|V_{us}| = |V_{cb}| = 1/2$ ,  $|V_{ub}| = 1/\sqrt{2}$ , and  $C_{\text{eff}} = \sqrt{2}$ . The frame pair was selected as the first nonzero- $J$  witness, not by an energy preference. Its up and down spectra come from independently chosen real column couplings. This demonstrates physical weak CP in a locally stable branch; it does not select one sign or reproduce observed quark mixing. Strong CP is not solved.

## A complete obstruction for the orthogonal axis branch

$$|V_{ij}| = 0 \quad \text{or} \quad |V_{ij}| \geq \frac{\sqrt{5}-1}{4} \approx 0.309$$

Every source column in this branch is a group-rotated coordinate ray. An overlap of two such rays is an entry of a group matrix, since its relative group product remains in  $3A_6$ . The implementation enumerates all exact squared entry values and compares them using a rational enclosure of  $\sqrt{5}$ . The smallest nonzero squared value is  $(3-\sqrt{5})/8$ , hence the smallest nonzero mixing magnitude is  $(\sqrt{5}-1)/4 = .3090169944$ . The ordering is certified by rational arithmetic, not inferred from a numerical minimum.

The entire orthogonal axis branch therefore has entries either zero or at least .3090169944. It cannot yield the observed nonzero  $|V_{ub}|$  near .004, under any up/down frame pair, group link or mass permutation. This is considerably stronger than rejecting the displayed CP witness. It is a dynamical realization of a residual-frame obstruction: the scalar model has now produced the relevant stationary directions and positive local masses. Nonorthogonal source choices and other source extrema lie outside the obstruction and require their own joint-vacuum calculations.

## The permitted interaction still moves the solved frame

$$H \Delta q = -\nabla O, \quad O = |\phi_{u1}^\dagger L \phi_{d1}|^2$$

The scalar EFT permits  $\epsilon |\phi_{u1}^\dagger L \phi_{d1}|^2$  under both family factors, all six shaping charges and CP. At the solved CP witness, the first source overlap probability is  $1/4$ . Its constrained source torque is nonzero, with squared norm  $r^4 \rho^6 B(1-B) = 1.875e-11$  in the declared units. Thus this interaction acts on an actual stable vacuum rather than on an assumed source frame.

The full 54-real-dimensional link/source Hessian is positive, with smallest eigenvalue about  $5.4e-7$ . Solving  $H \delta x = -\text{grad } O$  gives a first-order stationary-vacuum response. Canonical quark matching along that response changes both magnitudes and weak CP:  $dC_{\text{eff}}/d\epsilon$  is approximately 9.616 and  $dJ/d\epsilon$  approximately .03590. These numbers use a numerical Hessian and dimensionful EFT normalization; they are not experimental error bars, a radiative coefficient estimate or a naturalness scale. The retained  $\epsilon$  plus/minus  $1e-6$  cases are linear continuations, not freshly minimized finite- $\epsilon$  vacua.

A mechanism protecting a precise relative frame must consequently constrain this operator or account for its effects. The product family group by itself does not do so. Generic small deformations preserve a nearby local minimum, but they need not preserve its CKM quantities exactly. This distinguishes stability of a vacuum from protection of an observable relation.

## What the push establishes and what should follow

The earlier scalar-matching gap is now closed for a specified EFT: the determinant and sextic give a locally isolated nonzero link branch; a specified breaking sector gives nonzero sources and positive local spectra; a holomorphic cofactor replaces the conjugate-link reverse vertex; and physical weak CP is calculated from canonical quark matrices at a joint vacuum. The branch still fails the hierarchy and permits an explicit observable-changing cross-sector interaction. No target angle was supplied to obtain these conclusions.

The next useful search should explore nonorthogonal source vacua or new joint source/link contractions, while retaining the complete allowed breaking and kinetic operator spaces. It must derive their coefficients or demonstrate robustness across the independent ones. Adding a driving field or another representation is justified only if its operator census actually removes a demonstrated obstruction. The present calculation supplies concrete tests for that decision and avoids confusing another constructed golden root with a flavor prediction.

## Reproduction and relation to established work

Run `PYTHONPATH=python OPENBLAS_NUM_THREADS=1 python python/develop_valentiner_susy_vacua.py`. Its scientific receipt records exact tensor identities, the source critical-locus reduction, parameter variations, the finite frame census, canonical CP pairs, scalar stability bounds and the allowed-response calculation. Run `python python/tests/test_valentiner_susy_vacua.py` for independent original-polynomial, Hessian, cofactor covariance, holomorphy, full fermion-mass and CP-orbit checks. Numerical and exact statements are identified separately. No new Lean theorem or experimental fit is claimed.

The classical Valentiner sextic and the general idea of supersymmetric flavon alignment are established mathematics and model-building tools. This draft does not claim their discovery. Primary background: S. P. Martin, A Supersymmetry Primer, <https://arxiv.org/abs/hep-ph/9709356> ; F. Feruglio, C. Hagedorn and L. Merlo, Vacuum Alignment in SUSY A4 Models, <https://arxiv.org/abs/0910.4058> ; C. Hagedorn, M. L. Lopez-Ibanez, M. J. Perez, M. H. Rahat and O. Vives, Flavon vacuum alignment beyond SUSY, <https://arxiv.org/abs/2312.07430> ; M.-C. Chen et al., CP Violation from Finite Groups, <https://arxiv.org/abs/1402.0507> . The cited alignment papers concern their own models, not a proof of our coefficients or CKM relation. The exact branch, spectra, finite census and response reported here are calculations for this declared construction. Publication novelty requires comparison and independent review.

# The joint potential and the protection test

The nonorthogonal route is dynamically possible in the declared field content. This calculation enumerates the entire independent CP-even scalar polynomial space through degree six, activates every coefficient, and solves the full 54-real-field stationarity equations. It obtains positive local spectra, small nonzero mixing and physical weak CP. A shared-heavy-ray example also gives the ordering  $|V_{ub}| < |V_{cb}| < |V_{us}|$ . Neither example reproduces the measured CKM matrix, derives the quark-column hierarchy, selects a global vacuum or predicts a golden coefficient.

The decisive constraint comes from the allowed parameter space. On the retained nonorthogonal branch, independent scalar and quark couplings generate four independent physical observable directions, including after compensating the weights to hold all six quark masses fixed. Thus no exact nontrivial relation among the three mixing magnitudes and  $J$  holds throughout a neighborhood of this branch with these couplings free. Fully re-minimized one-percent scalar deformations directly change the effective coefficient, also after mass matching. The remaining task is to derive correlations among the interactions and quark weights, rather than extend an unconstrained potential search.

## Field content, order and conventions

Keep the preceding  $G_u \times G_d$  construction, with each factor the genuine  $3.A_6$  triplet group. The physical row link  $L$  transforms as  $g_u L g_d^\dagger$ . Three physical source vectors  $u_i$  transform under  $g_u$  and three  $d_i$  under  $g_d$ . Their chiral counterparts are conjugates. Each source has a distinct  $C_6$  shaping charge. CP acts through the previously checked involution  $X$ :  $L$  maps to  $X \text{ conjugate}(L) X^\dagger$  and each vector maps to  $X \text{ conjugate}(\phi)$ . The quark chain includes the independent holomorphic reverse cofactor vertex. Its tree matching and right-handed canonical normalization are retained.

The word complete here has a specified scope: every real CP-even polynomial scalar contraction through total field degree six, on the zero-quark, zero-Higgs flavor slice, under these two finite family groups and all six independent shaping factors. Constants are discarded. The  $F$  potentials induced by the earlier superpotential are kept as a separately specified positive completion: link degrees four, seven and ten, and source degree ten. This is a scalar symmetry-breaking EFT. General Kähler terms, all higher-degree quark operators, loop matching, electroweak breaking and a gauge/anomaly completion are not included in this census. There are infinitely many higher-order allowed terms; no claim of an all-order scalar action is made.

Use  $\rho=.1$  and express every link or source field in units of  $\rho$ , taking the reference mass scale  $\Lambda=1$ . Restoring it, the fields are  $\Lambda \rho z$  and  $V_{\text{physical}} = \Lambda^4 \rho^{10} V_{\text{scaled}}$ . A degree- $d$  coefficient  $c$  in the receipt means  $c \Lambda^{(4-d)} \rho^{(10-d)}$ . The raw real coordinates are  $\text{Re}(z)$ ,  $\text{Im}(z)$ ; canonical real squared masses are  $\Lambda^2 \rho^{8/2}$  times their scaled Hessian eigenvalues. This factor is necessary because a canonical complex scalar is  $(q+ip)/\sqrt{2}$ .

## The exact joint operator census

$$N_{\text{CP even}} = 7 + 1 + 41 + 7 + 207 = 263$$

The independent counts are as follows. The complex column counts polynomials before Hermiticity and CP; CP-even plus CP-odd real dimensions equal that complex dimension.

| Field degree | CP-even real terms | Complex dimension |
|--------------|--------------------|-------------------|
| 2            | 7                  | 7                 |
| 3            | 1                  | 2                 |
| 4            | 41                 | 41                |
| 5            | 7                  | 14                |
| 6            | 207                | 223               |
| Total        | 263                | 287               |

An independent exact character calculation obtains the complex dimensions. For link bidegree  $(p,q)$ , Cauchy's decomposition expands  $\text{Sym}^p(3 \otimes \bar{3})$  and its conjugate into Schur representations. Local Jacobi-Trudi characters are calculated in  $Q(\sqrt{5}, \omega)$ , and averaged exactly on each 1080-element group, with the balanced source characters multiplied in. Separate C6 charges require each source to occur equally often with its conjugate at these orders, except for an individual holomorphic sixth power. The six source sextics and their conjugates are added explicitly. The receipt retains every contributing multidegree sector, not just totals.

Independence is also certified without floating rank tests. The balanced contractions have integer coefficients in real coordinates. Evaluate them at 260 reproducible integer-coordinate points modulo the prime 1000003, using formal Gaussian residues and conjugation. Full-rank square evaluation minors establish exact independence of the 7 quadratic, 41 quartic and 199 balanced sextic polynomials. Their pivot-row witnesses are retained. Distinct source/link phase bidegrees separate the remaining source and determinant sectors. The two holomorphic link sextics are independent already on  $L = e_1 e_1$ -transpose, where  $I_6 = 1$  but  $\det(L)^2 = 0$ . Multiplication by the nonzero determinant polynomial preserves independence of the quadratic factors used at degree five. Together with the character dimensions and CP graph pairing, this establishes the complete independent real basis.

The explicit contraction basis is constructed independently from trace and Gram graphs. Local finite-group tensor counts agree with  $SU(3)$  through degree six on all balanced index sectors. The only additional finite-group contributions at this order are the individual source sextics and the link sextic. The group-specific sextic is therefore not used to remove any otherwise permitted balanced cross-sector interaction. CP pairs conjugate graph orientations. The two within-sector Gram triangles and the six link-dressed same-sector triangles give eight CP-odd balanced sextics; the holomorphic source and link pairs give eight more. Cubic and quintic conjugate pairs give respectively one and seven CP-odd terms. There are no CP-odd quadratic or quartic terms.

For an explicit basis, set  $M_u = L L^\dagger$ ,  $M_d = L^\dagger L$ ,  $S = \text{Tr } M_u$ ,  $T = \text{Tr } M_u^2$ ,  $R = \text{Tr } M_u^3$ ,  $D = \det L$ ,  $N_i = \phi_i^\dagger \phi_i$ ,  $Q_i = \phi_i^\dagger M_{\text{sector}} \phi_i$  and  $Q_2 = \phi_i^\dagger M_{\text{sector}}^2 \phi_i$ . Define  $G_{ij} = \phi_i^\dagger \phi_j$  only for sources in the same sector. The operator receipt provides a name, degree and coefficient for every contraction.

At degree two the basis is  $S$  and the six  $N_i$ . Degree three contains  $\text{Re } D$ . Degree four contains  $S^2$  and  $T$ , twelve  $S N_i$  and  $Q_i$  terms, all twenty-one norm products  $N_i N_j$ , and the six within-sector  $|G_{ij}|^2$ . Degree five contains  $S \text{Re } D$  and six  $N_i \text{Re } D$  terms.

At degree six the link-only terms are  $S^3$ ,  $S T$ ,  $R$ ,  $\text{Re } D^2$  and  $\text{Re } I_6$ . The twenty-four link-four/source-two terms are  $S^2 N_i$ ,  $T N_i$ ,  $S Q_i$  and  $Q_2^i$ . The seventy-eight link-two/source-four terms comprise twenty-one  $S N_i N_j$ , thirty-six  $Q_i N_j$ , six  $S |G_{ij}|^2$ , six real dressed Gram triangles and nine  $X_{ij}=|u_i \text{-dagger } L d_j|^2$ . The one hundred source-only terms are fifty-six triple norm products, thirty-six norm times within-sector overlap squares, two real Gram triangles and six  $\text{Re } F_6(\phi_i)$ . A redundant  $|\det L|^2$  is eliminated by Newton's trace identity. This gives  $5+24+78+100=207$  sextics.

All nine  $X_{ij}$  have independent coefficients. Their absolute squares are neutral under any additional Abelian phase assignments to the displayed fields: the charge of  $u_i \text{-dagger } L d_j$  cancels that of its conjugate. Ordinary phase charges alone therefore cannot forbid these norm-square operators, even if they constrain holomorphic vertices. A mechanism that relates or removes them needs an additional structural restriction, such as a justified interaction origin or a symmetry acting nontrivially on source labels. No such restriction is assumed here.

## A bounded potential with every coefficient active

The scaled base potential is the squared gradient of  $W_{\text{link}} = -32.4 \det L + I_6(L) + .2(\det L)^2$ , plus the sum of six source potentials  $\text{norm}(\text{grad } F_6(\phi_i))^2 - 36 \text{norm}(\phi_i)^2 - 48 \text{Re } F_6(\phi_i)$ . Its nonzero link  $L=I$  and all group-rotated coordinate sources reproduce the previously derived stationary branch. Sources need not be orthogonal to each other: each field independently occupies a stable axis orbit.

Add a real coefficient times every one of the 263 basis operators. The declared deterministic coefficient prescription uses seed 108066: small positive, distinct generic coefficients at the other terms, a positive .02 coefficient for  $S^3$ , independent positive within-sector overlaps near .5 and independent negative  $X_{ij}$  coefficients near -.5. These are chosen EFT inputs. Their ratios are not derived or adjusted to the golden relation. Every coefficient is nonzero in the three mixed cases; the unmixed case is retained as the baseline. A common factor 0, .25, 1 or 4 scales the added polynomial.

The positive pure-link sextic is a deliberate boundedness condition, not a flavor target. By Cauchy-Schwarz,  $|I_6(L)| \leq 16 S^3$ , and  $|\text{Re}(\det L)^2| \leq S^3/27$ . Also  $S T$  and  $R$  are bounded by  $S^3$ . The retained pure-link sextic has a strictly positive lower-bound coefficient .0194628392 after subtracting all these absolute contributions. The established smoothness of  $F_6$  implies a positive minimum of  $\text{norm}(\text{grad } F_6)^2$  on the unit sphere, so it gives degree-ten radial growth for each source. Every mixed contraction has weighted growth  $a/6 + b/10 < 1$ , where  $a$  and  $b$  count link and source fields. Young's inequality absorbs such terms into the positive link-six and source-ten bounds. All positive-factor potentials are consequently coercive and possess global minima. The calculations identify local minima; they do not classify those global minima.

## Nonorthogonal local vacua and physical CP

$$\text{Im}(G_{12}G_{23}G_{31}) = \frac{\sqrt{3}}{16}, \quad \det G = \frac{1}{8}$$

Start with the first full-rank axis-ray triple having a nonzero imaginary Gram triangle, and use it in both sectors. In the fixed enumeration its ray indices are 0,3,10. Exact field arithmetic gives  $G_{12} G_{23} G_{31} = (1+i \sqrt{3})/16$  and  $\det G = 1/8$ . The individual source directions are genuine

group-rotated stationary axes, not a copied CKM chart. Each is stable before adding the mixed operators. The phase invariant exists despite real potential and quark coefficients.

There is a useful exact CP identity for any shared full-rank column matrix  $C$ . Write  $H_u = C \text{diag}(a_i) C^\dagger$  and  $H_d = C \text{diag}(b_i) C^\dagger$ , with real weights, and  $G = C^\dagger C$ . Direct multiplication gives  $[H_u, H_d] = C A C^\dagger$ , where  $A_{ij} = (a_i b_j - b_i a_j) G_{ij}$  and  $A_{ii} = 0$ . Therefore

$$\det[H_u, H_d] = 2i \det(G) (a_1 b_2 - b_1 a_2)(a_2 b_3 - b_2 a_3)(a_3 b_1 - b_3 a_1) \text{Im}(G_{12} G_{23} G_{31}).$$

For the retained triple this is  $i \sqrt{3}/64$  times the three weight differences. The raw weighted matrices violate weak CP when that product is nonzero. On a scalar group-aligned link, the canonical up and down matching maps are strictly increasing spectral functions of these raw matrices, so their left projectors and physical  $J$  are unchanged. This proves the mechanism's CP content without identifying a complex entry with a physical phase. The perturbed link is handled by the full holomorphic mediator block and canonical matching, and CP is checked again there.

The quark-column inputs are  $(.007, .05, .9)$  in the up sector and  $(.02, .2, .85)$  in the down sector. These explicitly imposed hierarchies are not outputs of the scalar symmetry. The full 54-coordinate minimization, followed by Newton refinement, gives stationarity residuals below  $4e-11$  across the four cases. Their raw Hessian minima range from 108 at the baseline to 102.70 at factor four. The independent weights preserve three nondegenerate quark modes in each sector.

At factor one the matched magnitudes are  $|V_{us}| = .00139934$ ,  $|V_{cb}| = .01621072$  and  $|V_{ub}| = .00038814$ , with  $J = 8.64911e-9$  and standard phase 100.7496 degrees. The effective coefficient is 17.11066. These magnitudes escape the .309 lower bound of the orthogonal branch, but their ordering is not the observed Cabibbo hierarchy. They are a local capability example, not a data fit. The independent commutator determinant is nonzero, about  $1.38e-34$  in this dimensionful normalization. CP conjugation preserves potential energy and magnitudes and reverses  $J$ .

The nonzero source  $F$  backgrounds also require a matter stability check. Their weighted Frobenius norm gives an upper bound approximately  $7.55e-5$  on the source-induced quark-scalar bilinear operator norm. The specified positive universal matter breaking mass squared  $1e-4$  leaves a rigorous positive lower bound approximately  $2.45e-5$  after adding the nonnegative supersymmetric mass block. This check concerns the displayed matter action; arbitrary additional matter EFT operators are not included. Electroweak vacuum selection remains separate.

## Shared heavy directions and the ordering of the mixing angles

A second illustration keeps the dominant source ray common to both sectors and varies the two lighter down-sector rays. There are 946 unordered pairs to inspect; 902 pass the declared determinant cutoff .05, and 467 give nonzero  $J$  with  $V_{ub} < V_{cb} < V_{us}$  at the stated quark weights. These are numerical background diagnostics, not a complete classification of all nonorthogonal triples or a proof of energy selection.

The smallest- $V_{us}$  pair within this explicitly ordered sample uses down rays 18, 21, 10. This selection is phenomenological and is reported as such; no measured anchor, target phase or golden coefficient is fitted. Fully minimizing the same 263-coefficient potential from that

starting point gives  $|V_{us}|=.425322$ ,  $|V_{cb}|=.0216236$ ,  $|V_{ub}|=.0112854$  and  $J=6.19010e-5$ . Its stationarity residual is below  $9e-12$  and its raw Hessian minimum is 106.15. The phase is 41.2339 degrees and  $C_{\text{eff}}=1.22692$ . The largest angle remains too large and the coefficient is not golden. The source sectors are nonorthogonal and the shared heavy direction only becomes approximate after relaxation.

This demonstrates how shared dominant directions and hierarchical weights can suppress mixing involving the heavy family while retaining a larger light-family angle and physical CP. It does not establish that the present symmetry explains either the weights or the measured hierarchy. The same potential has other local branches and no cosmological occupation rule. The example supplies a concrete test case for a proposed source-label interaction.

## Why nonorthogonality changes the protection problem

$$H\Delta q = -\nabla O, \quad \text{rank} \frac{\partial(|V_{us}|, |V_{cb}|, |V_{ub}|, J)}{\partial c} = 4$$

There is an exact elementary theorem behind the numerical sensitivity. For a fixed full-rank column matrix  $C$ , let  $H(t) = \sum_i t_i c_i c_i^\dagger$ , with all  $t_i$  independently positive. If its unordered nondegenerate left eigenprojectors stay fixed on an open parameter set, then differentiating  $H$  with respect to each  $t_i$  shows that every rank-one matrix  $c_i c_i^\dagger$  is diagonal in that same eigenbasis. They therefore commute. Two nonzero rank-one projectors commute only when their vectors are parallel or orthogonal; full rank excludes parallel columns. The columns must be pairwise orthogonal. Conversely orthogonal columns share their projectors throughout every fixed mass-order region.

Thus nonorthogonal sources solve one problem and introduce another obligation: the quark-column coefficients affect the mixing frame. The scalar potential on the zero-matter/Higgs slice does not determine those coefficients. This theorem applies directly to the universal up mediator and to both sectors on a scalar aligned link. On the deformed link the fully matched numerical calculation establishes the response. The theorem concerns the full frame; it does not by itself rule out a specially protected single observable identity at another exceptional configuration.

At the actual factor-one vacuum, calculate the response to six same-sector overlap coefficients, all nine  $X_{ij}$  and all six logarithmic quark weights. Scalar variations solve  $H \delta x = -\text{grad } O$  using the full 54-dimensional Hessian. The resulting Jacobian of  $(|V_{us}|, |V_{cb}|, |V_{ub}|, J)$  has rank four. Because  $J$  is small, the raw matrix is poorly scaled. After dividing each observable by its nonzero reference magnitude and normalizing the parameter columns, its singular values are approximately 3.75815, 2.06946, 1.60725, .10196. The fourth direction is well separated from numerical roundoff in these coordinates.

The implicit function theorem supplies nearby scalar vacua, and the inverse function theorem applied to a nonsingular four-column observable minor supplies a neighborhood of physically independent observable values. On this regular local branch with the specified couplings free, an exact golden amplitude relation or a fixed CP phase therefore cannot be an identity. This is a local numerical-rank statement supported by explicit positive masses, not a theorem excluding every possible branch, new representation or correlated UV completion.

The stronger calculation holds the six quark singular values fixed. Write the six log-mass response as  $M_{\text{scalar}} \delta c + M_{\text{weights}} \delta_{\log} \lambda$ . The weight block is invertible, with singular values between .9845 and .9999. Set  $\delta_{\log} \lambda = -M_{\text{weights}}^{-1} M_{\text{scalar}} \delta c$  and substitute it into the mixing response. The remaining fifteen scalar directions still generate all four physical mixing directions. With the same fractional-observable and column

normalization, their singular values are 3.26373,1.59728,1.33898,.06274. Thus the freedom is not removed by calibrating all six quark masses. These reference masses are the illustrative masses of this vacuum, not experimental measurements.

## Fully re-minimized finite deformations

$$C_{\text{eff}} = \frac{|V_{ub}|(1 - |V_{ub}|^2)}{|V_{us}||V_{cb}|}, \quad C_{\text{eff}}^2 - 3C_{\text{eff}} + 1 = 0$$

The nominated coefficient is  $C_{\text{eff}} = |V_{ub}|(1 - |V_{ub}|^2)/(|V_{us}||V_{cb}|)$ . The golden condition requires  $C_{\text{eff}}^2 - 3C_{\text{eff}} + 1 = 0$  with  $0 < C_{\text{eff}} < 1$ . At the retained vacuum, the golden polynomial has nonzero derivatives along independent allowed directions. This is not an artifact of an off-vacuum source frame or an unnormalized quark matrix.

Perturb the  $X_{02}$  coefficient by plus and minus one percent of its absolute value, and solve the stationarity equations again.  $C_{\text{eff}}$  changes from 17.11066 to approximately 18.22351 and 15.99458. Perturbing the  $B_{02}$  same-sector overlap gives approximately 15.88999 and 18.32730. All four fresh vacua have positive Hessians and stationarity residuals below  $2e-10$ . Their receipt contains full observables and stability diagnostics. These are dimensionful EFT perturbations of the explicitly chosen coefficient values, not experimental error ranges or estimates of radiative corrections.

After re-solving each vacuum, independently refit all six positive quark-column weights to recover the baseline six masses. Their maximum log-mass residual is below  $2e-12$ . The  $X_{02}$  deformations still give  $C_{\text{eff}}$  approximately 18.22682 and 15.99157; the  $B_{02}$  deformations give 15.89963 and 18.31621. Since the tree-level zero-matter/Higgs scalar potential is independent of these quark weights, this compensation preserves scalar stationarity. This supplies finite examples of observable change at fixed masses, in addition to the local differential-rank test.

## The next interaction must constrain both sectors

An unconstrained joint-potential search can now be distinguished from a flavor prediction. The declared field content admits nonorthogonal stable vacua, weak CP and hierarchy mechanisms. Its complete low-degree operator space and independent quark weights leave all four physical mixing parameters adjustable on the tested branch. Finding a golden-looking point in that space would select couplings rather than explain a relation.

The next construction needs a justified non-Abelian action on source labels or a UV interaction whose unique invariant contractions correlate the quark-column weights and the nine mixed scalar coefficients. It must retain its allowed norm-square operators and kinetic corrections in the audit. A larger Abelian phase group cannot forbid the  $X_{ij}$ . A proposed M22-derived interaction must specify which of these actual contractions it relates, rather than supply an arbitrary matrix on a multiplicity space. New fields are useful only if their full operator space reduces the demonstrated physical freedom. No specific new representation is asserted here to achieve that reduction.

## Verification and established context

Run `OPENBLAS_NUM_THREADS=1 PYTHONPATH=python python python/develop_valentiner_joint.py` to reproduce the exact multidegree census, complete coefficient list, four mixed-potential cases, CP partner, shared-heavy-ray witness, 21-parameter response and four freshly minimized deformations. The field coordinates and every independent scalar coefficient are retained. The tests independently check all operator covariance under family generators, all six shaping factors and CP, every directional derivative, an independently contracted link F potential, the shared-column CP determinant identity, full stationary spectra, finite deformation replay and the eigenframe weight theorem.

Cross-coupling-driven reorientation and flavon alignment are established model-building problems. Relevant primary comparisons are M. Holthausen and M. A. Schmidt, Natural Vacuum Alignment from Group Theory: The Minimal Case, <https://arxiv.org/abs/1111.1730> ; and I. de Medeiros Varzielas, M.-S. Liu, A. Sengupta and J. Talbert, Residual Symmetries and Scalar Multiplet Vacuum Alignment in Non-Abelian Flavour Models, <https://arxiv.org/abs/2512.19789> . Their group models differ from this construction. The present results are the explicit operator census, nonorthogonal branch calculations, CP invariant and local freedom test for this specified EFT. Publication novelty and robustness beyond the declared truncation require independent review.

# CP-even kinetic freedom at fixed quark masses

## Result and assumptions

A complete Hermitian quark metric can be generated by nine CP-compatible polynomial covariants with real coefficients. When their CP triangle is nonzero, they span all nine real metric directions at the reference vacuum. Small positive changes of the bare right-handed metric can preserve a previously solved scalar vacuum, its superpotential and all six quark masses while changing every physical CKM parameter. This is an algebraic statement about allowed operators, rather than a claim that quantum corrections have a particular size. It makes a protecting model responsible for kinetic matching as well as scalar and Yukawa interactions.

The assumptions are linear unitary internal flavor transformations, flavor-covariant functions  $X$  and  $Z$  with the same left flavor representation, a full-rank  $X$  with distinct singular values, a nonzero CP triangle, and independent real coefficients for the complete allowed matter kinetic operator space.  $X$  and  $Z$  may have different right representations and ordinary phase charges: those cancel from  $R=X^\dagger X$  and  $S=X^\dagger Z Z^\dagger X$ . The degree bound below requires polynomial covariants. Matched analytic Yukawa functions obey the spanning argument but require an effective-field-theory expansion before a polynomial degree can be assigned. Degenerate spectra, CP-conserving vacua, coefficient restrictions imposed by a UV theory and additional restrictions beyond these internal transformation laws are outside this theorem.

The numerical demonstration uses the already minimized nonorthogonal joint vacuum from commit f76767e. The golden-angle test is a separately initialized attack fixture. It does not turn the nominated angle or golden coefficient into a dynamical prediction. The accompanying common-label construction supplies a justified interaction and a stable heavy-family branch, but has not been solved as a complete joint three-family model.

## Nine real polynomial covariants

$$R = X^\dagger X, \quad S = X^\dagger Z Z^\dagger X, \quad \det \mathcal{M} = \tau^3$$

Let  $R=X^\dagger X$  and  $S=X^\dagger Z Z^\dagger X$ . Both transform by conjugation on the right flavor space of  $X$ . Under generalized CP they transform into their complex conjugates conjugated by the corresponding unitary CP matrix. Fix the three positive distinct eigenvalues  $r_1 < r_2 < r_3$  of  $R$  at a reference vacuum as real numerical constants. Define  $P_i(R)$  as the product over  $j$  unequal to  $i$  of  $(R - r_j I)/(r_i - r_j)$ . The  $P_i$  are genuine matrix polynomials with fixed coefficients, rather than eigenprojectors recomputed as the fields vary. At the reference vacuum they are the three orthogonal rank-one spectral projectors.

Take the three  $P_i$  and, for each pair  $i < j$ , the two Hermitian covariants  $B_{ij} = P_i S P_j + P_j S P_i$  and  $D_{ij} = P_i S^2 P_j + P_j S^2 P_i$ . Every coefficient is real, every expression is Hermitian, and each expression has the required family and CP transformation law. In particular, there is no explicitly imaginary coefficient inserted to manufacture a missing phase direction.

In an  $R$  eigenbasis put  $\tau = \text{Im}(S_{12} S_{23} S_{31})$ . For the pairs (12),(13),(23), the real-coordinate determinants of the two complex off-diagonal entries  $S_{ij}$  and  $(S^2)_{ij}$  are respectively

$-\tau, +\tau, -\tau$ . The diagonal entries of  $S$  only add real multiples of  $S_{ij}$  and cancel from these determinants. The three diagonal projectors span the diagonal subspace. Consequently the nine-generator coordinate determinant is  $\tau$ -cubed in coordinates consisting of the three diagonals followed by real and imaginary off-diagonal entries. It is  $8\tau$ -cubed if both off-diagonal coordinates are multiplied by square-root two to obtain Hilbert-Schmidt orthonormal coordinates. This proves completeness when  $\tau$  is nonzero. The exact symbolic identities are independently checked in the driver and focused tests.

## An invariant certificate of physical CP

$$\text{Im Tr}(R^2 S^2 R S) = \Delta(R) \tau$$

The reference-basis condition can be written without choosing an eigenbasis:  $\text{Im Tr}(R\text{-squared } S\text{-squared } R S) = \Delta(R) \tau$ , where  $\Delta(R) = (r_2 - r_1)(r_3 - r_1)(r_3 - r_2)$ . This identity is checked by symbolic expansion of a general Hermitian  $S$ . A nonzero left-hand side certifies that the covariants do not admit a simultaneous CP-real frame.

If  $X$  and  $Z$  are the canonically normalized up and down Yukawa functions,  $S$  in the  $R$  eigenbasis is the up singular-value diagonal matrix times the down left mass-squared matrix times the same diagonal matrix. Its triangle imaginary part is the product of the three squared up singular values times the triangle imaginary part of the down matrix in the up mass basis. For nondegenerate spectra that latter quantity is proportional to the physical Jarlskog invariant and the down spectral Vandermonde. Thus nonzero physical weak CP supplies the required spanning condition for the matched Yukawa functions. For the finite polynomial covariants used in the explicit heavy-mediator example, the condition is checked directly; it is not inferred merely from the existence of a phase in their entries.

## Positive completion and finite degree

$$B(\Phi) = (I + T(\Phi))^2 + \eta I \geq \eta I$$

At the reference vacuum orthonormalize the three off-diagonal pairs using real Hilbert-Schmidt inner products. This uses fixed real normalization constants. Together with the three projectors the resulting  $Q_a$  form a real orthonormal Hermitian basis. Given any positive target right-handed metric  $B_0$ , choose a positive  $\eta$  strictly smaller than its least eigenvalue, set  $T_0 = \text{square-root}(B_0 - \eta I) - I$ , and expand  $T_0$  as the sum of the nine  $Q_a$  times the real coefficients  $\text{Tr}(Q_a T_0)$ . Use the same coefficients and fixed polynomial recipes at every field value to define  $T(\phi)$ .

The metric  $B(\phi) = (I + T(\phi))^2 + \eta I$  equals  $B_0$  at the reference point and is bounded below by  $\eta I$  at every field value. Positivity follows because  $T$  is Hermitian. This is a positive completion of the right-handed block. It is not a global positivity theorem for the entire Kahler metric, whose mixed matter-flavon blocks appear away from zero matter. Near zero matter backgrounds those mixed blocks vanish and positivity of the complete metric follows locally by continuity.

If  $X$  has flavor-field degree  $p$  and  $Z$  degree  $q$ , then  $R$  has degree  $2p$ ,  $S$  has degree  $2p+2q$ , and each interpolating  $P_i$  has degree at most  $4p$ . The highest-degree generator  $P_i S\text{-squared } P_j$  plus its adjoint has degree at most  $12p+4q$ . Squaring the Hermitian completion gives an upper bound  $24p+8q$  on the flavor-field degree of  $B$ . A matter bilinear adds two matter fields to the operator degree. The explicit down-sector example uses  $X = L C_d$  and  $Z = C_u$ , hence  $p=2, q=1$  and a maximum flavor-field degree of 56. High-degree coefficients and reference spectral

denominators can be large; no naturalness or loop-size estimate follows from this existence proof.

## Fixed masses with full heavy matching

$$B = Y_0^\dagger H_{\text{target}}^{-1} Y_0 - A^\dagger A, \quad YY^\dagger = H_{\text{target}}$$

The bare light right-handed quarks mix with heavy quarks through a fixed holomorphic null frame  $N=(I,-A)$ . Let  $Y_0$  be the corresponding unnormalized Yukawa matrix. With the old identity bare metric the effective metric is  $K_0=I+A^\dagger A$  and the matched Yukawa is  $Y_0 K_0$  to the power  $-1/2$ . Replace only the bare light-quark metric by  $B$ . The effective null-frame metric becomes  $K=B+A^\dagger A$ .

For any sufficiently nearby positive target left mass-squared matrix  $H_{\text{target}}$ , choose  $B=Y_0^\dagger H_{\text{target}}^{-1} Y_0 - A^\dagger A$ . Then  $K=Y_0^\dagger H_{\text{target}}^{-1} Y_0$  and, for invertible  $Y_0$ , the canonically matched Yukawa obeys  $Y Y^\dagger = H_{\text{target}}$  exactly. At the reference target this  $B$  is the identity, so positivity persists for a neighborhood of targets. Taking  $H_{\text{target}}=U H_{\text{down}} U^\dagger$  preserves all down masses. The up sector is untouched. Relative unitary rotations cover the four physical mixing directions after rephasings, at a generic three-family point.

Every such  $B$  can be realized by the CP-compatible positive polynomial completion above. In the chiral-superfield convention its transposed right-handed metric is used if required by the definition of the physical right-handed column; the positive spectrum and covariance statement are unchanged. This matching is performed before eliminating the heavy fields. It does not impose an arbitrary matrix after canonical normalization and call that matrix a symmetry-derived contraction.

At zero quark, heavy-quark and Higgs backgrounds, the additional Kahler term is a matter bilinear and vanishes. Its mixed derivatives vanish there as well, and its flavon block does not change. Therefore the existing scalar stationary point and flavor Hessian remain unchanged at this tree-level background. The claim does not include an electroweak vacuum calculation or a radiatively corrected scalar vacuum.

## Constructive checks on the solved joint vacuum

The driver reconstructs the unchanged up and down superpotential matrices and the complete down heavy null frame of the prior joint solution. It uses the polynomial covariants  $X=L C d$  and  $Z=C u$  on the bare down-quark label space. Six anti-Hermitian rotations in the up mass basis, each evaluated with both signs at magnitude 0.00001, produce twelve positive bare metrics. All twelve are realized with real polynomial coefficients and matched again through their heavy null-frame metrics.

The four singular values of the fractional-observable, column-normalized response matrix are 1.94061, 1.18728, 0.736859, 0.530491. These normalization-dependent numbers certify four independent observable directions; they are not physical sensitivity eigenvalues. The greatest logarithmic residual among all six quark singular values is below 7 times 10 to the power  $-13$  in double-precision SVD. The nine-basis orthonormality error and positive metric reconstruction errors are below 10 to the power  $-75$  in 90-digit arithmetic. After explicitly Hermitianizing the floating-point target mass matrix, the exact-matching residual is likewise checked at multiprecision. The driver records each metric, its eigenvalue bound, its nine real coefficients and all resulting observables.

A separate golden-chart fixture starts with  $C=(3\text{-square-root five})/2$ , physical phase 66 degrees,  $|V_{us}|=0.2253$  and  $|V_{cb}|=0.0409$ . Holding both anchors, the phase and all six masses fixed, two CP-compatible positive metrics change  $C$  by  $-0.1\%$  and  $+0.1\%$ . Their operator-norm distance from the identity is about 0.00343545, and their least eigenvalue exceeds 0.99657. This is a concrete allowed deformation of an initialized exact relation. It neither predicts these inputs nor establishes the typical size of kinetic corrections in a natural UV completion.

## A shared non-Abelian source label

$$Z_8 = Z - \frac{\text{Tr} Z}{3} I, \quad V_{\text{eff}} = -\frac{k^2}{M^2} \|Z_8\|^2$$

The positive interaction candidate replaces three independently charged source columns in each sector by a single matrix source with one common phase charge. Introduce a common label group  $H=3.A_6$  and sources  $A=(3_u, 3_{\bar{H}})$ ,  $B=(3_d, 3_{\bar{H}})$ , together with one right-handed  $H$  triplet per quark charge sector. The common group acts on source labels as a representation, rather than permitting three arbitrary column coefficients. The exact singlet multiplicity in triplet times conjugate triplet is one, so the corresponding quark-source contraction has one coupling per sector.

For the link  $L=(3_u, 3_{\bar{d}})$ ,  $Z=A^\dagger L B$  transforms by conjugation under  $H$ . The endomorphism representation decomposes as  $1+8$ . The two quadratic dressed channels are the singlet norm  $|\text{Tr} Z|^2$  and the traceless norm  $\text{Tr}(Z^\dagger Z) - |\text{Tr} Z|^2/3$ . Exact averaging over all 1080 group elements gives singlet multiplicities 1 in triplet times conjugate triplet, 1 in adjoint squared and 2 in adjoint cubed. These are representation-derived coupling counts.

A complex adjoint scalar  $S_H$ , with the phase charge of  $Z$ , permits  $M^2 \|S_H\|^2 - 2k \text{Re} \text{Tr}(S_H^\dagger Z_8)$ , where  $Z_8 = Z - (\text{Tr} Z)I/3$ . Eliminating  $S_H$  at this leading order gives  $-(k^2/M^2) \|Z_8\|^2$ . This correlates the two norm coefficients through the fixed  $-1/3$  trace-subtraction ratio. The ratio is a Clebsch coefficient and is not the golden coefficient. The source-link-mediator term is quartic in elementary scalar fields. Additional allowed mediator quartics and mediator norm couplings to the source and link bilinears must be included in a complete UV matching; the leading exchange is not a license to omit them.

## A stable heavy-family branch

The same sextic invariant tensor gives a matrix-source invariant  $I_6(A)$ . Include the second allowed holomorphic sextic through  $W_A = I_6(A) + c \det(A)^2$ , with real  $c$ . In dimensionless units consider  $V_A = \|\text{gradient } W_A\|^2 - (180+6t)\|A\|^2 + 2t \text{Re } W_A$ . The rank-one configuration  $A=\text{diag}(1,0,0)$  is stationary, as are its finite group rotations. The determinant begins at second order around rank one, so its square and its first three derivatives vanish there. Hence its coefficient  $c$  does not alter the local Hessian.

In canonically normalized real coordinates the eighteen mass coefficients are: radial  $1440+24t$ ; phase  $-36t$ ; four copies each of  $-90-6t$  minus and plus  $3\text{ square-root ten times } |24+t|$ ; and four copies each of  $-171-6t$  minus and plus  $3|24+t|$ . For  $-36 < t < -33$  all are positive and the energy  $-144-4t$  lies below the origin. At  $t=-34$  the energy is  $-8$ . At  $t=-34$ , the spectrum is 3 with multiplicity four,  $114-30\text{ square-root ten}$  with multiplicity four, 63 with multiplicity four,  $114+30\text{ square-root ten}$  with multiplicity four, 624 and 1224. An independent eighteen-dimensional finite-difference Hessian reproduces this spectrum.

With the unique universal quark-source contraction this branch has one heavy family and two exactly massless light families before rank lifting. It supplies a stable hierarchy starting point without assigning three independent column weights. The complete joint potential, its rank-lifting operators and its CKM output have not been solved on this new field content. The kinetic theorem still applies if the completed branch has full-rank nondegenerate spectra and the required nonzero CP triangle.

## What the result changes

A viable next UV construction must match its scalar potential, Yukawa contractions and matter metric together. The shared label group gives a justified contraction count and a heavy-family source branch, but internal symmetry with unrestricted allowed kinetic coefficients does not by itself protect an exact golden CKM equation at a CP-breaking vacuum. A finite-order prediction remains possible if omitted terms are controlled; an exact all-order relation requires stronger coefficient correlations. Merely adding phase charges, changing the group order or selecting a scalar stationary point does not supply those correlations for the operators constructed here.

The general importance of canonical normalization is established prior work. King and Peddie, Canonical normalisation and Yukawa matrices, Physics Letters B586 (2004) 83-94, arXiv:hep-ph/0312237, explicitly demonstrate changes in flavor textures. Espinosa and Ibarra, Flavour Symmetries and Kahler Operators, JHEP 0408 (2004) 010, arXiv:hep-ph/0405095, analyze its physical mixing effects and find subdominant quark effects under their phenomenological assumptions. Chen, Ramos-Sanchez and Ratz, A note on the predictions of models with modular flavor symmetries, arXiv:1909.06910, discuss additional Kahler parameters and predictivity. The contribution established in this repository is the explicit nine-polynomial spanning construction, its positive finite-degree completion, fixed-spectrum bare-mediator matching and reproducible application to the current joint vacuum. Publication priority has not been established by this targeted literature check.

## Reproduction and certificates

Run `OPENBLAS_NUM_THREADS=1 PYTHONPATH=python python python/develop_flavor_kinetic.py` from the repository root. The inputs are the retained joint-vacuum and exact invariant-tensor receipts; the output is `receipts/m22_interactions/flavor_kinetic.json`. The module `python/perfectpower/flavor_kinetic.py` contains the reusable interpolating covariants, positive completion and bare matching. The focused tests cover exact symbolic identities, nine-dimensional spanning, family/phase/CP covariance away from the reference point, positive completion away from the vacuum, independent heavy-null-space matching, the retained four-direction response, the fixed-anchor golden deformation and all eighteen source masses. A fresh extraction repeats the scientific receipt byte for byte. The complete monograph and source archive retain the previous calculations and their stated boundaries.

# Universal sources, rank lifting and physical CP

## The central construction

$$J = -2.67675 \times 10^{-5}, \quad m_{\text{scalar}, \min}^2 > 9.5 \times 10^{-7}$$

A specified CP-even joint potential now has a locally stable vacuum with three nonzero quark singular values in each charge sector and nonzero physical weak CP. The quark-source contractions have one universal column coupling per sector. Every allowed renormalizable scalar channel on the declared zero-matter/Higgs slice is present with a nonzero coefficient. The angular field, golden coefficient, CKM mixing anchors and observed quark masses are not input targets of this construction. The physical outputs come from minimizing the potential and then canonically matching the heavy quarks.

There are 35 complex flavor scalars, hence 70 real directions. At the retained solution the largest scaled stationarity residual is below 10 to the power -10. The smallest eigenvalue of the scaled real Hessian is about 7.92373; after restoring canonical field normalization, its least scalar mass squared is about 9.51209 times 10 to the power -7 in the chosen mass units. All 70 directions are massive and positive. These are numerical stationary-point and stability results, supported by independent derivative and symmetry checks, rather than an interval-arithmetic proof of an exact root.

The resulting Jarlskog invariant is about -2.67675 times 10 to the power -5. An independent 80-digit weak-basis invariant agrees with it, and the CP-conjugate vacuum has identical quark masses and mixing magnitudes with the opposite sign of J. The up singular values are approximately 0.000157929, 0.00664798, 0.120416; the down singular values are approximately 0.000000132725, 0.000128821, 0.0138156. The two light masses arise from interaction-driven rank lifting rather than three separately assigned column weights.

This benchmark is an existence construction, not a fit of the CKM matrix. It gives |V<sub>us</sub>| about 0.884742, |V<sub>cb</sub>| about 0.0384413 and |V<sub>ub</sub>| about 0.0751752, with standard phase about -1.30146 degrees and effective golden-chart coefficient about 2.19785. The agreement of the magnitude of J with an interesting scale does not establish phenomenological agreement. The nominated golden coefficient and 66-degree phase remain unforced.

## Fields and justified contractions

$$Y_f = -h_f (M_f^{-1} C_f)_u [I + (M_f^{-1} C_f)^\dagger (M_f^{-1} C_f)]^{-1/2}$$

The internal group is 3.A6<sub>u</sub> times 3.A6<sub>d</sub> times 3.A6<sub>H</sub>, together with C6<sub>A</sub> times C6<sub>B</sub> and the established generalized CP action. The matrix sources are A=(3<sub>u</sub>, 3<sub>bar\_H</sub>) and B=(3<sub>d</sub>, 3<sub>bar\_H</sub>). The link is L=(3<sub>u</sub>, 3<sub>bar\_d</sub>). One right-handed H triplet in each quark sector replaces three separately labeled singlets. The singlet in triplet times conjugate triplet has multiplicity one; the corresponding source-to-quark contraction therefore has a single column coefficient per sector.

A complex mediator S transforms as the H adjoint, with the source phase charge (-1, +1). For Z=A-dagger L B, its traceless part Z<sub>8</sub>=Z-(Tr Z)I/3 transforms in the same representation. The quartic scalar vertex is -2k Re Tr(S-dagger Z<sub>8</sub>). Together with M-squared ||S|| squared it gives,

at leading tree matching,  $-(k^2/M^2)||Z_8||^2$ . The trace-subtraction coefficient is fixed by the representation and equals  $-1/3$  inside the traceless norm. This enforces a real relation between dressed cross-sector coefficients without selecting either a golden ratio or a nominated angle.

The ordinary Dirac messenger chains have heavy triplets in the  $u,d,H$  family spaces. Their  $H$  messenger carries phase charge  $-B$  in the up sector and  $-A$  in the down sector. This permits the unique triplet-adjoint-triplet vertices  $S^\dagger$  and  $S$  to connect those heavy quarks to the corresponding bare right-handed  $H$  triplet. Both allowed orientations of the link and heavy-source vertices are retained. There are ten nonzero real mass/Yukawa coefficients per charge sector. Canonical bare matter metrics remain an independent input, and an all-degree or radiative UV completion is not asserted.

The source and link self-potentials use the recovered sextic tensor  $I_6$  and determinant contractions. Their higher-degree breaking terms are specified EFT interactions. The scalar mediator is not a new chiral quark family. Both heavy rows have nine independent rows and twelve right-handed columns. Each sector has nine heavy modes and a right-handed null space of dimension three. Invertible heavy blocks and canonical normalization are checked directly. Three left-handed quark modes are part of the declared triplet field content; their origin from the full global  $M_{22}$  mode problem is not established by this calculation.

The same scalar mediator now enters the fermion readout. Expanding the matched up chain gives an additional  $L B S^\dagger$  contraction alongside  $A$ ; the down chain gives  $A S$  alongside  $L B$ . The singlet in triplet times adjoint times conjugate triplet has multiplicity one. Eliminating  $S$  at leading order inserts the same traceless kernel  $(A^\dagger L B)_8$  into both scalar exchange and these quark matrices. Overall vertex strengths remain independent, but their component ratios and label transport are representation derived.

The three heavy masses are  $(0.9, 1.3, 2)$  for up and  $(0.8, 1.3, 2)$  for down. The forward/reverse link coefficients are  $0.71, 0.23$ ; forward/reverse source-to-heavy- $H$  coefficients are  $0.4, 0.2$ . Bare source and adjoint-quark coefficients are one, and the Higgs coefficients are  $0.6, 0.57$ . The inverse nine-by-nine heavy block and its complete null-frame metric are used without a small-source approximation.

## The complete scalar operator boundary

$$N_{\text{scalar}}^{\text{renorm}} = 4 + 1 + 25 = 30$$

For the three matrix fields  $L, A, B$ , the CP-even scalar counts through degree six are 3 quadratic, 1 cubic, 12 quartic, 3 quintic and 47 sextic contractions, totaling 66. The balanced contractions are 3, 12, 41 at degrees 2, 4, 6. They are generated by closed walks on the three flavor spaces and products of their traces. Full-rank finite-field evaluation minors modulo 1000003 certify independence of all 56 balanced polynomials. The remaining sectors are  $\text{Re det } L$ , its products with the three quadratic norms, the determinant-square and sextic-tensor contractions for each matrix field.

Completeness uses the invariant tensors at each flavor vertex. Balanced terms through this degree involve at most three triplet and three antitriplet indices. The restrictions of the  $SU(3)$  tensor-cube constituents remain inequivalent irreducibles of the finite group: the singlet, adjoint and symmetric cubic channels introduce no extra balanced contraction at these degrees. The separately charged pure source sextics and the previously certified bifundamental determinant/tensor sectors supply the unbalanced completion. This count belongs to the new shared-label field content; it does not identify its operators with the earlier 263-column-source

census.

With the adjoint included, the renormalizable scalar slice contains 4 quadratic, 1 cubic and 25 quartic CP-even contractions, totaling 30. The quartics split into 12 source/link contractions, 7 contractions of two mediator fields with a source/link bilinear, 5 pure mediator channels and the single linear-mediator dressed-source vertex. All are nonzero in the implemented benchmark. This is the complete stated flavor-scalar slice through degree four, not a complete all-degree EFT or a complete Standard Model scalar action. Higher operators must still be matched if an exact prediction is asserted.

The seven mediator-bilinear quartics are its norm times each of  $N_A, N_B, N_L$ , plus  $\text{Tr}(S\text{-dagger } S \text{ A-dagger } A)$ ,  $\text{Tr}(S \text{ S-dagger } A\text{-dagger } A)$ , and the analogous two B contractions. The two distinct adjoint cubic tensors give the two adjoint channels per source; the singlet gives its norm channel. Phase charges exclude a bare mediator square or cubic, and exclude other unbalanced products at these degrees.

## Five quartic channels with no arbitrary frame

$$\text{Sym}^2(8) = 1 + 5 + 5 + 8 + 8 + 9, \quad N_{\text{CP even}} = 5$$

Exact character averaging gives six singlets in  $\text{End}(\text{Sym-squared } 8)$ , one singlet in  $\text{Sym-cubed } 8$ , and two singlets in the unrestricted adjoint tensor cube. The symmetric-square representation has dimension 36 and splits into inequivalent channels of dimensions 1,5,5,8,8,9. It is multiplicity free. Consequently its six irreducible projectors are fixed by the representation; a freely chosen matrix acting on repeated copies is not used to define an interaction.

The numerical implementation constructs the projectors by a Reynolds group average over all 1080 elements. It independently identifies the  $\text{SU}(3)$  adjoint channel through the anticommutator map. The remaining channels are labeled by their inequivalent dimensions and by the second adjoint channel. Changing the temporary averaging seed leaves the canonical projectors unchanged within numerical precision. Generator covariance, idempotence and completeness are checked, with errors below  $10^{-11}$ .

Generalized CP exchanges the two five-dimensional channels and preserves the other four. The exact trace of CP on the six-dimensional invariant-quartic space is four; hence the CP-even dimension is  $(6+4)/2=5$ . The canonical CP-even channels are the singlet, the  $\text{SU}(3)$  adjoint, the second finite-group adjoint, the nine-dimensional channel and the paired five-plus-five channel. Their projectors have ranks 1,8,8,9,10. The pure mediator potential is a positive real linear combination of the squared norms of these projections of  $S$  tensor  $S$ .

The five coefficients in the benchmark are  $1/10, 1/5, 3/10, 2/5, 1/2$ . These coefficients remain independent allowed couplings; representation theory fixes the contractions and their component ratios, not these five relative strengths. The calculation uses every CP-even channel and does not set the extra finite-group channels to zero merely to preserve a selected frame.

## Controlled rank lifting

$$D = \frac{\alpha F + 30 Q \circ \bar{F}}{\alpha^2 - 900}, \quad \alpha = 33 + p$$

The isolated source branch is  $A = \text{diag}(1, 0, 0)$  and its finite group transforms. For  $W_A = I_6(A) + (1/5)\det(A)\text{-squared}$ , take  $V_A = \|\text{gradient } W_A\|^2 + 24\|A\|^2 - 68 \text{Re } W_A + p\|\text{adj } A\|^2$ . The adjugate norm is a nonnegative rank penalty, equal to half the difference between the squared total norm and the fourth-order Gram trace. It vanishes, with its

first derivative, on the rank-one branch and stiffens its transverse modes. The benchmark has  $p=30$  in each source sector.

In the lower two-by-two block around the unit axis source, write  $\alpha=33+p$ . The sextic tensor gives four holomorphic Hessian coefficients with modulus three and a fixed root-of-unity phase pattern. For the normalized phase matrix  $Q$  with rows  $(1, \omega^2)$  and  $(\omega, 1)$ , the linear normal response to a projected rank-one force  $F$  is  $D = (\alpha F + 30 Q \text{ times elementwise conjugate } F) / (\alpha^2 - 900)$ . Thus the response is the sum of two rank-one matrices fixed by the sextic tensor. Their relative orientation can make  $D$  rank two, supplying both previously zero light masses.

A direct calculation of the full 54-dimensional source/link Hessian reproduces this two-by-two response. Generic balanced cross-sector terms whose normal force vanishes at a unitary link cannot substitute for the dressed trace channel at this order. The light-mass ratio is not simply  $30/\alpha$ : it also depends on the overlaps between the forcing vectors and their tensor-defined conjugates. This geometric dependence is retained explicitly. The positive rank penalty provides a way to separate the two light scales without assigning separate quark-column couplings.

## The source CP audit

$$45^2 = 2025, \quad N_{\text{CP fixed}} = 289$$

The source geometry cannot be treated as uniformly CP conserving. An exact search in  $Q(\sqrt{5}, \omega)$  considers all 45 distinct axis rays in the left space and all 45 in the label space. It enumerates the declared CP action composed with every family transformation and permitted C6 phase compensation. Exactly 289 of the 2025 pairs are fixed by a transformation in this CP coset; 1736 are not. Every positive witness and the exhaustive candidate lists are recorded. A CP transformation may square to a residual family operation rather than the identity.

For a CP-fixed isolated stationary point, uniqueness under the implicit function theorem implies that CP-even deformations preserve CP on that local branch until its Hessian becomes singular or the branch terminates. A chiral scalar configuration can instead occur on other ray-pair branches. Quark CP must still be checked after the light degeneracies are lifted: a phase in a source or mediator is not sufficient. The full-mass benchmark has a nonzero weak-basis commutator invariant and therefore has physical weak CP, independent of any source-phase description.

The certificate concerns the declared CP coset and its family/phase equivalents. It is not a separate classification of every possible additional accidental antiunitary symmetry of every specially tuned potential. The nonzero quark CP invariant at the retained vacuum directly excludes CP conservation of its nondegenerate quark sector.

## Full mediator matching and a bounded potential

$$V_{\text{stab}} = 10^{-10} (N_L + N_A + N_B)^6$$

The mediator is minimized as eight complex fields together with the three matrix fields. It is not eliminated using its leading exchange while discarding allowed mass and quartic terms. Set the source field unit  $\rho=1/10$ , mediator mass  $M=10$ , and scaled exchange strength  $\epsilon=300$ . The underlying dressed quartic coupling is  $k=\sqrt{\epsilon} M$   $\rho^2=\sqrt{3}$ . The seven source/mediator quartic coefficients are  $1/10, 1/5, 3/10, 1/100, 1/50, 3/100, 1/25$ . The remaining scalar coefficients and source

self-interactions are specified in the module and receipt. Their values are fixed before the quark observables are evaluated.

In scaled variables the mediator/source-bilinear corrections carry  $\rho^2/M^2=0.0001$ . The mediator quartics carry  $\epsilon^2 \rho$  to the power ten divided by  $M$  to the power four. All are retained. A large  $\epsilon$  here compares the exchange with a weak high-degree source potential; it does not by itself mean an underlying quartic coupling of size 300. The physical mediator background and its mass are separately available from the field coordinates and scaling formula.

The polynomial also contains the positive degree-twelve invariant  $10$  to the power  $-10$  times  $(N_L+N_A+N_B)$  to the power six in scaled units. This radial stabilizer makes a global mathematical statement possible. The five positive mediator projectors sum to the identity, so their quartics bound the mediator norm to the fourth power below by a positive coefficient. Young's inequality absorbs the negative dressed vertex into half that quartic plus a negative constant times the squared total source norm. Negative holomorphic source terms grow at most as the cubed norm. The positive sixth power of the total norm dominates both, and the remaining positive mediator quartic dominates its independent large-field direction. The full polynomial is coercive and has global minima.

Coercivity does not certify the retained vacuum as the global winner, and large-field extrema need not remain in the validity range of the source EFT. The numerical result establishes a stable CP-breaking local branch. A cosmological occupation mechanism, global energy selection within a justified EFT domain, thresholds and radiative evolution are separate calculations.

## Fixed-mass protection and practical conclusion

$$\text{Im det}[H_u, H_d] = 2J \Delta(H_u)\Delta(H_d)$$

The independently computed commutator determinant agrees with the SVD-derived Jarlskog invariant. The CP partner is a stationary solution with equal energy, identical singular values and reversed CP-odd quantities. Fully re-minimizing after both signs of a one-percent change in a permitted source/link quartic preserves a stable full-rank weak-CP branch; its observables change. This demonstrates a neighborhood of interacting models, not a single input phase disguised as a vacuum solution.

The preceding kinetic theorem also applies to this completed nondegenerate CP-breaking quark sector. Twelve positive CP-compatible bare metrics are reconstructed from finite polynomial covariants and matched through the same heavy null space. They preserve all six masses to better than  $10$  to the power  $-9$  in ordinary singular-value extraction, while the response of  $|\text{Vus}|, |\text{Vcb}|, |\text{Vub}|, J$  has rank four. The scalar vacuum, its couplings and the universal source contractions remain fixed during that test. Thus the golden equation still needs a UV explanation of kinetic coefficients as well as the interaction construction developed here.

The central advance is a specified joint interaction that produces three massive quark families and physical CP using representation-derived source labels and universal quark couplings. Its complete renormalizable mediator sector, the canonical matching, source CP classification and stable vacuum are executable and reviewable. It provides a concrete dynamical starting point for the next search; it does not supply an observed CKM fit, an exact golden prediction or a global M22 origin of the three assigned chiral families.

## Reproduction

Run `OPENBLAS_NUM_THREADS=1 PYTHONPATH=python python python/develop_valentiner_adjoint_uv.py` from the repository root. It starts from the disclosed numerical source seed and re-minimizes all 70 coordinates; no CKM target is consulted. The optional `python/search_valentiner_rank_lifting.py` reproduces the bounded seed search from its fixed random seed and prescribed discrete cases. The exact CP-coset search is in `python/valentiner_source_cp.py`. The scalar trace certificates, quartic projector construction, analytic derivatives and heavy matching are in the adjacent modules.

Receipts retain the complete fields, input parameters, independent CP invariant, opposite-sign partner, re-minimized deformations and fixed-mass metrics. The validation report distinguishes exact character and finite-field statements from numerical stationary-point and stability evidence. A fresh extraction reproduces the scientific receipts byte for byte. The complete release retains all prior research and the updated monograph.

# Canonical quarks, currents and strong CP

## The canonical result

$$\bar{\theta}_{\text{tree}} = 0, \quad J = 1.13669 \times 10^{-6}$$

The same shared-adjoint scalar vacuum now admits a different, explicitly specified fermion assignment with physical weak CP and an exactly vanishing strong phase at tree level. The product-family representations enforce a universal bare Higgs contraction and forbid every Higgs-to-heavy Yukawa vertex at renormalizable order. All ten allowed renormalizable mass and Yukawa coefficients in each electric-charge sector are present and nonzero. The resulting Nelson–Barr mass structure is derived from this assignment, rather than imposed by setting an allowed renormalizable vertex to zero. The adjoint still connects the heavy source labels in both sectors. This changes the fermion representations of the previous construction; it is not a claim that canonical normalization repairs the previous mass matrices.

At the retained scalar background the new canonical benchmark has six nonzero light masses and a decoupling weak invariant  $J = 1.13669458335e-6$ . The CP partner has the opposite  $J$  and the same masses and mixing magnitudes. Neither branch uses a CKM target, a golden coefficient, an observed mass or a phase as an optimization input. The benchmark mixing magnitudes are  $|V_{us}| = 0.0458605886$ ,  $|V_{cb}| = 0.0260342476$  and  $|V_{ub}| = 0.0020441921$ . They do not constitute an observed CKM fit. The reference light Yukawa spectra are almost degenerate; a separate common-messenger-scale deformation demonstrates a canonical hierarchy mechanism, without claiming that its scales are dynamically selected or that its ratios reproduce observed quark masses.

The result has three precise parts: a representation-enforced renormalizable determinant protection, a canonical formula separating mass scales from mixing frames, and exact finite-scale correlations among charged currents, neutral currents and Higgs couplings. A constructive fixed-mass test also shows what the protection does not accomplish: positive allowed kinetic terms preserve the strong determinant phase while moving all four weak mixing directions. Thus protecting strong CP and predicting the golden relation are distinct mathematical obligations.

## Why the previous assignment needed a determinant audit

$$\det D(v) = v^3 \det M \det Y_0, \quad \bar{\theta}_{\text{old}} = -2.46264$$

In the previous completion, the twelve-by-twelve Dirac matrix has a Higgs row connecting the ordinary doublets to heavy singlets, followed by nine heavy mass rows. In the right-handed basis consisting of three bare quarks and nine heavy quarks, the Higgs row is  $T = [0, h, 1, 0, 0]$  and the heavy row is  $F = [C, M]$ . Put  $N = M^{-1} C$ . The exact zero-Higgs null frame is  $E = [I; -N](I + N^\dagger N)^{-1/2}$ . The dimension-four light Yukawa is  $Y = T E = Y_0(I + N^\dagger N)^{-1/2}$ , with  $Y_0 = -h N_u$ . This is exact in the source-to-heavy ratios and is a decoupling statement in the Higgs-to-heavy ratio.

Block elimination gives  $\det D(v) = v^3 \det M \det Y_0$ . The phase of this determinant is independent of the Higgs scale. More generally any positive Hermitian canonical metrics

multiply  $D$  from the left and right by inverse square roots with positive real determinants. They cannot change  $\arg \det D$ . A chiral phase rotation can make mass eigenvalues positive but shifts the microscopic QCD theta angle through the anomaly; it does not remove the invariant strong phase. With  $\theta_{\text{QCD}} = 0$  in the original real-coupling basis, the previous assignment gives  $\bar{\theta} = -2.46263764551$  radians. Its CP partner has the opposite phase. This is a tree-level result; no extra QCD mechanism was part of that assignment.

For an independent source-level check the nine-by-nine determinants reduce to six-by-six blocks. The up block is  $[[A, 0.71 L], [-0.2 B S^\dagger, 1.3 I - 0.04 B B^\dagger]]$ . The down block is  $[[ -0.2 A S, 0.71 L], [B, 1.3 I]]$ . The determinant coefficient in each sector is minus  $h^3$  times  $2^3$  times the determinant of its block. This removes the first heavy mass and its reverse link coefficient from the determinant audit. The reduced coefficients agree with the direct mass determinants to relative error below  $3e-12$ . Heavy decoupling and positive wavefunction normalization cannot suppress this phase.

## The new representation assignment

Retain the declared scalar group  $3_{A6_u} \times 3_{A6_d} \times 3_{A6_H} \times C6_A \times C6_B$  and CP, and the scalar fields  $L = (3_u, 3_{\bar{d}})$ ,  $A = (3_u, 3_{\bar{H}})$ ,  $B = (3_d, 3_{\bar{H}})$  and  $S = 8_H$ . Their phase charges are respectively  $(0,0)$ ,  $(1,0)$ ,  $(0,1)$  and  $(-1,1)$ . The ordinary left doublets and both bare right-handed quark triplets now transform as neutral  $3_u$ . In each charge sector introduce three vectorlike electroweak-singlet triplets  $D = (3_d; 0,0)$ ,  $H_A = (3_H; -1,0)$  and  $H_B = (3_H; 0,-1)$ . These are color triplets with charge  $2/3$  in the up sector or  $-1/3$  in the down sector. The Higgs is a flavor and phase singlet. Its conjugate occurs in the up Yukawa in the usual way.

The constant kinetic endomorphism of each irreducible triplet is a scalar by Schur's lemma. The distinct heavy family representations or phase charges forbid constant cross kinetic terms. Their positive constants can be absorbed into field normalization and the real mass and Yukawa coefficients. Canonical renormalizable bare metrics therefore define a justified convention for this field content. This does not exclude higher-degree kinetic covariants after flavor breaking.

The full list of fermion contractions is short enough to audit exhaustively. There are three real diagonal vectorlike masses. The bare right-handed triplet couples to  $D$  through  $L^\dagger$  and to  $H_A$  through  $A^\dagger$ .  $D$  and  $H_B$  couple through  $B$  and  $B^\dagger$  with independent real coefficients.  $H_A$  and  $H_B$  couple through  $S$  and  $S^\dagger$  with independent real coefficients. Finally the ordinary doublet and bare right-handed triplet have the unique Higgs contraction  $y I$ . There is no renormalizable source contraction from the bare right-handed  $3_u$  to  $H_B$ . There is no bare Higgs contraction from the left-handed  $3_u$  to any heavy right-handed representation. Both observations follow from the simultaneous family labels and phase charges.

The operator receipt enumerates all nine mass/source contractions using charges modulo six and certifies their multiplicities with the exact 1,080-element group. The exact triplet endomorphism multiplicity is one; the adjoint has no singlet; the triplet-adjoint-triplet singlet multiplicity is one. Thus the ten real coefficients do not hide arbitrary component matrices. Their overall values remain independent physical inputs. The scalar potential and its retained 70-real-field local minimum are unchanged. The quantum corrections from the changed fermions, global vacuum selection and Higgs-portal-induced vacuum shifts are not calculated here.

## A determinant protected by the interaction graph

$$\det D(v) = (vy)^3 \det M, \quad WMW^{-1} = (WMW^{-1})^\dagger$$

In the heavy order  $D, H_A, H_B$ , write  $M = [[m_D I, 0, b B], [0, m_A I, s S], [c B^\dagger, t S^\dagger, m_B I]]$  and  $C = [\lambda_L L^\dagger; \lambda_A A^\dagger; 0]$ . The exact full Dirac matrix is  $D(v) = [[v y I, 0], [C, M]]$ . Every displayed coefficient is present. The absence of its upper-right Higgs block is enforced at renormalizable order. Consequently  $\det D(v) = (v y)^3 \det M$  and is independent of the complex bare-heavy source column  $C$ .

The heavy interaction graph is a tree with two edges. For  $b, c, s, t$  positive, the constant diagonal matrix  $W$  with triplet weights  $\sqrt{c/b}$ ,  $\sqrt{t/s}$ , 1 makes  $W M W^{-1}$  Hermitian. Its two off-diagonal blocks are  $\sqrt{bc}$   $B$  and  $\sqrt{st}$   $S$  and their Hermitian conjugates. Therefore  $\det M$  is real exactly, for every scalar background. On the connected branch where this Hermitian matrix is positive definite, its determinant is positive. With positive  $y_u$  and  $y_d$ ,  $v$  positive and microscopic  $\theta_{\text{QCD}} = 0$ , the product of the up and down full determinants is positive and  $\bar{\theta}$  vanishes exactly at tree level. The positivity domain is a stated branch condition; a determinant crossing through zero can change its sign and is not covered by a blanket claim about all backgrounds.

The benchmark's Hermitian-similar heavy matrices lie inside this positive region. A separate 80-digit determinant check reconstructs the real coefficients as exact decimal rationals before multiplication; its imaginary residual is below  $1e-60$ . An independent 80-digit full-spectrum check uses the saved floating matrices and retains their minute rounding asymmetry explicitly. Positive Hermitian changes of any bare fermion metrics preserve the determinant phase because their canonical normalization determinants are positive. This determinant protection is stronger than the canonical-metric convention and does not require the weak mixing matrices to remain fixed.

This is a realization of the established Nelson–Barr mechanism. The contribution of this calculation is the explicit product-family assignment, complete contraction census and shared-adjoint matching at our retained vacuum. General Nelson–Barr determinant protection, singlet-induced nonunitary weak mixing and flavor-changing neutral currents are established physics, not claimed as new discoveries. Primary context is Cherchiglia and Nishi, Consequences of vector-like quarks of Nelson–Barr type, arXiv:2004.11318, and Morozumi, Shimizu, Takahashi and Umeeda, Effective theory analysis for vector-like quark model, arXiv:1801.05268. Their URLs are retained in the scientific receipt.

## Canonical mass scales and fixed mixing frames

$$y_i(\eta) = \frac{y\eta}{\sqrt{\eta^2 + \sigma_i^2}}, \quad P_i(\eta) = P_i(1)$$

The exact zero-Higgs null metric is  $K = I + N^\dagger N$ , where  $N = M^{-1} C$ , and the light Yukawa is  $Y = y K^{-1/2}$ . It is positive Hermitian in the common ordinary  $3_u$  frame. Complex noncommuting up and down metrics produce physical weak CP even though each determinant is positive. Their left spectral projectors give  $|V_{ij}|^2 = \text{Tr}(P_i^u P_j^d)$  in the decoupling limit. This gives a physical frame directly from the interaction and its canonical metric, with no free unitary matrix inserted in the readout.

Multiply the entire heavy  $M$  of one sector by a positive  $\eta$  while keeping  $C$  and  $y$  fixed. If  $\sigma_i$  are the singular values of  $M_0^{-1} C$ , the light Yukawa magnitudes are exactly  $y \eta / \sqrt{\eta^2 + \sigma_i^2}$ . The spectral projectors of  $K$  and  $Y$  remain unchanged because  $K = I + \eta^{-2} N_0^\dagger N_0$  is a scalar function of the same fixed Hermitian matrix. Independent positive scale changes in the up and down sectors therefore preserve their decoupling mixing magnitudes and weak  $J$ , provided the spectra remain nondegenerate. In the small- $\eta$  regime the

masses become inversely proportional to the  $\sigma_i$ . This proves a mass hierarchy mechanism and a separation of scales from frames for this universal Higgs assignment; it does not select  $\eta$  or predict the observed hierarchy.

The prescribed scan uses  $\eta = 1, 0.1, 0.01, 0.001, 0.0001$ , with no fit objective. It changes the six masses substantially while reproducing the same mixing magnitudes to below  $2e-13$ . The scalar vacuum is held fixed. The statement is a decoupling theorem: reducing heavy masses at a fixed physical Higgs scale can invalidate decoupling, so finite-scale observables must instead be obtained from the full spectrum below.

## Finite-scale pole normalization and physical currents

$$V = N_u^\dagger N_d, \quad Z_f = N_f^\dagger N_f, \quad h_{ij} = Z_{ij} m_j / v$$

For either fermion assignment define the full-rank heavy row  $F$  and Higgs row  $T$ , and let  $G = F F^\dagger$ . The exact left mass-squared matrix has blocks  $[[v^2 T T^\dagger, v T F^\dagger], [v F T^\dagger, G]]$ . At a light pole  $\lambda$  below the heavy gap, elimination of the singlet component gives  $v^2 [T T^\dagger - T F^\dagger (G - \lambda I)^{-1} F T^\dagger] q = \lambda q$ . The singlet component is  $r = -v (G - \lambda I)^{-1} F T^\dagger q$ . The normalized mode satisfies  $q^\dagger [I + v^2 T F^\dagger (G - \lambda I)^{-2} F T^\dagger] q = 1$ . Because this positive metric depends on the pole, one common exact three-by-three wavefunction metric cannot replace the full finite-scale calculation.

Expanding the pole equation yields the dimension-six left metric  $I + v^2 Z_1$ , with  $Z_1 = T F^\dagger G^{-2} F T^\dagger$ .  $Z_1$  is positive semidefinite because it is a Gram matrix. The light Hamiltonian at this order is  $v^2 K_{\text{left}}^{-1/2} H_0 K_{\text{left}}^{-1/2}$ , with  $H_0 = Y Y^\dagger$ . The implementation obtains the full twelve-by-twelve singular spectrum directly and compares it with this derivative expansion. At  $v = 0.03$  in the reference messenger units, the light-current magnitude approximation error is approximately  $7.23e-9$  and the maximum relative light-mass error approximately  $3.18e-9$ . Across  $v = 0.03$  and  $0.1$  the current error follows the expected fourth-order scaling. At very small  $v$  floating-point eigenframe errors set a numerical floor. No physical Higgs-to-messenger scale is fixed by this benchmark.

Let  $N_u$  and  $N_d$  be the ordinary-doublet rows of the three light left eigenvectors. The physical light charged current is  $V_{\text{light}} = N_u^\dagger N_d$ . The weak-isospin neutral currents are  $Z_u = N_u^\dagger N_u$  and  $Z_d = N_d^\dagger N_d$ , after stripping the standard electroweak couplings and universal electric-charge contribution. Generally  $V_{\text{light}}$  is not unitary. Its row deficit has the exact positive decomposition  $I - V_{\text{light}} V_{\text{light}}^\dagger = (I - N_u^\dagger N_u) + N_u^\dagger (I - N_d N_d^\dagger) N_u$ . The column version follows by exchanging sectors. The off-diagonal neutral-current entries are consequently part of the same matching prediction, rather than optional additional matrices.

With all light and heavy states included, the charged-current matrix is a rank-three partial isometry and its two products are the full neutral-current projectors. The Higgs coupling in the full mass basis obeys the exact identity  $h_{ij} = Z_{ij} m_j / v$ . It follows from the top rows of  $D R_j = m_j U_j$  and holds for all twelve modes in each sector. Tests check this relation, the neutral projectors, the positive deficit decompositions, the pole equation and the Higgs mass derivative independently. At finite scale the nine rephasing-invariant quartets need not have the same absolute value. A nonzero quartet certifies weak CP, while reporting one unitary  $J$  or extracting a unitary standard phase would discard part of the physical current information.

## The precise protection boundary

$$\Delta\bar{\theta} = 3\arg(y_d + c\det L/\Lambda^3)$$

The renormalizable assignment protects the full determinant phase, not the golden weak relation. On the ordinary right-handed  $3_u$  space use  $X = A\text{-dagger}$  and  $Z = B\text{-dagger} L\text{-dagger}$ . The previous nine-covariant construction provides CP-compatible real polynomial kinetic generators at this vacuum. Twelve positive polynomial metric deformations preserve all six decoupling masses while spanning four independent directions in  $|V_{us}|$ ,  $|V_{cb}|$ ,  $|V_{ub}|$  and  $J$ . The smallest normalized response singular value is 0.5567; the maximum mass log residual is  $1.34e-15$ . Because these metrics have positive determinants, they leave the protected tree strong phase zero. A protecting mechanism for the golden relation must still constrain this kinetic operator space.

An allowed Yukawa operator provides an explicit limit to the determinant protection: a real coefficient multiplying  $\det(L) \bar{Q}_L H d_R / \Lambda^3$  plus its Hermitian conjugate. The invariant  $\det L$  is neutral under the declared family and phase symmetries, and the bare universal Higgs Yukawa is independently invariant. Their product is therefore allowed and CP even with a real coefficient. It is dimension seven. It shifts the universal down Yukawa from  $y_d$  to  $y_d + c \det L / \Lambda^3$  and creates a strong phase  $3 \arg(y_d + c \det L / \Lambda^3)$ . No ordinary phase charge that retains both the bare Yukawa and the existing cubic  $\det L$  scalar interaction can forbid this product: invariance of the cubic requires the phase of  $\det L$  to be one.

At the retained background  $\text{Im } \det L = 3.17850793318e-6$  in reference messenger units. For coefficient one and cutoffs 1,10,100 in those units, this single operator creates phases approximately  $1.67e-5, 1.67e-8, 1.67e-11$  respectively. These are a scaling diagnostic, not a phenomenological cutoff determination. The leading expression is  $3 c \text{Im } \det L / (y_d \Lambda^3)$ . Quantum threshold contributions to strong CP and the complete higher-degree Yukawa operator space remain uncalculated. A vanishing tree determinant phase is not an all-orders solution of strong CP.

The fixed-background finite-Higgs spectra also do not establish the complete electroweak vacuum: Higgs portals allowed in the scalar action could shift the flavor condensates, and their coefficients were not part of the retained zero-Higgs scalar calculation. The determinant theorem holds throughout its positive branch regardless of those shifts, but the numerical mixing predictions would need to be recalculated. Global flavor-vacuum selection, the M22 origin of three ordinary doublets, observed masses and mixing, radiative stability and a protected golden relation remain separate requirements.

## Reproduction

Run `OPENBLAS_NUM_THREADS=1 PYTHONPATH=python python python/develop_valentiner_canonical.py` from the repository root. It reads the disclosed previous scalar vacuum and produces `receipts/m22_interactions/valentiner_canonical.json`, including the exact operator census, old determinant obstruction, new determinant checks, finite-scale spectra and currents, mass-scale scan, opposite CP branch, fixed-mass kinetic countermetrics and quality operator. No scalar minimization or CKM fit is hidden in this driver. The canonical matcher is general for a full-rank  $n$  by  $n+3$  heavy row with an invertible square heavy block and three ordinary doublets. The release includes the standalone paper, updated complete monograph, tests and complete tracked source archive.

# Electroweak completion, quantum matching and attainable spectra

## The completed calculations

The canonical Nelson–Barr assignment now has an explicitly minimized joint Higgs/flavor branch, a mass-basis one-loop threshold calculation using all 71 physical neutral real scalars, an exact higher-fermion covariant census through three scalar insertions, and a constructive hierarchy and CKM matching benchmark. The scientific receipt is ``receipts/m22_interactions/valentiner_quantum.json``; every field coordinate and real metric coefficient is retained. These are calculations on a specified action, rather than a derivation of its input couplings from observed flavor.

The selected real spurion is  $\epsilon = 10^{-5}$ . The joint branch has minimum scalar mass squared  $4.67875984 \times 10^{-14}$  in the declared flavor units and maximum scaled flavor gradient  $5.79 \times 10^{-9}$ . The neutral-scalar threshold is approximately  $+1.45 \times 10^{-14}$  radians. Its CP partner has the opposite phase within  $1.32 \times 10^{-15}$  radians. The finite-difference Hessian and double-precision eigenbasis set the numerical accuracy; the separate 60-digit trace check does not turn those input matrices into 60-digit data. We therefore use the conservative numerical statement  $|\delta \theta| < 2 \times 10^{-14}$  at this benchmark, without assigning significance to its last digits.

## A joint positive electroweak branch

Write the neutral Higgs component as  $H^0 = v + h/\sqrt{2}$ , with canonical real  $h$ , and retain the 35 complex flavor fields  $L, A, B, S$  of the canonical completion. The Higgs potential is  $-\mu_H^2 H^\dagger H + \lambda_H (H^\dagger H)^2$ , with  $\lambda_H = 0.13$ . Add all four renormalizable Higgs norm portals on this field content:  $(H^\dagger H)$  times  $\text{Tr}(L^\dagger L)$ ,  $\text{Tr}(A^\dagger A)$ ,  $\text{Tr}(B^\dagger B)$ , and  $\text{Tr}(S^\dagger S)$ . Their nonzero coefficients are respectively  $10^{-4}$ ,  $2 \times 10^{-4}$ ,  $3 \times 10^{-4}$  and  $4 \times 10^{-4}$ . The family representations and phase charges forbid further renormalizable Higgs portals with these scalars.

At a nominated  $v$ , minimize the full flavor potential including the portals. Fix the permitted Higgs mass parameter by  $\mu_H^2 = 2 \lambda_H v^2 + \sum (\lambda_i N_i)$ . This is ordinary electroweak scale matching:  $v$  is an input, not a prediction. In canonical real flavor coordinates  $\sigma$ , the 71-dimensional Hessian has the flavor block from the full portal-shifted potential, mixed entries  $\sqrt{2} v d(\sum \lambda_i N_i)/d \sigma$ , and  $h, h$  entry  $4 \lambda_H v^2$ . Positivity is tested on this full matrix, rather than on a fixed-Higgs flavor block.

The receipt scans  $\epsilon = 1, 0.1, 0.01, 0.001$  and  $10^{-5}$  with  $v = 0.03 \epsilon$ . Every scanned branch is stationary to the declared tolerance and has all 71 physical neutral eigenvalues positive. Three gauge Goldstone directions are removed by gauge fixing; they are not additional physical scalar minima. This establishes local stability on the tested branch, not a global vacuum selection theorem.

## A chiral spurion that preserves the mixing frame

An additional approximate  $Z_6$  has charges 1 for every heavy left-handed fermion, 0 for every heavy right-handed fermion, 2 for the ordinary left and right quarks, 0 for all flavor scalars and the Higgs, and 1 for epsilon. Thus the heavy mass operator requires epsilon, the heavy-left to bare-right source requires its conjugate, and a Higgs-to-heavy-right operator requires epsilon squared. A bare ordinary Higgs Yukawa remains neutral. Take epsilon real and positive, preserving CP in the coefficient basis.

For the declared benchmark all nine heavy mass/source coefficients per sector are multiplied by epsilon. Then  $M$  and  $C$  scale together,  $N = M^{-1} C$  is unchanged, and so is the canonical decoupling Yukawa. If  $v$  is also scaled by epsilon at a fixed scalar background, the entire 12-state Dirac matrix scales by epsilon. All left and right singular vectors, the full charged and neutral currents, and their CP quartets are exactly unchanged. The receipt checks this finite-scale identity numerically, including the appropriately rescaled pole metric. Portal-induced background shifts are included separately in the joint branch scan.

Small chiral breaking suppresses radiative communication to the flavor vacuum without suppressing the fixed-background weak phase. The general small-spurion strategy is established Nelson-Barr model-building practice; the present result is its explicit charge realization and full-matrix check on this shared-adjoint action. See Murai and Nakayama, *\*Revisiting the Minimal Nelson-Barr Model\**, arXiv:2407.16202, sections 2–3. No novelty is claimed for the general strategy.

## The one-loop determinant calculation

Let  $U^\dagger D V = \text{diag}(m_i)$ , with all 12 Dirac masses positive, and let  $O$  diagonalize the canonical 71-scalar Hessian. Each real scalar vertex becomes  $G_a = U^\dagger (\sum_b O_{ba} dD/d\sigma_b) V$ . At zero external momentum the finite chirality-changing mass threshold is

$$\delta D_{ij} = -\sum_{(a,k)} G_{a,ik} m_k G_{a,kj} B_0(0, m_k^2, m_a^2)/(16\pi^2), \quad B_0(0, x, y) = 1 - [x \log(x/\mu^2) - y \log(y/\mu^2)]/(x-y).$$

Use the continuous equal-mass limit  $B_0 = -\log(x/\mu^2)$ . The second vertex is  $G$ , not  $G^\dagger$ . This chirality structure permits a CP phase. The finite matching convention follows the zero-momentum limit of the neutral-scalar self-energy in appendix A of arXiv:2407.16202. The receipt uses  $\mu = 1$  in the stated flavor units. There is no extra color multiplicity on the open fermion self-energy line.

The leading determinant shift is  $\text{Im Tr}(D^{-1} \delta D)$ , evaluated in the mass basis as  $\sum_i \text{Im}(\delta D_{ii})/m_i$ . At  $\epsilon = 10^{-5}$  the up contribution is  $+2.22849 \times 10^{-14}$  and the down contribution  $-7.81399 \times 10^{-15}$  radians. The relative mass-correction norms are below 0.036. The UV phase coefficients are numerically below  $3 \times 10^{-17}$ . An independently accumulated 60-digit trace agrees with the double-precision contraction at the tested absolute tolerance.

All 70 flavor vertices are checked against finite differences of the original full mass matrix. The threshold also passes independent changes of left/right fermion basis and real scalar basis, exact conjugate-CP tests on generic matrices, and integral checks of  $B_0$  including extreme mass ratios. The full CP-transformed vacuum and independently evaluated Hessian provide the quoted partner check.

The remaining renormalizable one-loop diagonal phase factors vanish algebraically. Neutral gauge vertices have a Hermitian left current and a diagonal real right current. Charged  $W$  vertices give only Hermitian wavefunction terms. A neutral Goldstone vertex is  $i Z_{ij} m_j/(\sqrt{2})$

v), so its chirality-flip product is proportional to the real quantity  $-|Z_{ij}|^2 m_j^2$ . Charged Goldstone products similarly reduce to  $|V_{ij}|^2 m_j^2/v^2$  using the full 12-state charged current. These reality identities are tested before truncating to three light states. Higgs/flavor scalar mixing is already included in the 71-mode computation. Scalar tadpole shifts cannot generate a first-order determinant phase because the tree phase is identically zero throughout the positive heavy branch, not merely at one stationary point.

This closes the leading one-loop mass-phase calculation for the stated canonical-bare fermion action. It does not establish two-loop or all-loop protection, and it does not include higher fermion interactions in the loop vertices. A fully quantum-selected scalar vacuum would additionally require scalar Coleman-Weinberg terms and renormalized scalar inputs.

## Quantified fermionic vacuum feedback

The colored Dirac Coleman-Weinberg term is  $-3 \sum_i m_i^4 [\log(m_i^2/\mu^2) - 3/2]/(16 \pi^2)$ . Here the closed loop does carry three colors. Its canonical real-field gradient is  $-3 \sum_i m_i^3 [\log(m_i^2/\mu^2) - 1] \text{Re}(G_{a,ii})/(4 \pi^2)$ . The gradient is verified independently by finite differences.

At a fixed scalar background, scaling  $D$  and  $\mu$  by  $\epsilon$  scales this potential exactly by  $\epsilon^4$ . The measured ratio at  $\epsilon = 10^{-5}$  is  $9.99999999999927 \times 10^{-21}$ . Match the single allowed Higgs soft mass to hold  $v$  fixed; no flavor tadpoles are subtracted. The resulting linearized flavor-coordinate displacement from the fermion loop is below  $5.54 \times 10^{-18}$ . The Higgs soft-mass matching shift is  $9.34 \times 10^{-14}$ . These numbers quantify the fermionic contribution; they do not assert that scalar-loop feedback has been removed.

## The exact higher-fermion operator census

The group projector is computed over all 1,080 elements of each independent 3.A6 factor, in the exact coefficient field already used by the representation certificates. Its three group averages factor. Symmetric scalar powers use their cycle-index characters, including  $\chi(g^2)$  and  $\chi(g^3)$ , and phase charges are imposed modulo six. This counts commuting scalar insertions correctly. Sixteen distinct character-power signatures suffice for the exact sums.

The complete Higgs covariant counts by scalar degree 0,1,2,3 are 1,2,9,21; hence 33 independent complex covariants. The complete heavy-mass/source counts are 3,6,29,85; hence 123. These are complex intertwiner dimensions before choosing real CP-fixed coefficients. The anti-linear CP action determines the real coefficient basis; no claim is made that each displayed multidegree is individually CP fixed. The receipt retains each left/right endpoint, scalar multidegree and multiplicity.

One useful finite-group result is multiplicity eight for the  $H_A$  to  $H_B$  covariants with two  $S$  fields and one  $S$ -dagger. Ordinary continuous-SU(3) matrix and trace words span only five of those directions. A scan restricted to those words would therefore miss three permitted finite-group channels. The exact census prevents that omission without pretending that arbitrary continuous-SU(3) contractions are a complete 3.A6 basis.

An explicit degree-two heavy chord is  $L$ -dagger  $A$  between  $D$  and  $H_A$ ; another is  $B$   $S$ -dagger. Bare Yukawas admit  $\det(L)$  times the identity at degree three. Such interactions are retained in the counting and are not forbidden by the new spurion. The spurion suppresses Higgs-to-heavy vertices, but its common heavy mass factor cancels from relative corrections inside the heavy block.

For an explicit coefficient domain, the code gives a uniform determinant-phase certificate for *every* permitted higher Higgs operator of scalar degree one to three and heavy-mass/source operator of scalar degree two and three. Normalize each invariant tensor to Frobenius norm one and bound each coefficient magnitude by one. Cauchy-Schwarz bounds each vacuum contraction by the product of its scalar norms. Sum these bounds with the correct cutoff and spurion powers to bound  $r = ||D_{\text{reduced}}^{-1} \Delta D_{\text{reduced}}||$ . For  $r < 1$ ,  $|\delta \arg \det D| \leq -12 \log(1-r)$ . No cancellation or particular tensor basis is assumed.

At  $\Lambda = 10^{12}$  in the declared flavor units, the combined two-sector bound is  $2.936 \times 10^{-11}$  radians. This is a sufficient finite-EFT quality condition, deliberately conservative. The large cutoff is a stated condition, not a derived UV scale or a demonstrated phenomenologically viable completion. Operators beyond three scalar insertions and derivative operators are outside this certificate.

## Constructive hierarchical spectrum matching

For a chosen positive left-handed target  $H$  and  $N = M^{-1} C$ , a positive ordinary right-handed metric  $B$  satisfying  $B = y^2 H^{-1} - N^\dagger N$  gives  $YY^\dagger = H$  after exact decoupling matching. Choose  $y$  large enough to make  $B$  positive. At the retained CP-breaking vacuum the nine real polynomial kinetic covariants span every Hermitian metric, so  $B$  has a finite polynomial completion  $(I+T)^2 + \eta I$  that is positive for every flavor field value. The coefficients are real and therefore CP compatible. This proves attainability, rather than requiring a numerical fit to converge.

The benchmark uses illustrative dimensionless up Yukawas  $(10^{-5}, 0.003, 0.9)$  and down Yukawas  $(2 \times 10^{-5}, 0.0004, 0.02)$ , with bare coefficients 1.2 and 0.57. These hierarchical Yukawas are declared test inputs, not running quark mass measurements. For the CKM target use PDG 2026 equation 12.28 central parameters  $\sin(\theta_{12})=0.22517$ ,  $\sin(\theta_{13})=0.003763$ ,  $\sin(\theta_{23})=0.04189$ ,  $\delta=1.154$  radians. The constructed result has  $J = 3.15937163 \times 10^{-5}$  and reproduces all CKM magnitudes within  $4.45 \times 10^{-16}$ . The supplied target matrices are checked at 90 digits; the positive polynomial metric reconstruction errors are below  $10^{-65}$ .

This fitted kinetic action is distinct from the canonical-bare fermion action used for the loop threshold. Its field-dependent kinetic vertices must be included before any quantum claim is transferred to it. Positive metric determinants preserve the tree strong phase. Neither this inverse construction nor the spurion derives the golden CKM coefficient: the benchmark explicitly exposes the free real inputs used to match the hierarchy and mixing.

## Reproduction and release validation

Run `PYTHONPATH=python OPENBLAS_NUM_THREADS=1 python python/develop_valentiner_quantum.py``, then `PYTHONPATH=python python python/tests/test_valentiner_quantum.py``. The new suite contains 15 focused checks, supplemented by the 41 canonical, adjoint and kinetic checks. Fresh extraction and byte-identical scientific replay are recorded in `receipts/m22_interactions/valentiner_quantum_validation.json``. The standalone PDF and cumulative monograph contain this full chapter; the release archive contains every tracked source, receipt and document.

References: Murai and Nakayama, arXiv:2407.16202v2, appendix A; PDG, *CKM Quark-Mixing Matrix* (2026), section 12.4. Source URLs are retained in the scientific receipt.

# Hermitian sources and exact one-loop mass-phase cancellation

## The unified canonical result

A real Hermitian scalar nonet supplies a useful interaction constraint: every scalar variation of the heavy-to-light mixing block remains Hermitian. With a scalar universal heavy mass and Higgs Yukawa, the complete renormalizable fermion action then has a positive tree determinant and an exactly vanishing one-loop determinant phase. The cancellation holds for arbitrary real neutral-scalar mixing, including Higgs/flavor portals. The same canonical action realizes six hierarchical light masses and four finite-scale charged-current targets. No field-dependent quark metric is used in that benchmark.

The mass and current construction uses explicit fitted real source-center coefficients. It establishes compatibility of hierarchy, weak CP and one-loop mass-phase cancellation in one action; it does not derive the golden CKM relation. The scientific receipt retains those coefficients, all 128 branches of a separate mass-only affine search, and twelve fixed-mass deformations. The latter identify exactly where predictive freedom survives the one-loop theorem.

## A low-degree invariant basis with an exact determinant

$$\det \mathcal{W} = \chi^3, \quad \chi = \text{Im Tr}(R^2 T^2 R T)$$

Let  $R = A A^\dagger / r_{\text{scale}}$  and  $T = L L^\dagger / t_{\text{scale}}$  in the up-family triplet space, with fixed positive normalization constants. Both transform by conjugation. The following nine Hermitian matrix words have real coefficients and obey the elementary generalized CP transformation without an inserted imaginary unit:  $I$ ,  $R$ ,  $R$ -squared,  $T$ ,  $RT+TR$ ,  $R$ -squared  $T+T$   $R$ -squared,  $T$ -squared,  $RT$ -squared+ $T$ -squared  $R$ , and  $RT$ -squared  $R$ . Their elementary scalar degrees are respectively 0,2,4,2,4,6,4,6,8.

Use coordinates consisting of three real diagonal entries and the real and imaginary parts of entries 12,13,23. At a nondegenerate  $R$  eigenbasis define  $\Delta = (r_2-r_1)(r_3-r_1)(r_3-r_2)$  and  $\tau = \text{Im}(T_{12} T_{23} T_{31})$ . The determinant of the nine-word coordinate matrix is  $\Delta$ -cubed times  $\tau$ -cubed. Equivalently, if  $\chi = \text{Im Tr}(R\text{-squared } T\text{-squared } RT)$ , then  $\chi = \Delta \tau$  and the coordinate determinant is  $\chi$ -cubed. With orthonormal Hilbert–Schmidt off-diagonal coordinates the determinant is  $8 \chi$ -cubed. Nonzero  $\chi$  therefore provides a complete real Hermitian source-center basis through scalar degree eight.

The symbolic certificate factorizes the diagonal Vandermonde determinant  $\Delta$ , two off-diagonal coefficient determinants  $-\Delta$  and  $+\Delta$ , and three real pair determinants  $-\tau$ ,  $+\tau$ ,  $\tau$ . The row-reordering sign yields the stated result. A generic symbolic Hermitian  $T$  also verifies  $\chi = \Delta \tau$ . An independent exact integer fixture has  $\Delta = 48$ ,  $\tau = 11$  and determinant 147197952. Generic multiprecision fixtures verify covariance, CP and the determinant identity.

The first eight displayed words have rank eight on the exact fixture; the last word lifts it to nine. This is a constructive result within the displayed matrix-word class, not a completeness claim for every finite-group covariant at lower degree. General covariant rank saturation and its implications for flavor predictivity are established results of Grinstein, Lu, Merlo and Quilez, arXiv:2312.13349. The present contribution is the explicit determinant certificate, low-degree real word basis and its application to this canonical interaction. A positive quark metric can likewise be completed as  $(I + \sum c_a Q_a)^2 + \eta I$  for  $\eta$  positive, with elementary scalar degree at most sixteen; such higher kinetic terms are allowed, rather than eliminated by the new construction.

## Complete renormalizable fermion interactions

Keep the certified product-family scalar representations and their existing phase charges. Ordinary quarks and one additional vectorlike weak-singlet quark triplet per sector transform as the up-family 3 of 3.A6. Introduce two independent shaping symmetries  $Z2_{up}$  and  $Z2_{down}$ . The vectorlike triplet of sector  $f$  has both chiralities odd under  $Z2_f$ , with ordinary quarks even. Introduce a real scalar nonet  $H_f$  in  $1+8$  of the up-family group, and a real scalar singlet  $\eta_f$ ; both are odd under  $Z2_f$  and neutral under the other shaping factor. The nonet is intrinsically Hermitian, with CP acting as  $H_f \rightarrow X H_f^* X^\dagger$ . It is not a complex nonet with a Hermitian vacuum imposed afterwards.

The five independent real fermion coefficients per sector multiply the ordinary Higgs Yukawa  $y_f$ , the vectorlike mass  $m_f$ ,  $\eta_f$  times the heavy-left/ordinary-right bilinear,  $\text{Tr}(H_f)/3$  times that bilinear, and the traceless  $H_f$  times that bilinear. Exact triplet endomorphism multiplicity one and adjoint-triplet contraction multiplicity one justify the singlet and adjoint contractions. The receipt enumerates the ten two-sector terms and their shaping charges. These are all nonzero in the benchmark:  $y_{up} = 1.2$ ,  $y_{down} = 0.57$ ,  $m_{up} = m_{down} = 1$ ,  $g = 0.1$ ,  $c_s = 0.7$  and  $c_a = 0.8$ .

The shaping charges forbid a Higgs coupling to a heavy right-handed triplet, a constant heavy-left/ordinary-right mixing, the opposite-sector nonet or  $\eta$  insertion, and a nonet or  $\eta$  insertion in the vectorlike mass. Existing elementary L,A,B,S representations supply no further single-insertion contraction on these endpoints. Bare canonical kinetic terms are universal within each irreducible triplet; an ordinary/heavy cross kinetic term is forbidden by the shaping charge. Positive constant normalization factors can be absorbed into the five real coefficients.

Write  $C_f = g \eta_f I + c_s \text{Tr}(H_f) I/3 + c_a (H_f - \text{Tr}(H_f) I/3)$ . It is Hermitian for every scalar configuration and every real choice of the five coefficients. In ordinary/heavy ordering the six-state Dirac matrix has blocks  $D_f = [[v y_f I, 0], [C_f, m_f I]]$ . Its determinant is  $(v y_f m_f)^3$ , positive on the chosen positive real branch and independent of the source vacuum. This is the Nelson-Barr triangular determinant mechanism, not a new principle of tree strong-CP protection.

## Exact finite spectra and physical currents

$$h_i^2 = \frac{(a^2 - \ell_i^2)(m^2 - \ell_i^2)}{\ell_i^2}, \quad M_i = \frac{am}{\ell_i}$$

Diagonalize  $C_f = U_f \text{diag}(h_i) U_f^\dagger$ . Applying  $U_f$  on both chiralities leaves three real two-state blocks  $[[a, 0], [h_i, m]]$ , where  $a = v y_f$ . For any chosen positive light mass  $\ell_i$  smaller than both  $a$  and  $m$ , the exact inverse relation is  $h_i^2 = (a^2 - \ell_i^2)(m^2 - \ell_i^2)/\ell_i^2$ . Either sign of  $h_i$  is allowed.

The heavy partner mass is a  $m/\ell_i$ , without a small-flavon or small-vev expansion.

The ordinary-doublet weight obeys  $q_i^2 = a^2 (m^2 - \ell_i^2)/(a^2 m^2 - \ell_i^4)$ . Consequently the finite light charged current is  $\text{diag}(q_u) U_u^\dagger U_d \text{diag}(q_d)$ . It need not be unitary. The calculation retains all six fermions per sector, the neutral currents and positive light-current row and column deficits. Real coefficient changes in the five-term fermion action preserve the source eigenframes while altering masses and doublet weights; this statement excludes mass-ordering crossings and changes of the scalar source centers.

Use  $v = 0.03$  and illustrative up Yukawas  $(10^{-5}, 0.003, 0.9)$ , down Yukawas  $(2 \times 10^{-5}, 0.0004, 0.02)$ . These are test inputs, not measured running quark masses. Four target current quantities are obtained from PDG 2026 central unitary parameters  $\sin(\theta_{12})=0.22517$ ,  $\sin(\theta_{13})=0.003763$ ,  $\sin(\theta_{23})=0.04189$  and  $\delta=1.154$ :  $|V_{us}| = 0.2251684057717295$ ,  $|V_{cb}| = 0.04188970341421036$ ,  $|V_{ub}| = 0.003763$  and  $J = 3.159371628927255 \times 10^{-5}$ . Their use is explicit input matching.

The source eigenframe angles compensate the exact doublet weights, so those four quantities match in the finite current, rather than only in a unitary decoupling approximation. The largest relative current residual is  $2.67 \times 10^{-15}$ . The up and down mass residuals are below  $1.95 \times 10^{-13}$  and  $6.89 \times 10^{-15}$ . The CP-conjugate source produces  $-J$  with zero recorded double-precision partner residual. All source targets are realized by the nine real polynomial coefficients at 90-digit working precision. The largest coefficient magnitudes are approximately  $8.08 \times 10^{10}$  and  $2.00 \times 10^{10}$ . These large declared inputs preclude treating the construction as a demonstrated natural or perturbative scalar UV completion.

## An exact one-loop phase theorem

$$G_{IK}G_{KI} \in \mathbb{R} \implies \text{Im Tr}(D^{-1}\delta D) = 0$$

In the source eigenbasis the left and right fermion rotations are real two-state matrices. Every physical neutral real-scalar mode has a Hermitian source variation  $A$ , plus a real flavor-diagonal Higgs vertex. A mass-basis source vertex connecting states  $(i, \sigma)$  and  $(j, \tau)$  therefore equals a real two-state factor times  $A_{ij}$ . For  $i$  different from  $j$ , its chirality-flipping paired product  $G_{IK}G_{KI}$  is a real factor times  $|A_{ij}|^2$ . Within a single flavor,  $A_{ii}$  and every Higgs contribution are real. Every paired product is thus real separately, including interference between the Higgs and nonet portions of an arbitrarily mixed mode.

The one-loop diagonal chirality-changing self-energy is a sum of these paired products times real masses and real loop weights. Its imaginary part vanishes term by term. Hence  $\text{Im Tr}(D^{-1}\delta D) = 0$  exactly for every real neutral-scalar mixing matrix and positive scalar spectrum, without a small spurion or a cancellation between sectors. The same argument cancels the UV phase coefficient. Generic tests with 91 independently mixed modes give only floating-point errors of order  $10^{-15}$ ; these errors are checks of the analytic identity, not a predicted residual  $\theta$ .

Neutral gauge phase factors use Hermitian left currents and real diagonal right currents. The full charged and neutral Goldstone Ward products reduce to real squared-current magnitudes. Their reality is checked with all six states per sector. Charged gauge loops contribute Hermitian wavefunction factors. Scalar tadpole shifts preserve Hermitian  $C$  and the identically real tree determinant. These observations close the first-order one-loop mass-phase cancellation for the specified complete renormalizable fermion action and canonical bare metrics.

This theorem concerns the first-order one-loop determinant phase. Higher fermion or field-dependent kinetic vertices and two-loop corrections are outside its assumptions. It is not an all-loop strong-CP solution. One-loop-vanishing Nelson–Barr extensions already exist in the literature; see Alves and Nishi, arXiv:2506.03257 / JHEP 09 (2025) 162. No priority claim is made for that general possibility. The concrete result here is the real-nonnet interaction, termwise proof with arbitrary scalar mixing, and exact finite hierarchy/current construction in the same canonical action.

## A locally positive 91-scalar branch

Start from the portal-shifted 71-scalar Higgs/flavor stationary branch at  $v = 0.03$ . The four Higgs norm portals are  $10^{-4}$ ,  $2 \times 10^{-4}$ ,  $3 \times 10^{-4}$  and  $4 \times 10^{-4}$ . Its verified minimum scalar mass squared is  $9.40405681 \times 10^{-7}$  and maximum scaled gradient is  $9.64 \times 10^{-12}$ . For each sector form  $F_f = \sum c_{fa} Q_a(R,T)$ , using the explicit fitted real coefficients, and add  $\text{Tr}(H_f - \eta_f F_f)^2 + (\eta_f^2 - 1)^2$  to the base scalar potential.

At  $\eta_f = 1$  and  $H_f = F_f$  all new squares vanish. The old stationary point and its effective base potential are retained. This interaction respects the shaping charges and CP. Its maximum elementary scalar degree is eighteen; it is a specified scalar EFT, not a renormalizable scalar UV completion. The two nonets add eighteen real components and the two eta fields add two, giving 91 physical neutral real scalars.

The full quadratic form is  $\delta x^T A_0 \delta x + \text{norm}(\delta H - J \delta x)^2$ , where  $x$  consists of the old 71 coordinates and the two eta coordinates,  $A_0$  is the direct sum of the verified old Hessian and  $8I$ , and  $J$  is the source-center Jacobian. This congruence proves positivity for every finite  $J$ . A conservative conditional eigenvalue lower bound is  $\min(\lambda_{\min}(A_0), 1)/(1 + \text{norm}(J))^2$ , evaluated as  $4.70 \times 10^{-31}$ . The tiny bound reflects the large fitted coefficients; it is not a numerically resolved 91-state eigenvalue. Local positivity is conditional on the verified base Hessian and does not establish global or quantum vacuum selection. The one-loop theorem itself holds for arbitrary scalar mixing and does not depend on this particular squared-lock potential.

## Fixed-mass freedom inside the protected action

Rotate the down source by each of six independent anti-Hermitian flavor generators with either sign of a  $10^{-4}$  displacement, and recalibrate its nine real center coefficients. All twelve branches keep the old base vacuum stationary, preserve local positivity by the same congruence, retain the five real fermion coefficients, and retain all six finite light masses within  $6.40 \times 10^{-14}$  relative error. The tree and one-loop determinant-phase identities remain valid on every branch.

The four-observable response for  $(|V_{us}|, |V_{cb}|, |V_{ub}|, J)$ , after fractional response and column normalization, has singular values approximately  $(2.0097, 1.1586, 0.5964, 0.5131)$ . Thus all four current directions remain independent at fixed masses even while one-loop strong CP is protected. This is a concrete separation of strong-CP protection from CKM predictivity. The source-center coefficients, and the allowed higher kinetic basis, must be constrained by additional justified interactions before an observable golden relation can follow.

## A complete mass-only affine candidate calculation

$$q^3 f(t)^2 - d^2 g(t)^3 = 0, \quad t = a_2/a_1$$

A stricter candidate retains only  $C_f = a_0 I + a_1 R + a_2 T$ . At the fixed source vacuum and chosen Higgs/vectorlike coefficients, the six mass inputs determine a finite set of source centers without any CKM input. Let  $R_0, T_0$  denote traceless  $R, T$ , let  $c_i$  be the centered signed target eigenvalues, and set  $q = \sum c_i^2$  and  $d = \prod c_i$ . Define  $g(t) = \text{Tr}(R_0^2) + 2t \text{Tr}(R_0 T_0) + t^2 \text{Tr}(T_0^2)$ , and  $f(t) = \det R_0 + t \text{Tr}(R_0^2 T_0) + t^2 \text{Tr}(R_0 T_0^2) + t^3 \det T_0$ . With  $t = a_2/a_1$  the exact spectral constraint is  $q^3 f(t)^2 - d^2 g(t)^3 = 0$ .

This sextic is computed over rational numbers from explicitly declared decimal input fixtures. Exact Sturm counts and rational isolating intervals of width at most  $10^{-35}$  certify every real root. The reconstruction chooses  $a_1 = \text{sign}(d/f) \sqrt{q/g}$ ,  $a_2 = t a_1$ , and adjusts  $a_0$  to the mean eigenvalue. Pure- $T$  branches at infinity are separately checked and absent on the fixture. Four inequivalent signed eigenvalue patterns are included; an overall source sign leaves masses and current magnitudes unchanged. This equivalence is observable, rather than an assertion that all signs lie on one vacuum group orbit.

There are eight up branches and sixteen down branches, yielding 128 branch pairs. Their full finite currents, unitary decoupling charts, CP quartets and root intervals are retained. Eighty-digit source reconstruction checks accompany every branch. None reproduces the nominated  $|V_{us}|$  near 0.225 on this mass/source fixture; branches with  $|V_{us}|$  between 0.1 and 0.4 instead lie near 0.1444–0.1611. This excludes this precisely specified affine candidate, not other source potentials or the full framework. The restriction to three source words is not enforced against the other allowed center or kinetic operators. Its completed exhaustive calculation supplies a controlled baseline for testing a future interaction constraint.

## Reproduction and scientific scope

Run `PYTHONPATH=python OPENBLAS_NUM_THREADS=1 python python/develop_valentiner_hermitian.py`, then `PYTHONPATH=python python python/tests/test_valentiner_hermitian.py`. The release has eighteen focused checks and fifty-six related quantum, canonical, adjoint and kinetic checks. Fresh replay reproduces the scientific receipt byte for byte. Exact polynomial certificates, approximate scalar-vacuum inputs and fitted coefficient inputs are kept distinct. The standalone paper and cumulative monograph include this chapter in full.

References: Grinstein, Lu, Merlo and Quilez, arXiv:2312.13349; Alves and Nishi, arXiv:2506.03257, DOI 10.1007/JHEP09(2025)162; PDG, CKM Quark-Mixing Matrix (2026), equation 12.28. Primary URLs are retained in the scientific receipt. The calculations establish the stated conditional results; publication priority and a predictive physical UV completion require independent assessment.

# The Sommerfeld constant and the exact angle carrier

## The constants and angles being compared

Sommerfeld's fine-structure constant is the electromagnetic coupling  $\alpha$ , not the strong-CP determinant phase or the unitarity-triangle angle also conventionally called  $\alpha$ . In SI units  $\alpha = e\text{-squared}/(4 \pi \epsilon_0 \hbar c)$ . The latest available CODATA adjustment is 2022:  $\alpha(0) = 0.0072973525643$  with standard uncertainty 0.000000000011, and  $\alpha(0)\text{-inverse} = 137.035999177$  with standard uncertainty 0.000000021. These are measured references, not inputs used to calibrate the new phase-orbit calculation. NIST's current constants table and the PDG 2026 electroweak review retain these values. The coupling at nonzero momentum is scale dependent; a proposed exact formula must specify the physical limit, renormalization convention and threshold matching.

The nominated scalar angle  $\theta = 66$  degrees is  $11 \pi/30$  radians. It is distinct from the standard CKM phase  $\delta$ , the unitarity-triangle angles, and the strong-CP phase  $\theta\text{-bar}$ . The new Hermitian action protects  $\theta\text{-bar}$  through first-order one-loop order while allowing weak CP. That protection does not identify its scalar source frame with an independently locked 66-degree field. An equation connecting those fields to quark matrices is still necessary. Comparing decimal constants or using the degree number 66 inside a coupling formula would not supply that equation; physically meaningful angle formulas must survive changing angle units.

## The exact 66-degree algebraic relation

$$\theta = 11\pi/30, \quad 1 - 2\cos(12\theta) = (3 - \sqrt{5})/2$$

For  $z = \exp(11 \pi i/30)$ ,  $z$  has primitive order sixty. Its cyclotomic polynomial is  $\Phi_{60}(z) = z\text{-to-the-sixteenth} + z\text{-to-the-fourteenth} - z\text{-to-the-tenth} - z\text{-to-the-eighth} - z\text{-to-the-sixth} + z\text{-squared} + 1$ . Set  $x = z + z\text{-inverse} = 2 \cos(\theta)$ . The exact real minimal polynomial is  $P(x) = x\text{-to-the-eighth} - 7 x\text{-to-the-sixth} + 14 x\text{-to-the-fourth} - 8 x\text{-squared} + 1$ , with  $\Phi_{60}(z) = z\text{-to-the-eighth} P(z + z\text{-inverse})$ . The receipt verifies this polynomial identity symbolically and independently reuses the rational quotient-algebra carrier.

The twelfth power maps the selected root to  $\exp(2 \pi i/5)$ , so  $C = 1 - z\text{-to-the-twelfth} - z\text{-to-the-minus-twelfth} = 1 - 2 \cos(12 \theta) = (3 - \sqrt{5})/2 = \phi\text{-to-the-minus-two}$ . Its exact minimal polynomial is  $C\text{-squared} - 3C + 1 = 0$ . This is a genuine algebraic connection between the nominated angle and golden coefficient. It contains no electromagnetic coupling.

Moreover the positive-half-circle primitive-sixtieth-root angles yielding the smaller coefficient are exactly 6, 66, 114 and 174 degrees. All have negative partners with the same CP-even coefficient. The coefficient therefore loses the embedding information needed to distinguish 66 degrees. Our previously certified six-harmonic potential does select plus/minus 66 degrees among competitors, but its selected coefficient relations were constructed with that target known. A successful joint mechanism must justify the energy selector and transmit its selected phase to normalized quark matrices, not only repeat this quotient identity.

## A photon theorem for the protected canonical action

$$\sum_{i=1}^6 \log(\mu/m_i) = 6\log\mu - 3\log(vym)$$

Each complete up sector contains six Dirac states of electric charge  $+2/3$ , and each down sector six of charge  $-1/3$ . The ordinary doublet and vectorlike singlet components have the same electric charge within a sector. Consequently both chiral photon generators are  $Q_f$  times the six-dimensional identity. Independent left and right fermion diagonalization leaves each generator unchanged. The one-loop fermion photon vacuum polarization therefore depends only on the six singular masses; it cannot see their mixing eigenframes.

At spacelike momentum magnitude  $Q$  define the positive inverse-alpha screening contribution  $S_f(Q) = (2 N_c Q_f^2/\pi) \sum_i \int_0^1 dx x(1-x) \log[1+Q^2 x(1-x)/m_i^2]$ , with  $N_c = 3$ . Then  $\alpha_{\text{eff}}(Q)^{-1} = \alpha(0)^{-1} - \sum_f S_f(Q)$  at the stated one-loop accuracy, omitting other particle contributions. This expression is a perturbative free-fermion contribution in declared model units. It does not replace physical hadronic vacuum polarization, lepton and charged-gauge contributions, or electroweak matching required to relate the model to measured  $\alpha(0)$ . The numerical integral independently passes the small- $Q$  normalization  $S_f = N_c Q_f^2 Q^2/(15 \pi m^2)$  for one mass, as well as scale covariance and zero-momentum checks.

An even stronger logarithmic identity follows from the triangular determinant. For  $D_f = [[v y_f I, 0], [C_f, m_f I]]$ , the product of all six positive singular masses is  $|\det D_f| = (v y_f m_f)^6$ , independent of  $C_f$ . Hence  $\sum_i \log(\mu/m_i) = 6 \log(\mu) - 3 \log(v y_f m_f)$ . The complete one-loop logarithmic threshold,  $(2 N_c Q_f^2/(3 \pi))$  times this sum in the declared integration convention, is source-independent even when  $C_f$  changes its eigenvalues. This exact statement requires retaining both light and heavy partners. Heavy-only thresholds can depend on the light masses; their dependence cancels against the light-partner contribution in the all-state logarithmic sum. The all-state logarithmic result does not imply that a finite-momentum polarization is independent of varying masses.

## A continuous weak-phase orbit at fixed electromagnetic screening

Retain the preceding canonical action, illustrative six light masses,  $v = 0.03$ ,  $y_{\text{up}} = 1.2$ ,  $y_{\text{down}} = 0.57$  and universal vectorlike masses one. Fix  $C_{\text{up}}$  diagonal and construct  $C_{\text{down}} = U(\delta) \text{diag}(h_i) U(\delta)^\dagger$ , with the exact finite-mass inverse  $h_i^2 = (a^2 - \ell_i^2)(m^2 - \ell_i^2)/\ell_i^2$ .  $U(\delta)$  is a standard unitary three-angle chart with fixed  $\sin(\theta_{12})=0.22517$ ,  $\sin(\theta_{13})=0.003763$  and  $\sin(\theta_{23})=0.04189$ . Its  $\delta$  is varied continuously, not predicted. This is a counterexample family within the allowed source-center coefficient space.

At every point all six light masses, all six heavy partner masses, and all doublet weights remain fixed. The exact source-center covariant determinant permits each new Hermitian center to be reconstructed by nine real polynomial coefficients at the retained scalar vacuum. The squared-lock scalar EFT then keeps that vacuum stationary and locally positive by the previous congruence proof. The tree and one-loop strong-CP mass-phase theorem remains applicable. Thus the variation changes justified scalar-center inputs without changing the complete five-coefficient fermion action or introducing a new quark metric.

The receipt samples  $\delta = 0, 6, 42, 60, 66, 78, 90, 114, 138, 174, 180$  and  $-66$  degrees. It records all center coefficients, recovered unitary charts after removing exact finite doublet weights, finite-current CP quartets, complete spectra, and photon integrals at  $Q = 10^{-5}, 0.01, 1$  and  $100$  in model units. The largest relative mass change is  $3.78 \times 10^{-15}$ . The largest screening change is  $8.89 \times 10^{-16}$  in inverse-alpha units, and the maximum all-state logarithmic identity error is  $4.45 \times 10^{-16}$ . The analytic fixed-spectrum identity, rather than those numerical residuals, establishes invariance for the whole continuous orbit. Weak CP changes along that orbit, including its sign at the conjugate branch.

## The independent gauge boundary and the mechanism requirement

$$\mathcal{L}_Y = -\frac{k_0}{4} F_{\mu\nu} F^{\mu\nu}, \quad \alpha^{-1} = 4\pi k_0$$

In the unit-charge covariant-derivative normalization the gauge action contains  $-k_0 F_{\mu\nu} F^{\mu\nu}/4$ , where  $k_0 = 1/e^2$  and  $\alpha^{-1} = 4\pi k_0$ . This operator is invariant under every declared finite flavor group, shaping charge and CP transformation. Its positive coefficient is independent of the scalar source-center coefficients. Equivalently, above electroweak breaking,  $\alpha^{-1} = 4\pi (g_Y^{-2} + g_2^{-2})$  at tree matching. The finite flavor symmetry does not normalize either continuous gauge coupling.

A permitted scalar-dependent gauge kinetic interaction can create a dependence on scalar invariants, but it retains the independent constant  $k_0$ . Specifying its coefficient to make  $137.035999177$  emerge would match the observed coupling, not predict it. Conversely inserting  $\alpha$  into the source potential or fixing a source angle to  $66$  degrees would initialize the desired correlation. The new result precisely rules out using equal-charge one-loop mass thresholds as the missing frame-selection mechanism: complete photon screening remains unchanged along an allowed continuous weak-phase orbit, and its all-state logarithmic threshold is even independent of source eigenvalues.

A substantive extension must supply a justified UV gauge normalization or boundary condition, an interaction that fixes the source-center freedom, an embedding selector for the primitive-root angle, and their physical matching. A common scalar modulus alone supplies none of these coefficient relations automatically. Higher-loop weak/flavor corrections can carry mixing dependence and are not excluded by the one-loop theorem. They also require specified gauge and scalar inputs before they can predict a number. The current action therefore provides an exact golden-angle carrier and exact electromagnetic matching sum rules, not a derivation of the Sommerfeld constant from that angle.

## Reproduction and references

Run `PYTHONPATH=python OPENBLAS_NUM_THREADS=1 python python/develop_flavor_electromagnetic.py` and `PYTHONPATH=python python python/tests/test_flavor_electromagnetic.py`. All six new and eighteen related checks pass; fresh replay reproduces the receipt byte for byte. Both PDFs retain the calculation in full.

Primary references: NIST, 2022 CODATA constants table; PDG, Electroweak Model and Constraints on New Physics (2026), equations 10.19–10.20. Their primary URLs are retained in the receipt. The determinant and source-frame statements follow directly from the specified action. Standard QED screening and gauge-normalization freedom are established physics.

# Complete nonet interactions and invariant CKM geometry

## Result and declared theory

A complete renormalizable scalar basis, an isolated CP-breaking vacuum and a fixed-spectrum response calculation now belong to one specified minimal theory. Exact character arithmetic gives 60 independent real scalar coefficients before the Higgs, of which 28 are mixed quartics. The neutral Higgs adds two self coefficients and eight portals, giving 70 coefficients and 21 physical neutral real scalar coordinates. Numerical minimization produces nondegenerate source spectra and physical weak CP without supplying a nominated angle or CKM magnitude. The Hermitian fermion action retains its termwise first-order one-loop mass-phase cancellation for the calculated scalar eigenmodes.

The complete allowed coupling space does not predict the golden coefficient or the 66-degree standard phase. Four locally independent mixing directions survive even when all six light quark masses are held fixed. This is a constructive local statement on the delivered CP-breaking branch, not a classification of every branch. The repository's polynomial engine supplies an exact, rephasing-invariant test of the desired observable locus and derives a conditional elliptic slice of it. The interaction and the target locus are calculated separately so that algebraic target structure cannot be mistaken for dynamical prediction.

The fields are new minimal real Hermitian nonets, not the former complex source multiplets or the fitted degree-18 locking theory. Each sector contains  $\eta$ , the nonet trace coordinate  $s$  and a traceless Hermitian matrix  $A$ . Both singlets and all eight real components of  $A$  are odd under that sector's  $Z_2$ ; the other sector's  $Z_2$  acts trivially. Both sectors share the diagonal family group  $3.A_6$  and its previously certified generalized CP. Ordinary quarks are neutral triplets; vectorlike triplets carry their sector's odd charge. The Higgs is a family singlet and even under both charges. All kinetic metrics in this action are canonical bare metrics.

## Complete scalar census

Write  $A$  as a sum of eight orthonormal Hermitian basis matrices with real coordinates  $a$ . Let  $N$  be  $a \cdot a$ . The four quadratic invariant functions in either sector are  $\eta^2$ ,  $s^2$ ,  $\eta a$  and  $N$ . The quadratic adjoint covariants are  $\eta a$ ,  $s a$  and the traceless part of  $A^2$ . The symmetric-square adjoint vector  $w$  has 36 orthonormal coordinates. Its finite-group projectors have ranks 1, 8, 8, 9 and 10 after imposing CP. The rank-ten channel combines the inequivalent five-dimensional channels exchanged by CP. The two eight-dimensional channels are inequivalent representations, rather than an arbitrary multiplicity-space choice.

Exact character averages over all 1,080 elements are evaluated in  $Q(\sqrt{5}, \omega)$ . Symmetric square of the real representation  $8+1+1$  has four invariant singlets; its symmetric fourth power has twelve. The square of its symmetric-square character gives 29 mixed quartic singlets before CP. The difference of the two five-dimensional contractions is CP odd, leaving 28 even contractions. An independent numerical evaluation matrix for the entire displayed operator basis has rank 60. Numerical projector tensors are explicitly distinguished from the exact character dimensions.

The twelve self quartics per sector comprise the five degree-four monomials in  $\eta$  and  $s$ , the three degree-two singlet monomials times  $N$ ,  $\eta \text{Tr}(A^3)$ ,  $s \text{Tr}(A^3)$ ,  $N^2$ , and the self contraction in the extra finite eight-dimensional projector. The finite nine-dimensional self contraction equals  $9 N^2$  divided by 40, so it cannot replace the independent extra-eight contraction. That polynomial dependence is important when translating the representation decomposition into a scalar basis.

The mixed quartics comprise sixteen products of quadratic singlets, nine dot products of quadratic adjoint covariants and three additional cross-projector contractions in the extra eight, nine and CP-paired five channels. The singlet projector and ordinary  $SU(3)$  adjoint projector are already included in the first two sets. The singlet and adjoint multiplicities therefore matter: five adjoint-only channels do not constitute the complete nonet potential. Independent sector  $Z_2$  charges forbid mixed quadratic terms and all odd-degree scalar interactions. Eight Higgs portals multiply the four quadratic singlets in each sector by  $H^\dagger H$ .

## A bounded vacuum without flavor targets

The executable search uses a declared seeded family of potentials. Six unit radial quartics dominate; the sum of absolute values of all remaining quartic coefficients is 0.12. Every displayed normalized quartic is bounded in absolute value by the fourth power of the total nonet-coordinate norm. Consequently the quartic part is bounded below by  $7/150$  times that norm to the fourth power. All sixty coefficients are nonzero. The Higgs quartic is 0.13, all eight portals are nonzero, and their two singlet forms and adjoint coefficients are positive. In canonical neutral coordinates the joint quartic lower bound is 0.01625 times the fourth power of the combined norm. These bounds establish boundedness and existence of a global minimum; they do not identify the selected local branch as that global minimum.

The scan stops at the first locally positive, nondegenerate branch with absolute  $J$  above 10 to the minus five. This criterion requests spontaneous physical CP breaking, without comparing any result to a flavor target. The selected seed is 20261010, start zero. All seventeen minimizations, complete coefficients and canonical coordinates are retained in the receipt. A root refinement solves all stationary equations after adding Higgs portals. The neutral scalar Hessian is positive, including its shallow angular direction: its smallest eigenvalue is  $1.1262305e-05$ , the stationary gradient norm is below 10 to the minus twelve, and  $J=0.0002622534621$ . The exact generalized-CP transform gives an equal-energy partner with opposite  $J$  and the same masses and mixing magnitudes. CP invariance alone selects neither sign.

All quoted energies, coordinates, Hessian eigenvalues and vacuum responses are numerical. Exact character arithmetic, exact polynomial identities and the global analytic quartic bound have different guarantees. The retained search history is a reproducible existence calculation, not an exhaustive vacuum census.

## Physical quark matching and mass-phase protection

Each sector has five real renormalizable fermion coefficients. Its Hermitian heavy-to-light source is  $C=g \eta I + c s I + c a A$ . With  $a=v y$  and positive vectorlike mass  $m$ , the six-state Dirac matrix has blocks  $a I$ , zero,  $C$  and  $m I$ . Its determinant is  $(a m)^3$ , real and positive. CP sets the bare QCD theta parameter to zero; the spontaneous weak phase does not change this tree determinant.

In the eigenframe of  $C$  each family is a real two-state singular-value problem. This calculates the three light and three heavy masses, their ordering and the doublet weights without replacing

the finite theory by its asymptotic limit. The light charged-current matrix is  $\text{diag}(q_u) U \text{diag}(q_d)$ , where  $U$  is the unitary source-frame overlap. Dividing by the calculated weights recovers  $U$ . The invariant CKM polynomial identities below apply to this unitary frame; they must not be applied directly to a nonunitary finite light current. The illustrative vacuum has six nonzero, nondegenerate light masses and nonzero  $J$ ; it does not reproduce the measured hierarchy or CKM magnitudes.

For every neutral real scalar direction the variation of  $C$  is Hermitian. Diagonalizing the complete 21-coordinate scalar Hessian gives real orthogonal mixtures of both nonets, both eta fields and the Higgs. The paired chirality-flip products remain real for each eigenmode, giving zero imaginary first-order one-loop mass trace term by term. The numerical reality error at this branch is below 10 to the minus twelve. This verifies the existing algebraic theorem in the complete minimal scalar action. It does not establish two-loop protection, eliminate higher-dimensional fermion operators or derive the measured flavor parameters.

## Fixed-mass observable freedom

The implicit vacuum derivative for a scalar coefficient is minus the inverse Hessian times that operator's gradient. All 28 mixed coefficients are tested. To hold six signed source eigenvalues fixed, six independent allowed quadratic coefficients compensate the mixed perturbation. The six-by-eight quadratic compensator has full row rank. Holding the signed eigenvalues and Higgs expectation value fixed preserves all six light masses, all six heavy masses and the current weights in the exact finite block formula.

The four observables are the squared magnitudes of  $U_{us}$ ,  $U_{cb}$ ,  $U_{ub}$  and  $U_{td}$ . The fixed-spectrum response has four nonzero singular values: 133.153996, 20.7912106, 1.14673727, 0.00111955388. Four selected independent mixed directions are then varied in both signs and all stationary and eigenvalue constraints are solved again. These finite solutions preserve the complete masses to numerical precision, retain positive scalar curvature and agree with the implicit response. Their  $J$  values and golden residuals change. The maximum complete-mass change is below 10 to the minus fourteen. The independently computed four-column finite response has smallest singular value 0.00057542912, exceeding its finite/implicit spectral discrepancy 0.00011204845. The smallest response direction is much weaker than the others, so finite-difference and Hessian accuracy are retained in the validation rather than inferred from a rank threshold alone.

This calculation identifies actual permitted interactions that move physical flavor at fixed mass. It is stronger than varying an unconstrained source matrix or changing a phase convention. The present symmetries provide a complete action with spontaneous CP, but they leave the golden and 66-degree relations adjustable. Additional restrictions would have to constrain this displayed coupling space; merely reproducing a target stationary root would not remove these directions.

## Observable polynomial constraints

Set  $a=|U_{us}|^2$ ,  $b=|U_{cb}|^2$ ,  $c=|U_{ub}|^2$ ,  $d=|U_{td}|^2$  and  $t=1-c$ . Define  $K=abc(t-a)(t-b)$  and  $N=ab+c(t-a)(t-b)-d t^2$ . Then the exact unitary identity is

$$4t^2J^2 + N^2 - 4K = 0.$$

The standard CKM phase satisfies  $\cos(\delta)=N$  divided by twice the square root of  $K$ . This extraction uses magnitudes and  $J$ , without assigning unphysical quark phases. The scalar

vacuum angle, standard delta, unitarity-triangle gamma and strong theta are distinct quantities.

For the nominated depth  $C = \sqrt{c \text{ times } t} / \sqrt{ab}$ , put  $X = C$  squared. Golden depth  $C = \phi$  to the minus two obeys  $X$  squared minus seven  $X$  plus one equals zero. The observable necessary equation is

$$c^2 t^4 - 7abct^2 + a^2 b^2 = 0.$$

The small-root branch requires  $0 < ct \text{ squared} < ab$ . For  $\delta = 66$  degrees,  $x = 2 \cos(\delta)$  has minimal polynomial  $x$  to the eighth minus seven  $x$  to the sixth plus fourteen  $x$  to the fourth minus eight  $x$  squared plus one. Eliminating  $x$  gives

$$N^8 - 7N^6 K + 14N^4 K^2 - 8N^2 K^3 + K^4 = 0.$$

Branch conditions are  $K > 0$ ,  $N > 0$ ,  $J > 0$  and  $16K/25 < N \text{ squared} < 1681K/2500$ . Exact rational root isolation proves there is one phase root in the corresponding  $x$  interval  $[4/5, 41/50]$ . The receipt and tests reject the selected vacuum as a golden/66-degree prediction even though it satisfies the general unitary CP identity. Denominator-cleared equations without these branch conditions would admit unwanted roots.

## The polynomial machinery and the elliptic slice

Fix  $c$ ,  $X$  and a nontrivial standard phase independently. Write  $x = a/t$  and  $k = ct \text{ squared}/X$ . The remaining magnitude  $b$  is  $k/a$ , and the exact  $J$  relation becomes

$$(x/J)^2 = k c \sin^2 \delta x(1-x)(x-c/X).$$

For  $\lambda = c/X$  different from zero and one, the cubic has three distinct roots. This is a quadratic twist of the Legendre genus-one curve with  $j$ -invariant  $256(1-\lambda + \lambda^2)$  cubed divided by  $\lambda^2$  times  $(1-\lambda)^2$ . For  $0 < \lambda < 1$  the physical magnitude interval is  $\lambda < x < 1$ . The actual repository PolynomialCurve engine derives a two-dimensional de Rham connection from the cubic, and exact symbolic arithmetic checks the discriminant and coordinate substitution.

The fixed  $c$  slice is a conditional target locus, not a symmetry-derived selected vacuum. Its connection transports mathematical periods as  $\lambda$  changes; it does not select a quark frame or a gauge coupling. Rational slice examples are declared separately from the algebraic golden value. The machinery is useful for exact elimination, branch isolation and constraint geometry, while the allowed-interaction response supplies the physical prediction test.

There is also an exact restricted obstruction. At fixed nondegenerate spectra, four mixed traces  $\text{Tr}(RT)$ ,  $\text{Tr}(RT \text{ squared})$ ,  $\text{Tr}(R \text{ squared } T)$  and  $\text{Tr}(R \text{ squared } T \text{ squared})$  are independent magnitude coordinates, with Jacobian determinant  $\Delta(R)^2$  times  $\Delta(T)^2$ . Ordinary continuous-SU(3) trace quartics reduce to an affine function of those moments plus one squared moment. Their Hessian has rank at most one, so they cannot supply an isolated interior CP-breaking minimum on that four-dimensional slice. The finite-group quartics used above lie outside this restricted class. This distinguishes a necessary finite interaction from another unconstrained trace potential.

## Reproduction and verification

The scientific receipt is `receipts/m22_interactions/valentiner_nonet_joint.json`. The scalar module is `python/perfectpower/nonet_potential.py`; the target geometry module is `python/perfectpower/ckm_constraint_geometry.py`. Run `PYTHONPATH=python OPENBLAS_NUM_THREADS=1 python python/develop_valentiner_nonet_joint.py` to rebuild it. Focused tests cover exact counts, operator independence, analytic gradients, group and CP covariance, boundedness, the Higgs vacuum, finite currents, scalar-mode mass-phase reality, fixed-spectrum response and exact target/curve identities. Numerical replay is checked from a fresh archive extraction, with its environment and tolerances recorded in the validation receipt.

# Soft mediation, scalar-loop closure and protected flavor

## Central results

The source coefficients now have a constructive renormalizable mediator realization, and their scalar-loop closure is executable. Three finite-group mediator multiplets add 27 real coordinates to the existing 21-coordinate Higgs/nonet action. The resulting 48-scalar vacuum has a positive full Hessian, exactly the same source-frame CKM observables and finite quark masses, and the existing termwise first-order one-loop mass-phase protection. Their representations forbid new renormalizable quark bilinears, even when every permitted scalar interaction is admitted. This is a structural protection result for a class of scalar completions, rather than another fitted angular potential.

Exact character and generalized-CP coset arithmetic gives 251 permitted scalar coefficients on this field content. The constructed Gaussian benchmark chooses many of these coefficients to vanish; it does not suppress them by miscounting the operator space. Power counting identifies a radiatively closed soft-mediator restriction on the hard mediator couplings. Finite matching and low-energy running still generate calculable interactions. Six independent current coefficients move all four mixing magnitudes at fixed complete quark spectra, so the mediator realization does not enforce golden CKM depth or a 66-degree standard phase.

The scalar-loop compiler reconstructs the commutative, nonassociative one-loop multiplication of all 52 source quartics, with independent off-grid polynomial checks. Two radial quartics and just one finite-eight cross interaction generate the complete nine-operator adjoint-pair subalgebra. Combining the three finite channels into an SU(3) 27-plet instead gives a five-operator closed subalgebra and an enforced equality of their coefficients. That continuous-symmetry quartic limit cannot isolate an interior CP-breaking relative frame at fixed spectra. These results locate a concrete interaction boundary between protected coefficient relations and unconstrained flavor.

## Tree matching and the minimum mediator content

The kinetic action assigns one half the squared derivative to each  $\eta$ , each  $s$  and every orthonormal adjoint coordinate, together with canonical mediator coordinates and the usual Higgs kinetic term. This fixes the independent singlet normalization;  $H=s I+A$  is the nonet coordinate convention. Source kinetic metrics and their flavor couplings are declared inputs. The scalar-only one-loop quartic calculation uses this same canonical metric.

Let  $q$  contain the four sector-quadratic singlets  $\eta^2$ ,  $s^2$ ,  $\eta s$  and the adjoint norm. Let  $b$  contain the three quadratic adjoint covariants  $\eta A$ ,  $s A$  and the traceless part of  $A^2$ . The remaining quadratic currents are the finite eight-prime, nine and CP-paired five projections of the symmetric-square adjoint vector  $w$ . The five types of current exhaust the possible linear mediator couplings to two source fields.

For a positive Gaussian mediator mass, eliminating the mediator gives minus one half the squared source current divided by mass squared. A shared singlet has current  $g_u \cdot q_u$  plus  $g_d \cdot q_d$ . Its four-by-four mixed coefficient matrix is minus  $g_u g_d$ -transpose divided by mass squared, so all two-by-two minors vanish. A shared ordinary adjoint gives the same rank-one

structure for its three-by-three multiplicity matrix. A unique representation contraction leaves four singlet-current labels or three adjoint-current labels with independent real couplings; uniqueness does not select those multiplicity vectors.

One copy of each channel requires 36 real mediator coordinates and twenty cubic current coefficients. The mixed coefficient matrices have dimensions seven and five on their rank-one varieties; the three remaining scalar channels add three. Thus this tree restriction reduces the 28-dimensional mixed space to a generically fifteen-dimensional variety, of codimension thirteen. This dimension count is an exact algebraic statement on the specified Gaussian matching family. Direct source contacts and additional mediator interactions remain symmetry allowed.

Conversely, an arbitrary mixed matrix of rank  $r$  needs at least  $r$  copies of its mediator irrep. A singular-value decomposition constructs a positive-mass realization attaining this bound, including either sign of a cross coupling. Generic full-rank singlet and ordinary-adjoint matrices require four singlets and three ordinary adjoints. Together with one copy of each finite channel, the generic minimum is ten multiplets, or 55 real mediator coordinates. This universal construction reproduces all 28 mixed coefficients; it does not constrain them.

## A stable 48-scalar completion with protected fermion vertices

Use only the finite eight-prime, nine and CP-paired five mediators. Their dimensions are eight, nine and ten. For a source current  $J$  and a positive mass  $M$ , write the UV scalar potential as the target source potential plus  $M$  squared times the norm of  $S+J/M$  squared, divided by two. Expanding gives a source contact correction, a cubic  $S \cdot J$  and a positive Gaussian mediator mass. The contact correction cancels precisely the three mixed finite-source quartics, transferring them to mediator exchange. The other source operators remain explicit inputs.

The mediator stationary value is minus  $J/M$  squared. At that value the source vacuum and all its observables are exactly those of the original source potential. The full Hessian has source block  $H+D$ -transpose  $D/M$  squared, mixed block  $D$  and mediator block  $M$  squared  $I$ , where  $D$  is the current Jacobian. Its Schur complement is exactly  $H$ . Consequently every positive source Hessian has a positive full completion. The source potential's global lower bound and the added nonnegative square also establish boundedness; this does not certify the chosen local source minimum as the global vacuum.

The benchmark has 48 physical neutral real scalar coordinates and smallest Hessian eigenvalue approximately 1.126 times ten to the minus five. It retains  $J$  approximately 0.00026225346 and the same six light and six heavy quark masses. The current factors in the two sectors have different specified ratios, avoiding an accidental rank loss in a balanced factorization. Their values are declared inputs, without using a target angle or magnitude.

The quark triplet bilinear decomposes into a singlet and the ordinary adjoint eight. The finite eight-prime, nine and five-pair cannot appear in a renormalizable heavy-mass or heavy-to-light quark vertex. Thus the complete allowed renormalizable fermion action is unchanged by these 27 new scalar coordinates. All scalar mass eigenmodes may mix arbitrarily, but each quark source variation remains Hermitian. The paired mass-basis vertices are real term by term, preserving the first-order one-loop mass-phase result. This representation argument applies beyond the Gaussian potential to arbitrary allowed scalar coefficients, provided the vacuum and spectrum are defined. It does not extend the theorem to two loops or higher-dimensional fermion operators.

## Complete scalar operator census and soft closure

The mediator representation  $W$  is Symmetric square of eight minus the singlet and ordinary eight, of real dimension 27. Before CP the dimensions of  $W$  quadratic, cubic and quartic invariant spaces are 4, 17 and 99. Generalized CP reduces them to 3, 11 and 58. For each source sector the source-squared  $W$  and source-squared  $W$ -squared dimensions are 4 and 77 before CP, and 3 and 50 after CP. With the previously complete source basis and Higgs portals, the total is 251 real scalar coefficients.  $W$  has no singlet, so no  $W$  tadpole or Higgs-squared linear- $W$  coupling is available.

Every count, including the CP reduction, is computed exactly in  $Q(\sqrt{5})$ ,  $\omega$ . The method averages over the 1,080 family elements and over their antiunitary CP coset. On the Hermitian adjoint, an odd power of an antiunitary-induced real map has trace  $\text{Tr}(T$  to that power) minus one, while an even power has trace equal to the squared modulus of  $\text{Tr}(T$  to half that power), minus one. Here  $T$  is the unitary product  $U U^\dagger$ , with  $U=g X$ . The exact CP matrix's common normalization has squared value  $(3+\sqrt{5})/128$ , permitting the entire coset calculation without numerical trace rounding.

The Gaussian benchmark sets mediator self cubics, self quartics and quadratic-mediator/source-quadratic quartics to zero. They are allowed by the finite symmetry. Nevertheless, if  $W$  enters only through its positive mass and dimension-one source-current couplings, local ultraviolet power counting prevents divergent hard  $W$  quartic couplings from appearing. Every external  $W$  requires at least one dimension-one coupling; a divergent counterterm of dimension  $d$  has coefficient dimension  $4-d$  and contains no inverse mass factors. Pure  $W$  cubics, pure  $W$  quartics and  $W$ -squared/source-squared quartics cannot meet that condition. This restriction is a perturbative soft-current property, not an assertion that finite effective interactions vanish.

An exact two-source fixture checks the distinction directly. Its separated source quartics plus a Gaussian singlet current have no hard mediator quartic or source cross-quartic one-loop counterterm. After eliminating the mediator, the low-energy exchange quartic does run into additional light interactions. A complete UV prediction must include its matching and running; failure of a tree coefficient identity in the EFT does not by itself destroy UV predictivity. Source contact quartics still require their complete allowed counterterms, and additional hard mediator couplings would remove the soft restriction.

## Actual UV coupling response at fixed masses

The previous calculation varied effective mixed coefficients. This calculation varies the six actual finite-mediator cubic current coefficients while holding the UV source-contact quartics fixed. Each change induces correlated self and cross coefficients in the source EFT. Six independent quadratic coefficients compensate the signed source eigenvalues, preserving the complete finite quark masses and current weights.

The resulting four-by-six observable response has four nonzero singular values. Four independent current directions are varied in both signs, the source stationary and mass constraints are solved again, and the full 48-coordinate Hessian is reconstructed for each mediator action. Every selected solution remains locally positive and preserves the complete quark masses to numerical precision. Finite and implicit responses agree, with the discrepancy norm below the weakest finite singular value. The weakest direction is much smaller than the other three; the receipt retains that numerical separation explicitly.

The algebraic reason is also visible. For the finite-eight and five-pair currents, the two self-anisotropy coefficients and two cross coefficients have parameter Jacobian determinant minus  $(a-d-b-c)(a+d+b+c)$ . It is generically nonzero. The nine-dimensional channel supplies another independent cross coefficient. The remaining radial self changes can be compensated by allowed radial contacts. Representation safety protects the fermion structure, while those current coefficients still adjust the frame. The Gaussian mechanism therefore does not force the nominated golden and phase constraints.

## The ordinary adjoint changes the fermion boundary

The 76-scalar universal completion also includes singlet and ordinary-adjoint mediators. Those representations allow new heavy-mass Yukawa contractions. Setting them to zero reproduces the old protected action, but the old scalar-field charges do not forbid them. The termwise theorem cannot be transferred to the enlarged complete fermion action without checking those vertices.

An exact two-family block example has Hermitian source and heavy-mass vertex matrices, yet its heavy-state paired product is  $21/25$  minus  $3i/25$ . The imaginary part is nonzero despite both individual matrices being Hermitian. At the actual 76-scalar completed vacuum, an equivariant ordinary-adjoint heavy-mass coupling with real coefficient 0.05 gives a neutral-scalar first-order mass phase approximately minus 7.41 times ten to the minus eleven. The complete scalar Hessian and finite Dirac spectrum are used. This is an explicit neutral-scalar threshold, not an all-diagram strong-CP matching calculation or an experimental limit claim.

Thus finite eight-prime, nine and five-pair mediators retain the stated renormalizable protection automatically, while ordinary adjoints introduce a new allowed interaction that the previous theorem did not cover. Singlet heavy-mass couplings require their own analysis; this example does not assert that every additional allowed coupling produces a phase.

## Scalar-loop algebra and the enforced SU(3) relation

For canonical real scalars the scalar-only quartic beta function, multiplied by sixteen pi squared, is one half the trace of the squared potential Hessian. Polarizing this contraction defines a commutative, generally nonassociative product on quartic operators. The compiler reconstructs the 52-dimensional source algebra from polynomial Hessians and validates the identities at independent points. Its floating-point reconstruction agrees with rational coefficient candidates to better than ten to the minus ten; the rational table is a numerically validated representation, not an exact symbolic finite-group tensor proof. Single-field and eight-field radial coefficients independently fix the normalization.

Two separate radial quartics form a two-dimensional closed sector. Adding only the finite-eight cross term generates dimensions 3, 4, 8 and finally 9. The closure contains both self-anisotropies, the radial cross product, the ordinary adjoint quadratic-current product and all three finite cross channels. Starting with all three finite channels gives the same closure. In particular, omitting the nine and five-pair cross operators is not a scalar-loop invariant truncation of this finite-eight interaction.

The three finite projectors sum to the SU(3) 27 projector. A single unsplit SU(3) 27 mediator has one quadratic covariant per sector and enforces equal coefficients in the finite-eight, nine and CP-paired five channels. On two traceless Hermitian sources, the corresponding mixed invariant is  $\text{Tr}(AB)$  squared minus six fifths  $\text{Tr}(A \text{ squared } B \text{ squared})$  plus eleven fortieths  $\text{Tr}(A \text{ squared}) \text{Tr}(B \text{ squared})$ . Its self contraction is  $27/40$  times the squared adjoint norm. The scalar-loop

compiler confirms a five-dimensional closed subalgebra for this unified interaction and the two radial terms.

This is a genuine coefficient relation with a representation origin and a scalar-loop stability mechanism. It is not the golden relation. At fixed nondegenerate spectra the SU(3) trace quartics have moment-space Hessian rank at most one, so they cannot select an isolated interior CP-breaking relative frame. Splitting the 27 under the finite family group permits three independent masses and corresponding coefficient ratios. The 251-coefficient census and actual UV response show precisely which freedoms must then be constrained.

## Exact running identity and scope

For two real scalars with quartic coefficients  $a$ ,  $b$  and  $c$  multiplying  $x$  to the fourth,  $x$  squared  $y$  squared and  $y$  to the fourth, the scalar-only beta coefficients are  $72a^2 + 2b^2$ ,  $24b(a+c) + 16b^2$  and  $72c^2 + 2b^2$ . On the rank-one surface  $a = k u^2$ ,  $b = 2k u v$  and  $c = k v^2$ , the derivative of  $b^2 - 4ac$  is  $-128k^3 u^2 v^2 (u-v)^2$ . Generic rank-one exchange identities are therefore not preserved as those same tree relations by light-scalar running. The nontrivial radial case  $u=v$  is preserved.

The source model's scalar counterterm direction also changes its golden depth under a fixed-mass compensation. That response is a naturalness diagnostic; it is not a complete CKM renormalization-group evolution. Gauge terms, Yukawa running, Higgs contributions, field normalization and matching thresholds must be included before interpreting a flow as a physical CKM prediction.

## Reproduction and references

Run `PYTHONPATH=python OPENBLAS_NUM_THREADS=1 python python/develop_valentiner_mediator_closure.py`. The scientific receipt is `receipts/m22_interactions/valentiner_mediator_closure.json`; the reusable multiplication table is `nonet_scalar_loop_algebra.json` in the same directory. The modules `nonet_mediators.py` and `quartic_renormalization.py` implement constructive matching, exact operator counts, scalar-loop multiplication and generated-subalgebra searches. Focused checks cover current Jacobians, source contacts, positive Schur complements, actual fixed-mass UV deformations, the fermion boundary, exact soft/low-energy counterterms and independent polynomial evaluations.

The scalar-loop algebra viewpoint has prior literature: Nadia Flodgren and Bo Sundborg, One-loop algebras and fixed flow trajectories in adjoint multi-scalar gauge theory, arXiv:2303.13884. RG stability without an evident symmetry is also a real possibility, examined by Howard Haber and P. M. Ferreira, RG-stable parameter relations of a scalar field theory in absence of a symmetry, arXiv:2502.11011. Those works motivate testing closure rather than assuming every unsupported tree relation must fail. The 52-operator compilation, finite-group closure, exact 251-coefficient census and completed-vacuum response here are model-specific calculations; independent novelty review remains separate.

# Exact flavor tensors, running and mechanism selection

## Central results and declared scope

The complete minimal nonet action and its Gaussian finite-mediator extension now have exact scalar-loop proofs and a coupled one-loop evolution system. Every product of the 52 source quartics is proved coefficient by coefficient; adjoining all four real Higgs components gives a 61-quartic algebra with 1,891 proved products. The specified renormalizable soft-current class has 92 running nonconstant parameters. Its evolution includes gauge couplings, all eight allowed Yukawa coefficients, two universal heavy-quark masses, source masses, Higgs portals and nine mediator parameters. A complete finite census of coordinate-supported adjoint-pair subalgebras, exact current anomalous coefficients, leading heavy-scalar matching with canonical normalization, and finite quark-metric response accompany the evolution.

Three further results select mechanisms rather than nominate another angle. A pure traceless Hermitian adjoint with universal bare masses obeys the exact light-mass bound  $m_2/m_1 < 1 + m_2/m_3 < 2$ . A family-invariant CP-odd quartic gives an allowed dimension-eight CP-even Yukawa interaction that shifts the strong-CP mass phase. Finally, an explicitly calculated, locally stable CP-conserving stationary point lies below the retained CP-breaking branch of the complete minimal action. These are different statements: the mass bound is exact, the operator covariance is exact, and the stationary-point energy comparison is numerical with a resolved gap.

The target angle and golden CKM coefficient are not used in these new calculations. Neither is predicted. The electromagnetic gauge boundary remains free. Complete two-loop fermion matching, a complete higher-operator classification, a phenomenological threshold-matched CKM flow and a proof of the absolute global vacuum are not outputs of this chapter. Each approximation below is part of the result's definition.

## Exact tensor proof of the full scalar algebra

The earlier table obtained by numerical rational reconstruction is now independently certified. Use an unnormalized Hermitian adjoint basis with real off-diagonal entries equal to one, imaginary off-diagonal entries proportional to  $i \sqrt{3}$ , and diagonals  $(1, -1, 0)$  and  $(1, 1, -2)$ . The kinetic metric is  $G = \text{diag}(2, 6, 2, 6, 2, 6, 2, 6)$ . In this basis all invariant projector tensors lie in  $\mathbb{Q}(\sqrt{5})$ . Numerical projectors only propose integer coefficient pairs  $(a, b)/80$  representing  $(a + b \sqrt{5})/80$ ; the subsequent proof does not use a reconstruction tolerance.

Exact arbitrary-precision integer identities establish covariance under the four defining family generators and the specified generalized CP operation, metric idempotence, pairwise orthogonality, completeness, and ranks 1, 8, 8, 9, 10. This fixes the CP-even 5 plus 5-prime sum rather than an arbitrary multiplicity-space matrix. The source polynomial tensors have common denominator 240. Their kinetic Hessian contractions are computed symbolically. For every symmetric pair  $i, j$ , the complete polynomial identity  $\text{Tr}(G^{-1} \text{Hess } P_i G^{-1} \text{Hess } P_j)/2 = \sum_k A_{kij} P_k$  is checked, including all monomials generated outside the proposed basis support.

The 52-source table passes all 1,378 products and 19,227 nonzero monomial comparisons with zero exact residual. Adding H-dagger H squared and the eight source-quadratic Higgs portals gives 61 quartics, 1,891 products and 20,538 nonzero monomial comparisons. A finite-field evaluation matrix supplies an exact independence witness; tests also verify rank 61 for the extended basis. The finite-field argument proves characteristic-zero independence because the nonzero minor is evaluated at a prime with valid denominators and a specified image of  $\sqrt{5}$ . It does not replace the coefficient identities.

The proof lives in `python/perfectpower/exact_nonet_algebra.py`. `receipts/m22_interactions/nonet_exact_tensor_proof.json` stores the exact projector pairs and the source rank witness; `nonet_higgs_exact_loop_algebra.json` stores the extended rational multiplication table. These are exact Python certificates, not Lean kernel proofs. In physical normalization,  $16 \pi^2 \beta(V4) = \text{Tr}[(G^{-1} \text{Hess } V4)^2]/2$ . The familiar one-field coefficient 72 for  $x$  to the fourth and eight-component coefficient 128 for the squared adjoint norm provide normalization checks.

## Complete restricted mechanism census and current running

Consider the nine-dimensional adjoint-pair quartic space consisting of each sector's radial square and finite 8-prime self invariant, the product of the two radial norms, the ordinary adjoint overlap, and the three finite cross channels. Every one of its 512 coordinate-support subsets is tested against the exact multiplication table. Exactly 14 are closed, including the empty subspace. Among these subsets, any one containing an orientation-sensitive ordinary or finite cross channel must contain all nine operators. This is a complete coordinate-support classification within the stated ambient algebra. It does not classify arbitrary rotated subspaces or nonlinear invariant coupling varieties.

Exact generated-subalgebra calculations separately retain important rotated relations. Radial seeds close in dimension two. A single finite cross channel together with the two radial seeds generates dimension nine, as does the ordinary cross channel. Equal coefficients on all three finite channels implement the continuous  $SU(3)$  27-plet relation and generate a five-dimensional algebra. That algebra survives scalar renormalization, but the previously calculated fixed-spectrum orientation obstruction still prevents it from isolating the desired interior CP frame. Closure alone does not imply predictive mixing.

For the Gaussian UV completion, write the dimension-one currents as  $g_{uR} S_R q_{uR} + g_{dR} S_R q_{dR}$  and the mediator mass as  $M_R^2$ . The exact finite-8 self invariant acts on the three finite current irreps with eigenvalues  $k_R = (248/15, 16/5, -112/15)$ , ordered 8-prime, 9, and the CP-paired five-dimensional irreps. The identities are proved as exact field matrices, not inferred from sample vectors. With radial coefficients  $\lambda_f$ , finite-self coefficients  $\kappa_f$ , cross coefficients  $\lambda_R$  and adjoint Yukawa coefficients  $a_f$ , the one-loop coefficients are as follows, with a common factor  $16 \pi^2$  suppressed.

$\beta(g_{uR}) = (16 \lambda_u + k_R \kappa_u + 12 a_u^2) g_{uR} + 4 \lambda_R g_{dR}$ .  
Interchange  $u$  and  $d$  for  $\beta(g_{dR})$ .  $\beta(M_R^2) = 4(g_{uR}^2 + g_{dR}^2)$ .  
Differentiating the tree exchange coefficient  $-g_{uR} g_{dR}/M_R^2$  then gives its running, including the mass derivative. Generic finite-self anisotropy splits equal exchange coefficients because the three  $k_R$  differ. Vanishing anisotropy with unified cross coefficients is a genuine restricted exception. No conclusion excludes all possible tuned or nonlinear relations.

## An exact finite-mass exclusion for a pure adjoint source

The simplest source-label reduction removes the two singlets from each nonet. Its complete minimal scalar action has 15 coefficients, including 12 quartics and the Higgs; the corresponding nonconstant running counts are 24 without, or 33 with, the Gaussian finite mediators. This simplification cannot supply a strong hierarchy under universal canonical fermion matching.

Let  $D$  have blocks  $(a I_0; C, M I)$ , with  $a, M$  positive and  $C$  traceless Hermitian. For nondegenerate light singular values  $0 < m_1 < m_2 < m_3 < \min(a, M)$ , each squared source eigenvalue is  $f(m) = (a^2 - m^2)(M^2 - m^2)/m^2$ . The largest source magnitude belongs to  $m_1$ . Tracelessness in three dimensions forces that magnitude to equal the sum of the other two magnitudes, including either possible overall sign. The following two exact positive differences establish the bound without a large- $M$  approximation.

$f(m_1) - (m_2/m_1)^2 f(m_2) = (m_2^2 - m_1^2)(a^2 + M^2 - m_1^2 - m_2^2)/m_1^2 > 0$ . Likewise,  $(m_2/m_3)^2 f(m_2) - f(m_3) = (m_3^2 - m_2^2)(a^2 + M^2 - m_2^2 - m_3^2)/m_3^2 > 0$ . Taking positive square roots and imposing the trace condition gives  $m_2/m_1 < 1 + m_2/m_3 < 2$ .

Consequently this canonical universal pure-adjoint candidate cannot reproduce strongly separated first and second quark masses. Degenerate or zero source eigenvalues are boundary cases; they do not evade the nondegenerate hierarchy obstruction. Singlet shifts, nonuniversal heavy or bare Higgs matrices, or nontrivial quark metrics change the assumptions. They are possible mechanisms requiring their own justified contractions, rather than exceptions to this theorem. The symbolic factorization is executable and is checked independently against 80 finite singular-value examples and a traceful counterexample.

## All specified one-loop gauge, Yukawa and Higgs evolution

The minimal unbroken action has 24 real scalars: 20 real source fields and four Higgs components. Its colored Yukawa tensors use 24 Weyl flavor/weak indices per color: the three SM families and three vectorlike up and down pairs. Each charge sector has four real coefficients  $(y_f, g_{\eta,f}, g_{s,f}, a_f)$ . Lepton Yukawas are set to zero in this benchmark; the usual SM lepton representations remain in the gauge coefficients. The soft Gaussian fields have no new fermion vertices. This is the renormalizable class fixed by the previous representation audit, rather than the full 251-coefficient mediator scalar action.

An independent full Weyl-tensor calculation and an analytic eight-coefficient expression agree. Define  $b_f = g_{\eta,f}^2 + g_{s,f}^2$ ,  $k_f = b_f + 8 a_f^2/3$  and  $T = 9(y_u^2 + y_d^2)$ . The factor nine in  $T$  counts three universal families and three colors; this is not a single-top-family approximation. The one-loop coefficients, again suppressing  $16\pi^2$ , are  $\beta(y_f)/y_f = 3(y_f^2 - y_{\text{other}}^2)/2 + T + k_f/2 - 8 g_3^2 - 9 g_2^2/4 - h_f g_Y^2$ , where  $h_u = 17/12$  and  $h_d = 5/12$ .

For either singlet coefficient,  $\beta(g)/g = 21 b_f + 8 a_f^2 + y_f^2 - 8 g_3^2 - r_f g_Y^2$ ;  $\beta(a_f)/a_f = 3 b_f + 8 a_f^2 + y_f^2 - 8 g_3^2 - r_f g_Y^2$ , with  $r_u = 8/3$  and  $r_d = 2/3$ . The singlet ratio  $g_{\eta,f}/g_{s,f}$  is therefore constant at this one-loop order within this action. It remains an input ratio and is not the golden CKM relation. The universal heavy mass obeys  $\beta(M_f)/M_f = k_f/2 - 8 g_3^2 - r_f g_Y^2$ .

The source singlet anomalous dimension is the full two-by-two matrix  $18 g_i g_j$ ; retaining its off-diagonal entries matters. Each adjoint block is  $6 a_f^2 I_8$ . The Higgs block is  $(T - 9 g_2^2$

squared/4 - 3 gY squared/4) I4. Quartic evolution combines the exact scalar algebra, these complete wavefunctions, the colored fermion-box polynomial  $-6 \sum_f \text{Tr}[(C_f^2 + y_f^2 H^\dagger H) \text{squared}]$ , and the pure-gauge Higgs coefficient  $9 g^2$  to the fourth/8 + 3  $g^2$  squared  $gY$  squared/4 + 3  $gY$  to the fourth/8. Source and Higgs mass evolution and the separately running vacuum energy satisfy a direct one-loop Callan–Symanzik identity at independent field points; relative residuals are below 2.1 times 10 to the minus 15 at the retained benchmark.

There are 61 quartics, nine quadratic coefficients, eight Yukawa coefficients, two universal heavy masses and three gauge couplings: 83 nonconstant coefficients. Adding three mediator masses and six dimension-one currents gives 92. The Gaussian hard-zero restriction is preserved by the stated one-loop superrenormalizable power counting; finite threshold interactions are nevertheless generated. This count is not the complete generic mediator-potential count. Gauge beta coefficients above the six vectorlike thresholds are  $(-3, -19/6, 27/2)$ , in the order SU(3), SU(2), standard-normalized hypercharge. Below them the SM values are  $(-7, -19/6, 41/6)$ . The hypercharge convention is not the grand-unified normalization.

The benchmark uses gauge inputs  $(1, 0.65, 0.36)$ , universal Higgs coefficients  $(1.2, 0.57)$ , singlet coefficients  $(0.1, 0.7)$  and adjoint coefficients 0.8 in each sector. These are declared model inputs, not a measured flavor fit. The integrated 92-coefficient system records tree matching at each scale and stops at the first Higgs quartic zero,  $\log(\mu/\mu_0) = 0.5486383814$ . The initial  $16 \pi^2 \beta(\lambda_H)$  is  $-30.85623209$ . This is a calculated limitation of these inputs; changing free boundary coefficients can change it. It does not establish a physical CKM renormalization trajectory or a threshold-matched experimental exclusion.

## Matching and canonical normalization

The Gaussian heavy-scalar threshold has an exact two-by-two spectral expansion that fixes the leading matrix formulas. At  $\mu = M$ , with massless light sources and a large-heavy-mass expansion, the threshold coefficients are  $16 \pi^2 \Delta V_2 = -\text{Tr}(D D^\text{transpose})/2$  and  $16 \pi^2 \Delta V_4 = -\text{Tr}(D H_\text{EFT} D^\text{transpose})/(2 M^2)$ , where  $D$  is the heavy-light Hessian block of the source current and  $H_\text{EFT}$  is the light quartic Hessian. A direct spectral calculation checks the field-order expansion independently.

The same soft vertices give an adjoint wavefunction threshold  $16 \pi^2 \Delta Z_f = \sum_R \dim(R) g_{fR}^2/(4 M_R^2)$  I8. Canonical normalization adds  $-(\Delta Z_x) \cdot \text{grad}(V_4)/2$  to the quartic threshold. At the retained finite-mediator inputs the two wavefunction coefficients before division by  $16 \pi^2$  are 0.1827767223 and 0.1652805891. The complete canonical quartic threshold is projected onto the source basis and checked at independent points; its absolute fit residual is approximately 1.1 times 10 to the minus 12. Finite light-source masses, higher powers of fields/ $M$  and heavy-quark finite matching are outside this approximation.

The six vectorlike quarks also produce one-loop gauge decoupling,  $\Delta(1/g^2) = -\sum_i \Delta b_i \log(m_i/\mu)/(8 \pi^2)$ . Each heavy up or down family contributes  $2/3$  to the QCD coefficient; hypercharge contributions are  $16/9$  and  $4/9$  respectively. There is no SU(2) contribution because these vectorlike quarks are weak singlets. Matching-scale equality gives zero logarithmic threshold, and the signs and coefficients are tested independently. This matching changes a boundary value; it cannot remove the independent gauge kinetic coefficient. No Sommerfeld constant is derived.

## A dimension-eight CP-quality operator

The real adjoint contains a representation-theoretic CP-odd quartic that was excluded from the CP-even scalar potential but must be considered inside Yukawa interactions. Let  $w_f$  denote the symmetric square of  $A_f$ , and form  $O4 = w_u\text{-transpose} (P5 - P5\text{-prime}) w_d$ . Exact  $Q(\sqrt{5})$  arithmetic proves family covariance, odd parity under the same generalized CP operation, and  $(P5 - P5\text{-prime})^2 = P5 + P5\text{-prime}$  in the kinetic metric. The single-sector polynomial vanishes identically; the mixed polynomial does not. It is neutral under both ordinary sector sign charges because each source appears twice.

With a real coefficient  $c$ , the interaction  $i c O4 \bar{Q}_L H d_R / \Lambda^4$  to the fourth + Hermitian conjugate is allowed by the specified family and CP symmetries. The coefficient function satisfies  $(i O4)(\text{CP fields}) = (i O4(\text{fields}))^*$ ; therefore the full Yukawa operator is CP even despite  $O4$  being CP odd. Its canonical dimension is eight. On the retained vacuum  $O4 = 0.003462945590$  in declared source units. The universal down Higgs coefficient becomes  $y_d + i c O4 / \Lambda^4$  to the fourth. The triangular determinant gives the exact tree mass phase  $\Delta\theta = 3 \arctan[c O4 / (y_d \Lambda^4)]$ .

For  $c = 1$  and  $y_d = 0.57$ , the phases at  $\Lambda = 1, 10, 100, 1000$  are approximately  $0.0182258, 1.82260 \times 10^{-6}, 1.82260 \times 10^{-10}$  and  $1.82260 \times 10^{-14}$  radians. Requiring this single contribution to be below a specified reference tolerance of  $10^{-10}$  gives  $\Lambda > 116.1911$  in these model units. That tolerance is an illustrative calculation parameter. The cutoff, its relation to physical energy units and the coefficient are not derived. This is one explicit quality operator on the current nonet content, not the earlier complex-link determinant operator or a census of all higher interactions. The renormalizable first-order protection remains valid within its scope.

## Higher kinetic interactions retain four physical directions

Both  $R = A_u^2$  and  $T = A_d^2$  are sign-neutral and CP-compatible source covariants. The existing nine-word Hermitian basis built from  $R, T$  has degrees  $0, 2, 4, 2, 4, 6, 4, 6, 8$ . In orthonormal Hermitian coordinates its determinant is  $8 \chi^3$ , where  $\chi = \text{Im Tr}(R^2 T^2 R T)$ . At the retained nonet vacuum  $\chi = 1.904868255 \times 10^{-8}$ , so these words span all Hermitian quark metrics. Arbitrary infinitesimal Hermitian metric directions can therefore be realized by justified polynomial covariants with real coefficients at this fixed background.

The physical test uses the exact finite six-by-six fermion matrix, canonicalizes a positive down-right metric and computes the light charged-current overlaps by singular vectors. Six Hermitian metric variations are compensated by three diagonal terms so all three down light singular values remain fixed; the up sector and scalar vacuum remain unchanged. The response of  $(|V_{us}|^2, |V_{cb}|^2, |V_{ub}|^2, \text{Im}[V_{us} V_{cb} V_{ub}^* V_{cs}^*])$  has four nonzero row-normalized singular values, approximately  $(1.67360, 0.99510, 0.41462, 0.19219)$ . Two-sided finite variations preserve all six light masses with errors below  $10^{-12}$  and positive metrics above  $0.99$ .

The heavy masses are allowed to change and their changes are recorded. The earlier six-current test fixed complete light and heavy spectra; this higher-kinetic test makes the narrower light-mass statement. Positive completion by  $(I + \sum_i c_i Q_i)^2 + \epsilon I$ ,  $\epsilon > 0$ , has scalar degree at most sixteen. A positive metric does not change the tree determinant phase. These higher kinetic vertices are not included in the renormalizable first-order loop theorem. A predictive mechanism must constrain them or supply a justified expansion that

suppresses them.

## Resolved vacuum selection and quantum feedback

Twenty-four renewed all-source minimizations and six minimizations on the generalized-CP fixed subspace are performed with the Higgs temporarily fixed. CP-fixed candidates are then refined using the full 21-coordinate Higgs/source stationarity equations. The best retained stable CP-conserving candidate has a positive full Hessian minimum above 4 times  $10^{-4}$ , a gradient below  $10^{-8}$ , a CP-fixed residual below  $10^{-8}$  and negligible rephasing-invariant  $J$ . Its full-action energy is below the nominated CP-breaking branch by 0.01419555290. The complete coordinates, energies and residuals are stored.

This strict comparison proves that the retained CP-breaking branch is not the global minimum of its declared action. The earlier local-stability result is unaffected. A Gaussian square completion preserves all EFT branch energies at  $S = -J/M^2$ , so the same comparison applies to that completion at tree level. The absolute global winner over every branch is not proved; an all-field search also finds other lower candidates. A physical symmetry-breaking history cannot be inferred from a local branch alone.

The leading one-loop minimal-action vacuum gradient includes all four Higgs components, the real-source Hessian, twelve Dirac eigenvalues including color, and W/Z terms in Landau gauge at fixed  $\overline{MS}$  inputs  $\mu = 1$ . The linear diagnostic  $-H^{-1} \text{grad}(V_{CW})$  has norm 15.5158 times the tree background norm. Its size invalidates a small-displacement treatment for this benchmark. It is not a trustworthy prediction of the new vacuum position. Counterterm choices, changed input coefficients or a new mechanism can modify the result; the present calculation does not silently retune them. The scalar calculation concerns the minimal action and does not include the additional Gaussian mediator modes in the full UV Coleman–Weinberg potential.

Expanding the determinant of the one-loop-corrected fermion matrix gives first-order phases near  $10^{-17}$  and iterated quadratic terms near  $10^{-19}$  radians. These cancellations do not constitute a complete two-loop result: genuine two-loop self energies and finite wavefunction matching are omitted. The receipt labels the determinant expansion explicitly.

The leading high-temperature quadratic potential is calculated from  $\text{Tr}(H_{\text{scalar}}) T^2/24$ , the colored fermion mass-square trace and the gauge Higgs term. All 24 thermal Hessian eigenvalues are positive at the benchmark, with minimum approximately 0.99998 in the coefficient convention. This supports asymptotic restoration in the leading minimal-action expansion. It is neither a resummed transition calculation nor a choice of CP sign. The extra UV mediators become thermally active above their own thresholds and are not counted in this minimal-action thermal result.

## Reproducibility and the next mechanism's obligations

Run `PYTHONPATH=python OPENBLAS_NUM_THREADS=1 OMP_NUM_THREADS=1 python python/develop_flavor_frontier.py`. It regenerates three scientific receipts: the exact source proof, full Higgs loop algebra, and `flavor_frontier.json`. The dedicated tests independently verify exact identities and reject a tampered rational coefficient, compare Weyl-tensor and analytic Yukawa evolution, compare fermion boxes to mass-tensor traces, check Callan-Symanzik normalization, re-evaluate finite mass bounds, construct metric responses, and re-evaluate the lower CP-conserving stationary point in the full action. Validation records actual test counts and fresh replay hashes separately; receipt generation is deterministic under the declared numerical environment.

The substantive mechanism-selection conclusions are now precise. Eliminating source singlets alone is excluded by the finite-mass hierarchy theorem. Selecting only one coordinate-supported orientation channel is incompatible with exact scalar-loop closure. Continuous SU(3) unification remains a real relation but lacks the required isolated interior frame. The renormalizable phase theorem requires a separate higher-operator quality mechanism. The retained tree benchmark requires a different vacuum-selection or input structure, and its perturbative vacuum expansion is uncontrolled at  $\mu = 1$ . An eventual interaction must correlate source labels, quark and mediator vertices, canonical metrics and thresholds while escaping these explicit tests. No target angle or gauge constant is promoted to a prediction by algebraic closure alone.

The primary general-RGE reference is Schienbein, Staub, Steudtner and Svirina, Revisiting RGEs for general gauge theories, arXiv:1809.06797, <https://arxiv.org/pdf/1809.06797>. Its treatment of non-diagonal scalar wavefunctions is relevant to this calculation. For context on Nelson-Barr quality and vacuum questions, see Murai and Nakayama, Revisiting the Minimal Nelson-Barr Model, arXiv:2407.16202v2, <https://arxiv.org/html/2407.16202v2>. The exact projector, restricted-census, pure-adjoint mass-bound and numerical nonet tests here are repository calculations; literature context is not a claim of independent novelty review.

# Metastable tree decay and conditional CP walls

The current release turns the conditional wall interface into a calculation on the published flavor action. It consumes the lower CP-conserving competitor already found by the other flavor session, rather than searching for new stationary points or changing its coefficients. At the pinned main-branch commit 22e4edf06dbdeea0f9245dbb06669ed57b7906fc, the nominated CP-breaking branch has energy -4.196243686677212, while the retained stable CP-conserving competitor has energy -4.210439239580322. The gap is 0.01419555290310992. Both are locally stationary to approximately  $10^{-15}$  in the declared numerical action, and their reported smallest physical Hessian eigenvalues are positive. The absolute global minimum remains unclassified. This benchmark therefore calls for a metastable-decay calculation, not a claim of stable walls between global CP-breaking vacua.

The new reusable module is `python/perfectpower/flavor_decay.py`, with the driver `python/develop_flavor_cosmology.py` and eighteen focused tests in `python/tests/test_flavor_decay.py`. The scientific receipt is `receipts/flavor_cosmology/tree_decay.json`. It retains exact rational coefficients for the radial trial profiles, the input-file hashes, every chosen endpoint and the full 48-coordinate mediator-lifted path. Run `PYTHONPATH=python OPENBLAS_NUM_THREADS=1 OMP_NUM_THREADS=1 python python/develop_flavor_cosmology.py` from the repository root. No source couplings are retuned, no stationary points are re-minimized, and no quark matching is repeated. The earlier 104-check Node calculation remains an independent conditional interface.

The numerical source tensors, coefficients and endpoint coordinates are converted to exact binary fractions. Five exact evaluations along a source line reconstruct its quartic potential by rational Newton interpolation. This arithmetic is exact for the binary-rational realization of those arrays. It does not turn floating-point finite-group projectors into exact algebraic-number tensors, or approximate stationary coordinates into proven exact minima. Thirty-three independent ordinary floating-point evaluations per candidate check the reconstructed polynomial against the existing full potential. This distinction matters: the radial action integrals below are exact, but the physical identification with a certified bounce remains conditional on the false-vacuum and reduced-variational hypotheses.

Let  $\gamma(q)$  be a polynomial path in canonically normalized real fields, with  $\gamma(0)$  the supplied false vacuum and  $\gamma(1)$  a lower-energy configuration. Let  $W(q) = V(\gamma(q)) - V(\gamma(0))$ . A compact  $O(4)$  trial profile has  $q=1$  inside a core, a smooth shell, and  $q=0$  outside. Put  $u=r/R$ . The core is  $0 \leq u \leq L$ ; in the shell write  $t=u-L$  between zero and one and set  $q(t) = 1 - 3t^2 + 2t^3$ . This taper has zero derivative at both joins, so the profile is continuously differentiable. Outside  $u=L+1$  the field equals the false background exactly. No exponential tail cutoff or divergent outside energy is introduced.

The complete action of this profile is

$$S_4(R,L) = 2\pi^2 [T(L) R^2 + U(L) R^4],$$

where  $T(L) = \frac{1}{2}$  times the integral from zero to one of  $(L+t)^3$  times the squared norm of  $d\gamma(q(t))/dt$ . The potential term is  $U(L) = W(1)L^4/4$  plus the integral from zero to one of  $(L+t)^3 W(q(t))$ . Both are rational polynomials in  $L$  whenever the supplied field-path and

potential coefficients are rational.  $T$  has degree at most three and  $U$  has degree at most four. The core-volume contribution ensures  $U < 0$  for sufficiently large  $L$  whenever  $W(1) < 0$ . This makes a negative-action escape configuration available without pretending the Euclidean bounce is a global action minimum.

For a retained shape with  $T > 0$  and  $U < 0$ , the action as a function of size starts at zero, reaches a positive maximum and then becomes negative. The maximum lies at  $R_{\text{crit}}^2 = -T/(2U)$ , with  $S_{\text{max}}/\pi^2 = -T^2/(2U)$ . Its derivative under radial scaling vanishes there and its scaling second derivative is negative, as expected for the escape direction. The code evaluates these quantities as exact fractions. It uses a numerical one-parameter search only to choose a rational  $L$  and then recomputes the retained action exactly. It does not certify global optimality even within the shape family.

The appropriate variational interpretation is an upper estimate for the least decay barrier, under the usual reduced-problem or mountain-pass existence assumptions and an exact metastable exterior. Every such trial profile produces a route from the false configuration to negative action; the least separating saddle cannot have a larger minimax level than that route. This is not a lower bound on the bounce action. Consequently a large trial action cannot establish a long lifetime. The familiar scale-reduced formulation originates in Coleman, Glaser and Martin, Action Minima among Solutions to a Class of Euclidean Scalar Field Equations, Communications in Mathematical Physics 58 (1978) 211-221, DOI 10.1007/BF01609421. The multifield symmetry and reduced-problem existence assumptions are treated by Blum, Honda, Sato, Takimoto and Tobioka,  $O(N)$  Invariance of the Multi-Field Bounce, arXiv:1611.04570, <https://arxiv.org/abs/1611.04570>. The present radial-profile construction and its integration are an adapter to this established framework.

A significant practical reduction comes from correctly using the internal symmetry. The driver evaluates the 1080 existing family matrices on the already-published lower endpoint, together with both sector-sign transformations. This supplies 4320 symmetry-related endpoint images; it does not produce 4320 independently minimized vacua. The source coefficients remain identical. Four nearest images and the original endpoint are retained as candidate straight paths. A central group image can repeat the same real-adjoint endpoint, so the retained list is not claimed to classify distinct physical vacua. A farther endpoint might have a better potential barrier; this finite nearest-image selection is deliberately a bounded path search, not an exhaustive optimization.

With the original endpoint, the retained compact-profile trial action is approximately  $2.5305352605 \times 10^{10}$ . The best retained symmetry-related endpoint yields approximately  $3.3396009270 \times 10^7$ , a reduction by a factor of 757.7358241. Its squared source distance is 3.210476017821027; the original endpoint's squared distance is 11.166145333203302. Its core-to-shell ratio  $L$  is 45.94751799. These are dimensionless trial actions for the specified tree potential. Their numerical size is not a cosmological lifetime, a measured decay rate, or a statement that this action is near the actual least bounce. The result shows why symmetry-aware field-path selection materially affects even a conservative decay calculation.

The mediator completion introduces a further geometric issue. The positive Gaussian square allows the scalar potential to match the source potential exactly at  $S = -J(x)/M^2$ , but this equality does not erase the mediator kinetic energy when  $x$  varies across a bubble. Following that trough gives the induced field-space metric

$$ds^2 = dx^T dx + dJ^T dJ/M^4.$$

The driver constructs all 27 mediator coordinates as exact quadratic polynomials along the source line, retaining the 21 source/Higgs coordinates. The complete trial bubble therefore uses 48 canonical real fields. Its potential along the path is unchanged, while its kinetic integral increases by one half the radial integral of the squared mediator derivatives. The positive-square contribution is evaluated independently at seventeen path points; it is negligible at floating-point reconstruction precision. Potential matching and kinetic matching are thus kept separate.

At any fixed path and retained shape, the completion changes the candidate action by the exact factor  $[1 + \Delta T(L)/T_{\text{source}}(L)]^2$ . The extra kinetic integral is nonnegative, so this factor cannot decrease the action of that particular trial profile. This is not a theorem that the true UV least bounce is greater than the EFT least bounce: UV paths can leave the trough, and the least path has not been solved. The statement concerns the explicitly retained trough-following profiles. Treating Gaussian potential matching alone as sufficient for tunneling would omit this positive geometric cost.

The mass scan keeps the EFT coefficients fixed and scales the current couplings in the existing Gaussian realization proportionally to  $M$ . Then  $S = -J/M^2$  scales as  $1/M$ , and the induced kinetic correction scales as  $1/M^2$ . For masses 5, 10, 20, 100 in the stated model units, the separately shape-optimized trough-profile action ratios to the best source candidate are approximately 1.0002861286, 1.0000715283, 1.0000178818, and 1.0000007153. At  $M=10$  the lifted action is approximately  $3.3398398030 \times 10^7$ , an increase of 0.00715283 percent. This scan demonstrates decoupling within the same completed tree construction. It does not vary the mediator mass while incorrectly freezing its current normalization, and it includes no mediator-loop Coleman-Weinberg correction.

The CP interpretation also becomes sharper. Direct evaluation of the supplied generalized-CP partner reproduces the false-branch energy, and CP covariance is checked on the false point, chosen lower point and their midpoint. CP-even matching or subsequent CP-even thermal evolution cannot select a weak-CP sign merely by splitting this conjugate pair. If the CP-conserving lower basin ultimately occupies the universe, it removes the physical weak-CP order parameter instead of predicting its observed sign. Whether this basin is populated requires nucleation dynamics, initial conditions and an adequately controlled action; the local CP-breaking benchmark alone supplies none of those.

The upstream quantum calculation is a substantial limit on physical interpretation. At its fixed  $\mu=1$  inputs, the linearized one-loop vacuum displacement is 15.51581387 times the tree-background norm. This signals an uncontrolled small-displacement expansion. A tree trial action cannot be presented as the quantum-corrected tunneling exponent for that benchmark. The current result deliberately imports and displays that warning alongside its own action rather than fitting counterterms or changing couplings to conceal it. A controlled quantum action is an input the mechanism session must eventually supply before physical decay claims become meaningful.

The upstream leading high-temperature minimal-action Hessian is positive, supporting asymptotic restoration within its declared expansion. That does not provide a transition temperature or an  $O(3)$  nucleation action. The new calculation is zero-temperature, flat-space  $O(4)$ , with gravity omitted. A finite-temperature extension requires the resummed effective potential, the thermally active mediator content, possible friction and the maximum post-inflation temperature. Inferring thermal bubble rates by dividing the present  $O(4)$  action by temperature would be incorrect. Likewise, reinstating a stable-wall calculation requires a candidate whose CP-breaking vacua actually lie on the global branch.

The eighteen Python tests independently integrate the compact shell with numerical quadrature, check rational interpolation, endpoint smoothness, action-scaling stationarity and the negative-action escape, reject invalid cores and exterior energies, prove scale cancellation on exact fixtures, and check the increased kinetic cost of extra fields. Receipt tests verify the published energy ordering, exact retained action replay, full-potential sample agreement, the 48-field lift, mediator-current matching, monotone decoupling in the retained scan, and CP degeneracy. The scientific receipt replays byte for byte in a fresh process under the specified single-thread numerical environment. These are focused mathematical and action checks; no full repository regression or new Lean theorem is claimed for this push.

The continued coordination boundary is explicit. The other flavor session owns exact tensor closure, constrained mechanisms, running, quantum corrections, physical matching and vacuum discovery. This release consumes its c9010fb conclusions and changes only the cosmology module, driver, focused tests, cosmology receipt, plotting/rendering scripts and this monograph. It commits directly to the main working branch after checking the latest head. The arithmetic and Lean streams remain intact. Future cosmology calculations can reuse the same radial compiler on a new action without borrowing protection, hierarchy or vacuum claims from incompatible branches.

The advance is a reusable decay-path compiler tied to the repository's actual published action, together with its mediator-induced kinetic geometry and exact radial witnesses. It sharpens what has to be measured next: a controlled action and least-path calculation, a dimensionful normalization and prefactor, and a viable cosmological occupation history. The conditional CP-wall and constrained-bias results from the preceding release remain useful for future globally CP-breaking candidates; they are retained below with their original hypotheses.

This advance consumes the published flavor conclusions without changing their mechanisms, matching calculations, operator algebra, scalar vacua, or cumulative flavor monograph. It adds exact rational tools for a conditional cosmological question: if the chosen flavor branch really supplies degenerate physical CP-breaking global vacua, can a proposed cosmological history avoid their walls, and can a permitted explicit bias remove them without undoing specified first-order flavor protections? No dimensionful wall tension, transition temperature, gravitational-wave spectrum, or cosmologically viable repository benchmark is obtained in this release.

The implementation is `research/flavor_cosmology/cp_wall_tools.mjs`; its dependency-free test is `research/flavor_cosmology/test_cp_wall_tools.mjs`. Its examples and validation record are `receipts/flavor_cosmology/conditional_wall_tools.json`. Run `node research/flavor_cosmology/replay.mjs` to reproduce that receipt; replay retains the historical validation record and regenerates the scientific examples. Run `node research/flavor_cosmology/test_cp_wall_tools.mjs` from the repository root. These additions are isolated from the active scalar-loop and mechanism work. The exact mathematical tools were executed in the available V8 runtime; 104 assertions passed; the standalone replay reproduces the receipt byte for byte in V8. The workspace execution server failed to initialize, so neither the delivered Node command, Python, Lean, nor the full repository suite ran in this session. This difference is recorded rather than attributed to another session's tests.

The coordination snapshot is pinned to `a5d0f04aa6c8f666dfd9428747abbbdf3811fe50`, published at 2026-10-06 19:04 UTC. The arithmetic stream there adds elliptic division-fibre theorems, coordinate normalization, halving transport and rational-root completeness. Its predecessor `ff5d04ab56e417bbc359bfe2c11264ac566cc4d1` supplies exact elliptic witnesses and rational division fibres. The published flavor stream is `8f18f0a037d7936f7fee9aca1a2a2959c19190e`, published at 18:36 UTC: the protected

soft-mediator completion and scalar-loop closure. The user reports that a separate flavor session now owns the full source-quartic exact proof, constrained mechanisms, coupled matching and running, leading quantum and higher-operator responses, extended vacuum tests, physical parameters, and its publication. All of those are excluded from this branch's file edits. The monitoring scope is published GitHub changes and user-supplied session updates; uncommitted work and private session execution are not observable.

The existing 48-scalar completion has a positive full local Hessian, complete finite quark masses and nonzero physical weak CP in its chosen benchmark, with stated first-order mass-phase protection. Its 251-coefficient operator census and fixed-mass response do not establish a golden depth or a predicted 66-degree phase. The earlier exact field-content CP intertwiner is an anti-linear involution on the representations that pass its consistency checks, not a class-inverting operation valid for arbitrary added representations. Those are upstream conclusions in docs/VALENTINER\_SOFT\_MEDIATION\_AND\_LOOP\_CLOSURE.md and docs/VALENTINER\_CP\_AND\_GLOBAL\_MODES.md, consumed at the pinned commit. Their current numerical vacuum is not promoted here to a classified global minimum.

Let  $C$  denote a consistent CP operation on the full theory and let  $v$  be a homogeneous stationary field configuration. If the scalar potential and kinetic metric obey  $V(C\phi)=V(\phi)$  and the corresponding isometry relation,  $Cv$  is stationary with the same energy and the same canonically measured fluctuation spectrum. An internal-symmetry-invariant CP-odd observable  $O$  obeys  $O(g\phi)=O(\phi)$  and  $O(C\phi)=-O(\phi)$ . If  $O(v)$  is nonzero,  $v$  and  $Cv$  cannot be in the same orbit of the internal unitary group: assuming  $Cv=gv$  gives  $O(v)=-O(v)$ . For a quark-matched example, a nonzero Jarlskog invariant with nondegenerate spectra is such a diagnostic. It proves orbit distinction, not a count of all global minima. Distinct orbits alone do not prove disconnected vacuum components: a continuous degenerate path could join them. Isolated global minima after gauge reduction supply the additional hypothesis. A positive physical Hessian establishes local isolation; the repository's remaining global search must establish global minimality.

Counting all 1080 Valentiner matrices as independent physical vacua would therefore be wrong if an internal gauge redundancy identifies their configurations. Conversely, quotienting by ordinary internal transformations does not justify erasing spontaneous CP walls. CP reverses spatial orientation, and gauged parity/CP is not ordinary internal discrete gauge symmetry. McNamara and Reece, Reflections on Parity Breaking, arXiv:2212.00039, analyze exactly stable walls in the gauged-parity setting. A gravity embedding and its orientation structure are not supplied by this finite flavor representation calculation. This release leaves that embedding explicit rather than treating the phrase gauged CP as a wall-removal mechanism.

There is an exact energy result independent of any numerical minimizer. In a CP-invariant quantum theory with a compatible measure, regulator and renormalization prescription, CP-even corrections preserve equality of conjugate branch energies. Thresholds, perturbative running, higher-loop terms and CP-even higher-dimensional operators cannot alone create a CP-conjugate vacuum bias. They may change the common vacuum energy, wall profile or transition order; they do not select a sign. The thermal version holds for a CP-symmetric state, including zero CP-odd chemical potentials. A baryon asymmetry, imposed CP-odd background or explicit CP-breaking spurion changes this premise and must be included as a physical input. Full quantum CP consistency for the repository's enlarged field content is not established by its restricted elementary intertwiner alone.

QCD also supplies no automatic loophole. At a CP-symmetric bare  $\theta$  and in a CP-symmetric medium, its free energy obeys  $E(\theta)=E(-\theta)$ . Conjugate branches with opposite induced strong phases therefore have equal QCD energies. In the illustrative susceptibility expansion  $E(\theta)=\chi \theta^2/2$ , adding an independent explicit common  $\theta_0$  gives

$E(\theta_0 + \theta_s) - E(\theta_0 - \theta_s) = 2 \chi \theta_0 \theta_s$ . That bias relies on the explicit input  $\theta_0$  and vanishes if  $\theta_s$  is zero. It is not a mechanism that follows from protected Nelson–Barr matching. Asadi, Homiller, Lu and Reece, Chiral Nelson–Barr Models: Quality and Cosmology, arXiv:2212.03882, discuss the interaction between strong-CP quality, inflation and reheating in concrete models.

The constructive wall result begins with two supplied degenerate global minima and a continuously differentiable path  $\gamma(t)$ ,  $0 \leq t \leq 1$ , in canonically normalized real fields. Write  $W(t) = V(\gamma(t)) - V_{\min}$  and  $K(t) = \text{the Euclidean squared norm of } \gamma'(t)$ . Provided  $W$  is nonnegative, minimization of the one-dimensional spatial parametrization of that fixed path gives a candidate interface tension equal to the integral of  $\sqrt{2 W K}$ . The true infimum of the interface tension cannot exceed the value from this path. Cauchy–Schwarz then gives the useful exact squared bound

$$\sigma_{\min}^2 \leq \sigma_{\text{path}}^2 \leq 2 \left( \int_0^1 W(t) dt \right) \left( \int_0^1 K(t) dt \right).$$

One sees the first step by introducing  $s = dt/dz > 0$  in the static energy: its density integrated over  $t$  is  $K s/2 + W/s$ , minimized pointwise at  $s = \sqrt{2W/K}$ . Endpoint singularities or interior zeros are understood by limiting finite-energy profiles. Additional paths can lower the upper bound; no lower bound follows. The argument assumes the declared degenerate global minima. Metastable minima with a deeper third branch do not by themselves define a stable cosmological wall.

For rational polynomial inputs, the code computes both integrals exactly with BigInt fractions. It checks that  $W$  and its derivative vanish at both endpoints, divides  $W$  by  $t$  squared times  $(1-t)$  squared, and checks nonnegative Bernstein coefficients of the residual polynomial. Nonnegative Bernstein coefficients certify nonnegativity on the unit interval; failure is deliberately inconclusive. This is not a complete polynomial positivity algorithm. The code accepts a caller-supplied  $W$ , so it does not independently verify that  $W$  is the composition of a multivariate repository potential with  $\gamma$ . Nor does it certify the endpoints' global minimality. Those hypotheses are written into each receipt. Algebraic endpoints and a nonconstant field metric require further exact-field or interval adapters.

For the canonical one-field benchmark  $V(a) = (a^2 - 1)^2/4$  and  $\gamma(t) = 2t - 1$ ,  $W = 4t^2(1-t)^2$  and  $K = 4$ . Their integrals are  $2/15$  and  $4$ . The exact upper bound is  $\sigma^2 \leq 16/15$ . The known kink has  $\sigma = 2\sqrt{2}/3$ , hence  $\sigma^2 = 8/9$ ; this gives an independent strict check on the variational bound. These are dimensionless toy data. No repository tension has been assigned this value. With general positive  $\lambda$  and  $v$ , the same canonical quartic has tension  $2\sqrt{2}\lambda v^3/3$ , illustrating why a dimensionless flavor Hessian cannot fix an astrophysical wall tension without a physical normalization and scale.

An explicit bias has a separate, precisely bounded algebraic question. Let  $O_i$  be specified CP-odd operators and  $c_i$  their small real coefficients. To first order, the relaxed branch-energy difference is  $b^T c$ , where  $b_i = 2 O_i(v)$ , up to the chosen sign of which branch is subtracted. The derivative of a relaxed energy is the operator value at its unperturbed stationary point; the first-order vacuum shift does not contribute because the unperturbed potential gradient vanishes. The vacuum shift itself is  $-H^{-1}$  times the summed  $c_i$  gradient  $O_i$ . A protected observable  $Q$  has first-order response equal to its direct coefficient response minus gradient  $Q^T H^{-1}$  gradient  $O_i$ . Physical matching must provide that direct response and the properly gauge-reduced Hessian; the present module does not calculate either.

Suppose rows of  $A$  collect responses required to vanish to first order, for example a supplied strong-CP response and specified CKM responses. Choose an explicit positive definite coefficient

metric  $R$  and a budget  $c$  transpose  $R c \leq \rho$  squared. The mathematical problem is to maximize  $|b \text{ transpose } c|$  subject to  $A c = 0$ . Discard redundant rows of  $A$  by exact elimination. Set  $x = R \text{ inverse } b$  and

$$z = R \text{ inverse } b - R \text{ inverse } A \text{ transpose } (A R \text{ inverse } A \text{ transpose}) \text{ inverse } A R \text{ inverse } b.$$

Then  $A z = 0$ ,  $z \text{ transpose } R z = b \text{ transpose } z \geq 0$ , and the exact maximum squared bias is  $\rho$  squared times  $b \text{ transpose } z$ . For a nonzero result, the maximizing coefficient vector is  $\rho z / \sqrt{b \text{ transpose } z}$ . This follows by taking the  $R$ -orthogonal projection of  $x$  onto  $\ker A$  and applying Cauchy-Schwarz in the  $R$  metric. The module gives the rational projected direction, exact squared support and all constraint residuals. A vanishing support is a first-order impossibility result under the supplied constraints and coefficient budget, not an all-orders theorem that a cosmology is impossible.

The exact criterion has a useful sharp interpretation. Because  $R$  is positive definite, no allowed first-order bias exists exactly when  $b$  lies in the row span of  $A$ . If a CP-odd operator distinguishes the conjugate vacua but its bias functional is fully tied to the responses being protected, forbidding those responses also forbids that bias. If  $b$  has a surviving component in the constrained subspace, an algebraic first-order direction exists. Its physical size, radiative stability, higher-order strong phase, naturalness and full thermal dynamics remain upstream model calculations. Treating a cancellation  $A c = 0$  as a symmetry-derived Wilson-coefficient relation would exceed this result.

The hand-worked example uses  $R = \text{diag}(1, 2, 3)$ ,  $b = (2, 2, 2)$ , and the response constraint  $c_1 + c_2 = 0$ . Its projected direction is  $(0, 0, 2/3)$ , so the unit-budget squared maximum is  $4/3$ , versus  $22/3$  without the constraint. A duplicate constraint  $2c_1 + 2c_2 = 0$  changes nothing. In a two-operator example with  $R = I$  and  $b = (2, 2)$ , the same response constraint removes all bias: the support is exactly zero. Twenty-four additional positive definite rational examples check that the projected direction satisfies every original constraint, that its  $R$  norm equals its bias pairing, and that adding constraints cannot enlarge the support. No rows in these examples are claimed to be the actual repository strong-CP derivative.

The thermal adapter is deliberately a transparent toy. For a single real CP-odd field with  $V(a, T) = \lambda(a^2 - v^2)^2/4 + c T^2 a^2/2$ , positive  $\lambda$ ,  $c$  and  $v^2$  give  $T_c^2 = \lambda v^2/c$ . Below  $T_c$  the conjugate minima have  $a^2 = v^2 - c T^2/\lambda$  and remain exactly energy-degenerate. The example  $\lambda = 2$ ,  $v^2 = 9$ ,  $c = 3$  gives  $T_c^2 = 6$ . The actual 48-scalar system has independent portals and mixed thermal masses. Its critical temperature, possible symmetry nonrestoration, transition order, tunneling and nonthermal restoration are open. In a thermal restoration scenario, maximum temperature after inflation matters; the final reheating temperature alone can miss an earlier hotter epoch. Preheating and inflationary fluctuations likewise need their own calculation.

For bookkeeping only, `classicalScalingScreen` records an explicitly declared horizon-curvature estimate:  $\rho_{\text{wall}} = \text{area} \cdot \sigma \cdot H$  and  $H_{\text{ann}} = \Delta V / (\text{annihilation} \cdot \sigma)$ , with reduced Planck mass  $M$ . Equating wall energy to  $3 M^2 H^2$  gives  $H_{\text{dom}} = \text{area} \cdot \sigma / (3 M^2)$ . In this convention, annihilation before both a supplied deadline  $H_d$  and wall domination asks for  $\Delta V$  greater than  $\text{annihilation} \cdot \sigma \cdot \max(H_d, H_{\text{dom}})$ . All quantities, area and annihilation factors are caller inputs. This is a phenomenological screen, not a rigorous cosmological bound or a simulation.

Combining that declared screen with the exact coefficient envelope yields a conditional test: its required  $\Delta V$  squared must be below  $\rho$  squared  $b \text{ transpose } z$  to have any linearized

coefficient direction within the chosen budget. A certified tension upper bound can supply a conservative sufficient threshold within this screen because both threshold terms increase with sigma. An upper bound that fails the test cannot establish impossibility: the true tension might be smaller. Likewise, a support zero under homogeneous first-order response constraints does not exclude quadratic-order bias, inflationary removal or a different field content. Classical estimates are not promoted to universal dynamics. Babichev and collaborators, arXiv:2504.07902, find different bias dependence in their simulated real-scalar model, reinforcing the need to calibrate a physical network before claiming a gravitational-wave prediction.

The bounded handoff to the active mechanism session is now concrete. It can supply a globally classified pair, a dimensionful canonical metric and potential, physical CP consistency for the complete content, a set of allowed CP-odd operators if explicit breaking is admitted, and their direct plus relaxed strong-CP/CKM response rows. The separate cosmology layer can then consume these inputs to compute path tension bounds, constrained bias support and thermal-history conditions without editing its loop algebra or redoing its matching. If exact CP must remain fundamental, the explicit-bias route is unavailable and the remaining cosmological questions are inflationary removal, restoration avoidance and the actual spacetime realization of CP. These are distinct from choosing a locally stable flavor frame.

The contribution is the executable rational interface between supplied flavor protection constraints and cosmological bias, together with a conservative polynomial-path tension bound and a recorded concurrency boundary. CP-domain-wall existence, their standard cosmological problem, variational wall mechanics and constrained quadratic optimization are established mathematics and physics. No claim of a new fundamental wall mechanism, a new CKM prediction or publication priority is made. References are McNamara and Reece, <https://arxiv.org/abs/2212.00039>; Asadi, Homiller, Lu and Reece, <https://arxiv.org/abs/2212.03882>; Saikawa, <https://arxiv.org/abs/1703.02576>; and Babichev et al., <https://arxiv.org/abs/2504.07902>.

# Finite scalar matching and the phase-protection criterion

## The finite matching result

Gaussian scalar mediation correlates a potential, a derivative expansion and finite pole normalization. This chapter derives those correlations exactly and evaluates them on the existing nonet/Higgs branch. All 21 light scalar poles obtained from the momentum-dependent source operator agree with the full 48- and 76-coordinate UV matrices to approximately 5 times  $10^{-14}$  in squared-mass units. This closes the finite tree scalar pole and residue problem for the specified Gaussian completions, including unequal positive mediator masses in the general implementation. It does not identify source-field normalization with quark normalization or derive a mixing angle.

The second result is a precise boundary on the mass-phase theorem. First-order phase cancellation remains intact. An exact rational spectral-kernel example and an independently evaluated finite B0 example show that the quadratic term in the determinant of the one-loop-corrected mass matrix can nevertheless be nonzero. Separating Higgs and source scalar modes protects that restricted resummed determinant, but the specified nonzero Higgs and source Yukawas regenerate Higgs portals at one loop. Thus this particular protection mechanism has an exact renormalization test, rather than an assumed extension of first-order reality.

The Gaussian derivative metric also connects directly to the other session's cosmology tools. Its latest merged calculation lifts the actual metastable-decay trial paths to all 48 canonical fields; the pole and residue results here give the corresponding finite-spectrum description. A rigorous wall-tension sandwich and an exactly classified two-field example are implemented in both SymPy and the existing BigInt polynomial-path engine. The example has genuine global minima; no cosmological tension is assigned to the retained quark-flavor branch, which the preceding chapter proved is not globally selected by its action.

## Exact scalar poles and canonical residues

$$F(z) = H - zK(z), \quad Z(z) = -F'(z)$$

On the stationary Gaussian valley  $S_a = -J_a/M_a$  squared, write  $H$  for the source/Higgs Hessian and  $D_{ai} = \text{partial}_i J_a$ . The full canonical UV Hessian has blocks  $H + D\text{-transpose diag}(1/M_a \text{ squared}) D$ ,  $D\text{-transpose}$ ,  $D$ , and  $\text{diag}(M_a \text{ squared})$ . Its static Schur complement is exactly  $H$ , so its stationary energies and the signs of its light static curvatures reproduce the original source potential. Its finite pole masses are different because heavy fields carry canonical kinetic terms.

For a timelike squared momentum  $z$  below every mediator threshold, eliminating the heavy fluctuation gives  $F(z) = H - z K(z)$ , where  $K(z) = I + D\text{-transpose diag}[1/(M_a \text{ squared} (M_a \text{ squared} - z))] D$ . The full characteristic determinant factors into  $\det[\text{diag}(M_a \text{ squared}) - z I] \det F(z)$ . Exact symbolic checks establish the one-source identity for general  $h, d, M, z$  and a polynomial identity with two source coordinates and three unequal mediators. The general formula follows directly from the same block Schur complement. No field-to-heavy-mass expansion is used for the pole equation.

The two-derivative tree metric is  $K(0) = I + D\text{-transpose diag}(1/M_a \text{ to the fourth}) D$ . At a root  $z$ , the physical residue metric is  $Z(z) = -\partial_z F(z) = I + D\text{-transpose diag}[1/(M_a^2 - z^2)] D$ . If  $F(z)v = 0$ , its heavy component is  $-\text{diag}[1/(M_a^2 - z^2)] Dv$ . The full vector is canonically normalized precisely when  $v\text{-transpose } Z(z) v = 1$ . Distinct full eigenvectors are orthogonal even though their projected source vectors use different pole metrics. Degenerate source modes are handled by the full nullspace basis; individual eigenvectors inside a degenerate subspace have no invariant identity.

These formulas distinguish a zero-momentum curvature, a local derivative approximation and a finite UV pole. For a positive source Hessian whose largest eigenvalue is below the smallest mediator squared mass, every ordered root obeys  $\text{lower\_bound}_i \leq \text{exact\_pole}_i \leq \text{two\_derivative\_pole}_i \leq \text{bare\_curvature}_i$ . A useful lower bound divides the ordered eigenvalue of  $H$  by  $1 + \text{norm}(D)^2 / [M_{\min}^2 (M_{\min}^2 - \lambda_{\max}(H))]$ . Here  $\text{norm}(D)$  is the spectral norm. It follows from the positive matrix bound on  $K(z)$  and the min-max principle. The momentum-dependent eigenvalues of  $F(z)$  decrease strictly because  $-\partial_z F$  is positive; each branch therefore has one root in the stated below-threshold interval. The numerical solver independently checks the resulting roots, full canonical vectors and collective light-subspace projector against the full matrix.

At the existing finite-only completion, the tree metric's largest eigenvalue is 1.0011338172 and its up/down source kinetic block has spectral norm 0.0003900162. The local two-derivative approximation's maximum relative pole error is 0.0000666455. For the universal completion the corresponding numbers are 1.0012836495, 0.0004148295 and 0.0000781621. The full finite poles agree to 4.98 times 10 to the minus 14 and 2.85 times 10 to the minus 14 respectively. These small errors support this derivative expansion at the declared mediator mass 10; they are not bounds for arbitrary masses or changed coefficients.

In a mostly-minus Minkowski convention, integrating out canonical Gaussian scalars gives the Lagrangian contribution  $+J\text{-transpose } (M^2 + \Box)\text{-inverse } J/2$ . Its zero-derivative effective potential is  $-J^2/(2M^2)$ ; integration by parts gives the positive kinetic term  $(\partial J)^2/(2M^4)$ . The higher derivative terms are fixed by the same inverse operator. Potential coefficients, scalar derivative coefficients and scalar residues therefore cannot be chosen independently within this completion. The former tree matching of a quartic potential is recovered as the zero-derivative limit. The new tree derivative term is distinct from the preceding chapter's one-loop wavefunction threshold.

## What first-order mass-phase reality does and does not enforce

For a canonical triangular fermion matrix  $D = (a I, 0; C, M I)$ ,  $C$  Hermitian and  $a, M$  real positive, each neutral real-scalar vertex has the form  $G_\alpha = (b_\alpha I, 0; H_\alpha, 0)$ , with  $b_\alpha$  real and  $H_\alpha$  Hermitian. The established paired-vertex theorem gives  $\text{Im Tr}(D\text{-inverse } \delta D) = 0$  for arbitrary real scalar mixing and positive scalar masses. This remains a first-order statement in the one-loop mass correction.

Every corrected vertex contribution still has zero right column block, so the determinant of  $D + \delta D$  depends only on its corrected upper-left three-by-three block. In the basis diagonalizing  $C$ , a real spectral kernel  $f_\alpha(D D^\dagger)$  has upper contractions  $L_\alpha(C)$  and  $K_\alpha(C)$ , both real functions of  $C$ . The upper mass correction is proportional to the sum of  $b_\alpha^2 L_\alpha(C) + b_\alpha K_\alpha(C) H_\alpha$ . The second term need not be Hermitian because its two Hermitian factors need not commute. Its trace is real; its higher trace powers need not be.

An exact resolvent kernel makes this distinction algebraically transparent. For  $f_s(X) = (X+sI)^{-1}$ ,  $L_s = a(M^2+s)/\text{den}$  and  $K_s = sC/\text{den}$ , with  $\text{den} = (a^2+s)(M^2+s)+sC^2$ . Take  $C = \text{diag}(2/5, 7/10, 11/10)$ ,  $a = 1/10$ ,  $M = 1$  and scalar vertices consisting of a Higgs coefficient  $1/5$ , a real 12 off-diagonal source coefficient  $1/5$  and an imaginary Hermitian 12 source coefficient  $1/5$ . Mix their three scalar fields with the rational orthogonal matrix having rows  $(1, 2, 2)/3$ ,  $(2, 1, -2)/3$  and  $(2, -2, 1)/3$ ; the squared masses are  $3/10, 4/5, 7/5$ . The linear determinant phase coefficient vanishes exactly. The quadratic coefficient is the nonzero rational  $-5152/727803505$  before the loop-normalization factor. Exact expansion of the full three-by-three determinant verifies the coefficient.

The resolvent is a structural example, not a substitute for the physical loop function. Evaluating the actual finite B0 function for the same data gives a quadratic determinant phase  $-4.480088664908940366$  times 10 to the minus 10 radians and a resummed phase  $-4.481809518794191284$  times 10 to the minus 10 radians. The full six-by-six SVD calculation agrees with an independent 80-digit divided-difference evaluation of the upper mass block. Its relative correction norm is below 0.000533; its first-order phase is zero analytically. Formal scaling of the one-loop correction verifies the quadratic approach to the zero-correction limit.

The independent four-scalar example imposes an additional structural restriction: the Higgs Hessian row couples only to a diagonal source, with zero direct Higgs mixing into either off-diagonal source. The diagonal source mixes with the real and imaginary off-diagonal sources in the remaining positive Hessian. The first-order phase is below 10 to the minus 16, while the resummed one-loop matrix phase is approximately  $1.4070541$  times 10 to the minus 10 with relative correction norm below 0.02. This shows that a Higgs row aligned to a commuting source direction is not alone enough to extend the first-order theorem through propagation across the scalar sector.

Both examples are spectral tests. A complete invariant scalar potential with those Hessians, its stationary vacuum and all gauge/fermion sectors have not been supplied. More importantly, the quadratic determinant term of a one-loop-corrected matrix is only one contribution at the next perturbative order. Genuine two-loop self energies and the remaining matching terms can contribute at that order. These examples disprove the inference that first-order reality automatically protects the resummed determinant; they do not calculate a complete two-loop strong-CP prediction for the repository model.

## A protective candidate with an exact running test

If no scalar mass eigenmode contains both a Higgs vertex and a source vertex, source modes have  $b_\alpha = 0$  and do not correct the top block. Higgs modes have  $H_\alpha = 0$  and correct it only by real functions  $L_\alpha(C)$ . The corrected top block is then Hermitian and commutes with  $C$ . Its determinant ratio remains real and positive whenever the normalized correction norm is below one. This protects every power in that restricted determinant resummation. It remains a statement about one-loop matrices; it does not address all diagrams at arbitrary loop order.

The candidate separation is not stable under the stated Yukawa interactions. At zero Higgs portals, pure source quartics and the Higgs self quartic share no scalar-loop product, while off-diagonal source wavefunctions cannot generate a source/Higgs product from those separated potentials. The colored fermion box does generate them. For each real charge sector with Higgs coefficient  $y$ , source singlet coefficients  $e, t$  and adjoint coefficient  $a$ , its four source-quadratic portal beta coefficients are  $-36 y^2 e^2$ ,  $-36 y^2 t^2$ ,  $-72 y^2 e t$  and  $-12 y^2 a^2$ . These are  $16 \pi^2$ -normalized coefficients multiplying  $H^\dagger H$  times  $\eta^2$ ,  $s^2$ ,  $\eta s$  and the adjoint norm respectively.

All four vanish exactly only if  $y = 0$  or  $e = t = a = 0$  in that sector. The diagonal coefficients establish this directly by positivity for real couplings. Nonzero Higgs masses together with nonzero source mixing therefore prevent this zero-portal subspace from closing under the specified running. Ordinary phase charges cannot forbid these neutral products while allowing their factors. Additional particles, a different interaction structure or justified cancellation relations can alter the conclusion; none is supplied by setting these portals to zero as a boundary input. The new result specifies exactly what a protection mechanism must change.

## Connecting the scalar metric to conditional CP walls

$$\sigma_{\text{EFT}} \leq \sigma_{\text{UV}} \leq \sigma_{\text{valley}}$$

The merged cosmology layer supplies exact polynomial-path bounds for canonical full fields. Gaussian matching provides the previously missing derivative contribution on its valley: for any path  $x(t)$ , the canonical speed squared is  $\text{norm}(x\text{-prime})^2 + \sum_a (J_a\text{-prime})^2 / M_a$  to the fourth. This is precisely the induced tree metric  $K(0)$ , evaluated on the path. A source-only canonical path omits that positive term.

When the supplied endpoints are degenerate global minima of the source potential and the Gaussian completion, every full static profile has energy at least that of its projected canonical source profile: the heavy-field kinetic energy and the Gaussian potential square are nonnegative. Conversely, the Gaussian valley is an admissible full-field path. Taking infima gives  $\sigma_{\text{canonical\_EFT}} \leq \sigma_{\text{full\_UV}} \leq \sigma_{\text{valley}}$ . The last inequality is an upper-bound construction, not an assertion that a true wall stays on the valley; heavy gradients can favor departure. The endpoints must be global. The quark benchmark's lower CP-conserving competitor prevents applying this stable-wall conclusion to its retained CP-breaking branch.

The exact two-field example is  $V = \lambda(x^2 - v^2)^2 / 4 + M^2(S + g x^2 / M^2)^2 / 2$  with canonical  $x, S$ . Its only global minima are  $x = \pm v$ ,  $S = -g v^2 / M^2$ , from simultaneous vanishing of the two positive squares. CP acts as  $x \rightarrow -x$  and  $S \rightarrow S$ . The canonical source wall has tension squared  $8 \lambda v^6 / 9$ . On the polynomial valley path  $x = v(2t-1)$ , the cosmology engine's Cauchy bound is  $\sigma^2 \leq 16 \lambda v^6 / 15 \times [1 + 4 g^2 v^2 / (3 M^4)]$ .

For  $\lambda = v = g = 1$  and  $M = 3$ , the full UV tension obeys  $8/9 \leq \sigma_{\text{full}}^2 \leq 3952/3645$ . SymPy integration and the independently implemented BigInt path engine return the same rational upper bound. The latter receives both canonical path coordinates, including  $S(t)$ , rather than an unexplained metric input. The potential's composition and global endpoints are verified explicitly for this example. These are dimensionless toy results, not a flavor wall tension or thermal-history prediction. The other session's conditional bias optimization and restoration tools remain applicable only when their physical hypotheses are supplied.

## Reproducibility and mechanism consequence

Run `PYTHONPATH=python OPENBLAS_NUM_THREADS=1 OMP_NUM_THREADS=1 python python/develop_flavor_spectral_matching.py` to regenerate `receipts/m22_interactions/flavor_spectral_matching.json`. Run `node research/flavor_cosmology/gaussian_wall_bridge.mjs` for the independent rational bridge, and `node research/flavor_cosmology/test_cp_wall_tools.mjs` for the merged cosmology module's 104 assertions. The dedicated Python tests independently check unequal-mass Schur identities, finite pole bounds, residue derivatives, degenerate modes, the full completion spectra, high-precision  $B_0$  phases, separated scalar modes and regenerated portals. Validation records full related-test counts, receipt hashes and fresh-extraction results separately.

The new mechanism obligation is specific: a proposed separation must survive the portal operators forced by its own quark couplings, or supply another exact structure that constrains the higher determinant terms. Gaussian scalar exchange already enforces derivative correlations and permits controlled finite matching, but that scalar relation does not itself fix quark kinetic operators, cross-sector flavor coefficients, golden CKM or a gauge boundary. The first-order phase theorem, new finite pole formulas and new sequestering exclusion coexist without promoting a partial determinant expansion to a full two-loop prediction.

The finite  $B_0$  convention and neutral-scalar mass-threshold normalization are inherited from `python/perfectpower/flavor_quantum.py` and the preceding Hermitian-source chapters. The general-RGE primary reference remains <https://arxiv.org/pdf/1809.06797>. Conditional wall mechanics and the literature context are recorded in `docs/FLAVOR_COSMOLOGY_MONOGRAPH.md`. All new spectral and exact rational calculations here are executable repository results; no independent publication-priority review is asserted.